

In this document, we present extended tables for the journal paper that give the coverage, mean time, and mean number of Q -values for selected algorithms per problem, split by heuristics. The tables are associated with the journal paper's sections thus:

7.4 What are the best versions of CG-iLAO?* Compares different versions of CG-iLAO*.

- Tab. 1 – ROC
- Tab. 2 – PDB
- Tab. 3 – LMCut

7.5 Comparing CG-iLAO to the State-of-the-art* Compares CG-iLAO* to iLAO* and LRTDP.

- Tab. 4 – ROC
- Tab. 5 – PDB
- Tab. 6 – LMCut

These results and more are also available in CSV format, where we give all the algorithms considered in this paper with additional statistics such as heuristic calls and partial SSP sizes.

	CG-iLAO* _{ied}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}	
BW	04-01	5 (0.2 ± 0.0) (4.4 ± 0.4)	5 (0.2 ± 0.0) (5.4 ± 0.8)	5 (0.2 ± 0.0) (5.0 ± 1.0)	5 (0.4 ± 0.0) (4.8 ± 0.1)	5 (0.3 ± 0.0) (3.9 ± 0.2)
	04-02	5 (0.2 ± 0.0) (2.9 ± 0.2)	5 (0.2 ± 0.0) (4.5 ± 0.5)	5 (0.2 ± 0.0) (4.6 ± 1.5)	5 (0.3 ± 0.0) (3.8 ± 0.2)	5 (0.3 ± 0.0) (3.5 ± 0.2)
	04-03	5 (0.2 ± 0.0) (0.5 ± 0.2)	5 (0.1 ± 0.0) (1.4 ± 0.4)	5 (0.2 ± 0.0) (2.1 ± 1.1)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)
	04-04	5 (0.2 ± 0.0) (7.4 ± 0.1)	5 (0.2 ± 0.0) (10.8 ± 1.7)	5 (0.2 ± 0.0) (9.5 ± 1.1)	5 (0.6 ± 0.0) (10.2 ± 0.5)	5 (0.4 ± 0.0) (7.9 ± 0.7)
	04-05	5 (0.2 ± 0.0) (2.0 ± 0.1)	5 (0.2 ± 0.0) (3.3 ± 0.7)	5 (0.2 ± 0.0) (3.6 ± 1.3)	5 (0.3 ± 0.0) (3.0 ± 0.1)	5 (0.2 ± 0.0) (2.2 ± 0.1)
	04-06	5 (67.5 ± 1.3) (834.4 ± 8.8)	5 (70.1 ± 5.3) (1041.7 ± 10.9)	5 (59.8 ± 4.4) (1066.7 ± 11.8)	5 (206.3 ± 3.6) (1053.8 ± 9.6)	5 (253.5 ± 3.2) (1000.0 ± 7.6)
	04-08	5 (17.8 ± 1.3) (266.3 ± 7.4)	5 (16.2 ± 1.0) (266.2 ± 12.6)	5 (15.0 ± 0.9) (291.2 ± 16.2)	5 (44.3 ± 3.3) (245.2 ± 2.7)	5 (29.4 ± 1.8) (224.8 ± 4.6)
	04-09	5 (14.4 ± 1.2) (173.8 ± 4.4)	5 (13.8 ± 1.4) (194.2 ± 6.0)	5 (13.0 ± 1.0) (227.5 ± 3.3)	5 (33.5 ± 1.9) (187.3 ± 2.8)	5 (69.5 ± 1.9) (184.9 ± 4.0)
	04-10	5 (170.9 ± 9.9) (2250.5 ± 21.5)	5 (172.7 ± 6.9) (2906.7 ± 34.8)	5 (143.2 ± 4.9) (3029.4 ± 37.0)	5 (874.8 ± 77.0) (2812.4 ± 15.3)	5 (1474.2 ± 18.1) (2765.3 ± 19.6)
	08-01	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)	5 (0.2 ± 0.0) (6.3 ± 1.6)	5 (0.2 ± 0.0) (2.2 ± 0.3)	5 (0.2 ± 0.0) (2.0 ± 0.2)
	08-02	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)	5 (0.2 ± 0.0) (5.6 ± 1.8)	5 (0.3 ± 0.0) (2.2 ± 0.3)	5 (0.2 ± 0.0) (2.0 ± 0.2)
	08-03	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)	5 (0.2 ± 0.0) (5.7 ± 1.9)	5 (0.3 ± 0.0) (2.2 ± 0.3)	5 (0.2 ± 0.0) (2.0 ± 0.2)
08-04	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)	5 (0.2 ± 0.0) (6.7 ± 1.4)	5 (0.3 ± 0.0) (2.2 ± 0.3)	5 (0.2 ± 0.0) (2.0 ± 0.2)	
08-05	5 (361.1 ± 19.4) (6615.8 ± 53.7)	5 (351.0 ± 6.5) (8267.2 ± 152.7)	5 (323.6 ± 27.2) (8730.4 ± 50.6)	3 (1204.1 ± 15.1) (8032.0 ± 82.6)	5 (809.0 ± 34.2) (8067.9 ± 36.6)	
08-06	5 (354.8 ± 18.0) (6615.8 ± 53.7)	5 (348.2 ± 11.2) (8267.2 ± 152.7)	5 (320.9 ± 36.6) (8720.8 ± 45.8)	4 (1235.1 ± 26.4) (8025.4 ± 53.5)	5 (822.0 ± 38.5) (8067.9 ± 36.6)	
08-07	5 (359.9 ± 24.3) (6615.8 ± 53.7)	5 (370.4 ± 20.8) (8267.2 ± 152.7)	5 (322.0 ± 26.5) (8731.9 ± 28.8)	4 (1217.9 ± 22.7) (8029.1 ± 58.7)	5 (807.2 ± 9.6) (8067.9 ± 36.6)	
08-08	5 (358.6 ± 9.0) (6615.8 ± 53.7)	5 (363.2 ± 25.2) (8267.2 ± 152.7)	5 (327.9 ± 15.5) (8732.3 ± 47.4)	4 (1213.9 ± 8.0) (8044.8 ± 63.6)	5 (813.0 ± 9.5) (8067.9 ± 36.6)	
08-09	5 (321.0 ± 36.5) (708.9 ± 26.8)	5 (293.9 ± 31.8) (912.6 ± 31.9)	5 (293.8 ± 41.0) (1022.8 ± 21.9)	5 (1215.1 ± 32.0) (878.9 ± 16.3)		
08-10	5 (338.1 ± 30.5) (708.9 ± 26.8)	5 (327.6 ± 44.6) (912.6 ± 31.9)	5 (335.3 ± 70.0) (1028.8 ± 16.1)	5 (1222.7 ± 34.4) (878.9 ± 16.3)		
08-11	5 (352.6 ± 17.8) (708.9 ± 26.8)	5 (334.3 ± 20.5) (912.6 ± 31.9)	5 (268.8 ± 55.5) (1030.4 ± 13.8)	5 (1228.6 ± 50.9) (878.9 ± 16.3)		
08-12	5 (297.5 ± 40.9) (708.9 ± 26.8)	5 (270.0 ± 19.7) (912.6 ± 31.9)	5 (292.6 ± 43.2) (1030.1 ± 17.3)	5 (1218.2 ± 72.8) (878.9 ± 16.3)		

results for **ROC** continued on next page

		CG-iLAO* _{tid}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
Coresec	01	5 (0.7 ± 0.0) ⟨100.9 ± 0.7⟩	5 (0.7 ± 0.0) ⟨94.6 ± 1.1⟩	5 (0.7 ± 0.0) ⟨98.9 ± 2.0⟩	5 (8.1 ± 0.8) ⟨71.4 ± 0.5⟩	5 (15.5 ± 1.5) ⟨71.4 ± 0.5⟩
	02	5 (2.1 ± 0.1) ⟨331.1 ± 3.8⟩	5 (2.1 ± 0.1) ⟨310.4 ± 3.1⟩	5 (2.2 ± 0.0) ⟨331.4 ± 1.4⟩	5 (23.6 ± 1.8) ⟨231.1 ± 2.3⟩	5 (44.4 ± 3.6) ⟨231.1 ± 2.3⟩
	03	5 (5.9 ± 0.1) ⟨1161.1 ± 8.6⟩	5 (5.9 ± 0.1) ⟨1099.9 ± 4.8⟩	5 (6.4 ± 0.2) ⟨1221.6 ± 2.1⟩	5 (69.2 ± 4.8) ⟨854.5 ± 3.3⟩	5 (134.2 ± 9.9) ⟨854.5 ± 3.3⟩
	04	5 (16.2 ± 0.4) ⟨3024.4 ± 12.0⟩	5 (16.0 ± 0.2) ⟨2871.6 ± 13.2⟩	5 (17.1 ± 0.2) ⟨3154.2 ± 7.8⟩	5 (181.5 ± 12.3) ⟨2210.8 ± 2.6⟩	5 (344.0 ± 27.3) ⟨2210.8 ± 2.6⟩
	05	5 (397.4 ± 14.3) ⟨67262.5 ± 60.6⟩	5 (413.1 ± 15.9) ⟨64252.3 ± 112.7⟩	5 (451.1 ± 23.0) ⟨70127.3 ± 46.1⟩		
Elev	04-01	5 (0.2 ± 0.0) ⟨31.0 ± 1.9⟩	5 (0.2 ± 0.0) ⟨34.2 ± 2.2⟩	5 (0.2 ± 0.0) ⟨37.8 ± 3.3⟩	5 (0.8 ± 0.1) ⟨24.2 ± 0.3⟩	5 (0.8 ± 0.1) ⟨24.5 ± 0.7⟩
	04-02	5 (0.1 ± 0.0) ⟨1.5 ± 0.1⟩	5 (0.1 ± 0.0) ⟨2.1 ± 0.2⟩	5 (0.1 ± 0.0) ⟨4.4 ± 3.7⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.1⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩
	04-03	5 (0.2 ± 0.0) ⟨34.9 ± 2.2⟩	5 (0.2 ± 0.0) ⟨38.0 ± 1.8⟩	5 (0.2 ± 0.0) ⟨44.6 ± 3.1⟩	5 (0.8 ± 0.1) ⟨30.7 ± 0.5⟩	5 (0.8 ± 0.1) ⟨31.1 ± 0.5⟩
	04-04	5 (0.1 ± 0.0) ⟨20.7 ± 0.1⟩	5 (0.1 ± 0.0) ⟨22.8 ± 0.4⟩	5 (0.2 ± 0.0) ⟨25.4 ± 0.6⟩	5 (0.6 ± 0.0) ⟨19.5 ± 0.4⟩	5 (0.6 ± 0.1) ⟨19.7 ± 0.3⟩
	04-05	5 (0.1 ± 0.0) ⟨12.8 ± 0.2⟩	5 (0.1 ± 0.0) ⟨14.3 ± 0.5⟩	5 (0.1 ± 0.0) ⟨16.0 ± 0.7⟩	5 (0.6 ± 0.0) ⟨11.1 ± 0.6⟩	5 (0.5 ± 0.1) ⟨11.3 ± 0.4⟩
	04-06	5 (1.1 ± 0.0) ⟨323.6 ± 23.7⟩	5 (1.2 ± 0.0) ⟨339.7 ± 18.4⟩	5 (1.5 ± 0.1) ⟨416.8 ± 19.7⟩	5 (8.2 ± 0.6) ⟨366.5 ± 3.1⟩	5 (7.8 ± 0.8) ⟨356.3 ± 2.2⟩
	04-07	5 (1.2 ± 0.1) ⟨359.7 ± 7.7⟩	5 (1.2 ± 0.1) ⟨380.5 ± 13.1⟩	5 (1.5 ± 0.1) ⟨464.3 ± 12.1⟩	5 (7.7 ± 0.7) ⟨336.3 ± 20.1⟩	5 (7.7 ± 0.6) ⟨340.1 ± 20.4⟩
	04-08	5 (3.5 ± 0.1) ⟨1847.7 ± 22.4⟩	5 (3.7 ± 0.1) ⟨1950.9 ± 16.2⟩	5 (4.0 ± 0.1) ⟨2218.8 ± 8.9⟩	5 (19.6 ± 1.2) ⟨1831.2 ± 8.2⟩	5 (26.6 ± 7.2) ⟨1831.6 ± 9.4⟩
	04-09	5 (2.8 ± 0.1) ⟨1212.9 ± 21.4⟩	5 (2.9 ± 0.1) ⟨1290.6 ± 17.5⟩	5 (3.2 ± 0.1) ⟨1589.7 ± 11.6⟩	5 (17.7 ± 1.8) ⟨1213.5 ± 9.9⟩	5 (17.8 ± 2.3) ⟨1199.7 ± 16.9⟩
	04-10	5 (5.5 ± 0.2) ⟨4483.0 ± 74.3⟩	5 (5.7 ± 0.2) ⟨4588.2 ± 99.1⟩	5 (7.0 ± 0.2) ⟨5931.3 ± 29.9⟩	5 (21.2 ± 1.7) ⟨4781.4 ± 57.5⟩	5 (26.3 ± 3.3) ⟨4815.7 ± 32.1⟩
	04-11	5 (324.5 ± 30.9) ⟨181799.6 ± 2298.6⟩	5 (347.0 ± 35.3) ⟨184142.2 ± 1767.0⟩	5 (396.4 ± 10.5) ⟨225230.6 ± 2044.2⟩	1 (925.8±) ⟨189243.6±⟩	1 (918.0±) ⟨190354.1±⟩
	04-12	5 (75.0 ± 2.1) ⟨35569.6 ± 539.2⟩	5 (79.8 ± 1.9) ⟨36463.0 ± 550.7⟩	5 (93.8 ± 2.3) ⟨42070.4 ± 101.5⟩	5 (441.6 ± 25.4) ⟨36068.3 ± 203.7⟩	5 (640.1 ± 211.5) ⟨36432.5 ± 277.0⟩
	04-13	5 (1471.3 ± 112.9) ⟨1142941.6 ± 10195.4⟩	5 (1505.3 ± 110.0) ⟨1124113.1 ± 5931.4⟩			
	04-14	5 (163.5 ± 5.5) ⟨84851.3 ± 1605.0⟩	5 (174.5 ± 5.8) ⟨86241.8 ± 1638.0⟩	5 (192.4 ± 3.6) ⟨102397.4 ± 1408.5⟩	3 (774.4 ± 35.4) ⟨78528.9 ± 17.8⟩	4 (771.5 ± 41.2) ⟨77213.4 ± 285.5⟩
	04-15	5 (407.8 ± 23.7) ⟨233426.6 ± 1388.6⟩	5 (415.2 ± 16.4) ⟨235101.4 ± 1406.1⟩	5 (456.1 ± 14.7) ⟨268864.1 ± 991.7⟩		1 (942.4±) ⟨232247.5±⟩

results for **ROC** continued on next page

	CG-iLAO* _{tied}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
ExBW	04-01	5 (0.1 ± 0.0) (0.1 ± 0.0)	5 (0.1 ± 0.0) (0.1 ± 0.0)	5 (0.1 ± 0.0) (0.3 ± 0.2)	5 (0.1 ± 0.0) (0.1 ± 0.0)
	04-02	5 (0.1 ± 0.0) (0.1 ± 0.0)	5 (0.1 ± 0.0) (0.1 ± 0.0)	5 (0.1 ± 0.0) (0.3 ± 0.1)	5 (0.1 ± 0.0) (0.1 ± 0.0)
	04-03	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.4 ± 0.2)	5 (0.1 ± 0.0) (0.1 ± 0.0)
	04-04	5 (0.2 ± 0.0) (35.8 ± 3.3)	5 (0.2 ± 0.0) (36.5 ± 2.8)	5 (0.2 ± 0.0) (28.4 ± 1.2)	5 (0.3 ± 0.1) (1.2 ± 0.0)
	04-05	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)
	04-06	5 (0.6 ± 0.0) (63.0 ± 0.4)	5 (0.6 ± 0.1) (58.9 ± 4.6)	5 (1.5 ± 0.2) (26.0 ± 2.5)	5 (1.4 ± 0.2) (56.4 ± 0.5)
	04-07	5 (11.0 ± 0.3) (453.4 ± 8.5)	5 (10.8 ± 0.2) (455.0 ± 6.4)	5 (19.7 ± 0.7) (663.5 ± 36.2)	5 (31.5 ± 1.0) (436.1 ± 3.6)
	04-08	5 (0.9 ± 0.0) (4.7 ± 0.1)	5 (0.8 ± 0.2) (4.6 ± 1.6)	5 (1.9 ± 0.2) (17.6 ± 2.4)	5 (0.9 ± 0.0) (2.9 ± 0.0)
	04-09	5 (41.1 ± 0.9) (3535.5 ± 25.5)	5 (39.2 ± 0.9) (3284.1 ± 159.0)	5 (68.8 ± 4.1) (4791.6 ± 177.8)	5 (151.2 ± 12.2) (4036.9 ± 155.2)
	04-10	5 (239.9 ± 6.1) (27692.1 ± 291.6)	5 (236.5 ± 9.2) (27102.2 ± 454.4)	5 (378.5 ± 15.7) (34211.6 ± 422.8)	5 (576.3 ± 22.0) (20710.8 ± 111.6)
	08-01	5 (0.1 ± 0.0) (0.7 ± 0.0)	5 (0.1 ± 0.0) (0.7 ± 0.0)	5 (0.1 ± 0.0) (1.3 ± 0.2)	5 (0.2 ± 0.0) (0.6 ± 0.0)
	08-02	5 (0.2 ± 0.0) (17.1 ± 0.4)	5 (0.2 ± 0.0) (25.7 ± 2.8)	5 (0.2 ± 0.0) (38.5 ± 0.6)	5 (0.1 ± 0.0) (0.5 ± 0.0)
	08-03	5 (0.2 ± 0.0) (19.3 ± 0.3)	5 (0.2 ± 0.0) (22.6 ± 7.8)	5 (0.3 ± 0.0) (45.4 ± 1.2)	5 (0.4 ± 0.0) (38.2 ± 0.8)
	08-04	5 (0.3 ± 0.0) (27.3 ± 0.2)	5 (0.3 ± 0.0) (27.6 ± 0.3)	5 (0.4 ± 0.0) (24.3 ± 0.4)	5 (0.7 ± 0.1) (6.2 ± 0.1)
	08-05	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.3 ± 0.0)	5 (0.1 ± 0.0) (1.1 ± 0.2)	5 (0.1 ± 0.0) (0.1 ± 0.0)
	08-06	5 (0.3 ± 0.0) (37.8 ± 0.7)	5 (0.3 ± 0.0) (36.9 ± 1.2)	5 (0.6 ± 0.0) (21.1 ± 1.5)	5 (0.7 ± 0.0) (29.1 ± 10.9)
	08-07	5 (0.4 ± 0.1) (3.1 ± 0.5)	5 (0.5 ± 0.1) (5.3 ± 1.3)	5 (1.3 ± 0.2) (14.3 ± 1.7)	5 (0.8 ± 0.0) (4.6 ± 0.6)
	08-08	5 (50.7 ± 4.1) (8422.5 ± 75.0)	5 (51.3 ± 3.6) (7748.8 ± 842.4)	5 (90.5 ± 5.1) (8230.8 ± 319.8)	
	08-09	5 (5.6 ± 0.2) (457.4 ± 9.4)	5 (6.6 ± 0.2) (914.2 ± 48.4)	5 (10.1 ± 0.6) (1081.6 ± 50.3)	5 (47.0 ± 1.9) (965.5 ± 8.9)
	08-10	5 (165.5 ± 13.7) (79741.2 ± 2529.6)	5 (204.0 ± 67.7) (78362.9 ± 3960.4)	5 (277.8 ± 42.1) (99607.6 ± 3640.8)	5 (696.3 ± 26.1) (69297.9 ± 175.0)
	08-11	5 (386.5 ± 14.2) (10930.6 ± 90.2)	5 (391.7 ± 8.4) (11340.7 ± 190.3)	5 (645.6 ± 36.0) (16008.6 ± 517.2)	

results for **ROC** continued on next page

		CG-iLAO* _{tie}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
PARC-N	4, 1	5 (9.3 ± 0.3) ⟨1027.5 ± 10.0⟩	5 (9.2 ± 0.4) ⟨719.2 ± 9.4⟩	5 (8.8 ± 0.6) ⟨898.1 ± 13.6⟩	5 (18.9 ± 0.9) ⟨765.7 ± 3.8⟩	5 (19.4 ± 1.1) ⟨765.7 ± 3.8⟩
	4, 2	5 (14.0 ± 0.3) ⟨1678.7 ± 36.8⟩	5 (13.8 ± 0.5) ⟨1094.9 ± 4.3⟩	5 (13.1 ± 1.1) ⟨1332.5 ± 30.3⟩	5 (25.9 ± 2.1) ⟨1033.5 ± 8.7⟩	5 (26.1 ± 0.5) ⟨1033.5 ± 8.7⟩
	4, 3	5 (19.2 ± 0.8) ⟨2902.4 ± 82.8⟩	5 (18.6 ± 1.0) ⟨1676.8 ± 23.2⟩	5 (18.7 ± 1.7) ⟨2108.2 ± 116.2⟩	5 (36.0 ± 8.3) ⟨1528.3 ± 4.8⟩	5 (33.4 ± 2.7) ⟨1528.3 ± 4.8⟩
	5, 1	5 (104.9 ± 3.9) ⟨12478.9 ± 325.1⟩	5 (104.3 ± 2.3) ⟨7946.2 ± 53.6⟩	5 (97.0 ± 2.6) ⟨9992.6 ± 239.1⟩	5 (203.2 ± 30.1) ⟨8597.7 ± 21.2⟩	5 (201.9 ± 18.8) ⟨8597.7 ± 21.2⟩
	5, 2	5 (161.0 ± 8.7) ⟨20373.5 ± 406.7⟩	5 (155.9 ± 7.8) ⟨12420.7 ± 58.6⟩	5 (146.0 ± 5.3) ⟨14824.3 ± 471.7⟩	5 (279.6 ± 40.1) ⟨11977.5 ± 56.0⟩	5 (276.3 ± 23.9) ⟨11977.5 ± 56.0⟩
	5, 3	5 (218.8 ± 12.9) ⟨36113.3 ± 1005.8⟩	5 (204.7 ± 13.4) ⟨18934.2 ± 96.6⟩	5 (202.4 ± 13.0) ⟨23139.8 ± 578.3⟩	5 (304.9 ± 14.7) ⟨17792.1 ± 78.4⟩	5 (327.4 ± 11.8) ⟨17792.1 ± 78.4⟩
PARC-R	4, 1	5 (33.6 ± 2.0) ⟨6146.0 ± 49.9⟩	5 (33.5 ± 2.3) ⟨5817.6 ± 33.8⟩	5 (31.7 ± 0.8) ⟨6661.3 ± 14.6⟩	5 (74.2 ± 2.0) ⟨6390.8 ± 13.8⟩	5 (73.2 ± 2.3) ⟨6390.8 ± 13.8⟩
	4, 2	5 (84.4 ± 1.4) ⟨14664.1 ± 69.2⟩	5 (85.2 ± 4.4) ⟨14490.7 ± 71.7⟩	5 (86.8 ± 5.8) ⟨16827.7 ± 81.3⟩	5 (200.6 ± 9.8) ⟨16967.5 ± 42.2⟩	5 (192.9 ± 7.0) ⟨16967.5 ± 42.2⟩
	4, 3	5 (119.3 ± 4.3) ⟨21937.1 ± 56.3⟩	5 (119.2 ± 4.1) ⟨21833.5 ± 144.0⟩	5 (130.9 ± 3.3) ⟨24052.6 ± 167.0⟩	5 (274.3 ± 6.2) ⟨24076.3 ± 56.9⟩	5 (298.4 ± 43.1) ⟨24076.3 ± 56.9⟩
	5, 1	5 (498.8 ± 107.5) ⟨62262.2 ± 641.7⟩	5 (450.7 ± 14.3) ⟨58383.3 ± 118.4⟩	5 (410.8 ± 7.2) ⟨64622.4 ± 199.3⟩	5 (930.3 ± 10.5) ⟨65668.5 ± 201.2⟩	5 (987.8 ± 112.7) ⟨65668.5 ± 201.2⟩
	5, 2	5 (1227.0 ± 23.2) ⟨159572.7 ± 498.4⟩	5 (1225.0 ± 31.1) ⟨157581.7 ± 403.4⟩	4 (1166.7 ± 15.4) ⟨162551.1 ± 227.4⟩		
Rand	04-03	5 (0.3 ± 0.0) ⟨0.6 ± 0.1⟩	5 (0.3 ± 0.0) ⟨0.6 ± 0.1⟩	5 (0.3 ± 0.0) ⟨1.0 ± 0.9⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	04-04	5 (0.5 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.5 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.5 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.3 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.0 ± 0.0⟩
	04-05	5 (13.3 ± 4.1) ⟨9.8 ± 4.9⟩	5 (15.5 ± 7.8) ⟨11.0 ± 6.0⟩	5 (87.4 ± 44.4) ⟨156.3 ± 46.4⟩	5 (19.0 ± 4.4) ⟨11.0 ± 3.0⟩	5 (19.1 ± 4.0) ⟨11.0 ± 3.0⟩
	04-06	5 (1.5 ± 0.1) ⟨2.4 ± 0.0⟩	5 (1.5 ± 0.1) ⟨1.3 ± 0.1⟩	5 (1.5 ± 0.1) ⟨1.1 ± 0.1⟩	5 (0.7 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.6 ± 0.0) ⟨0.0 ± 0.0⟩
	04-07	5 (0.6 ± 0.3) ⟨1.3 ± 1.2⟩	5 (0.6 ± 0.3) ⟨1.8 ± 1.5⟩	5 (0.7 ± 0.3) ⟨1.8 ± 1.8⟩	5 (1.2 ± 0.0) ⟨3.2 ± 0.0⟩	5 (1.2 ± 0.0) ⟨3.2 ± 0.0⟩
	04-08		5 (1471.1 ± 122.8) ⟨811.2 ± 5.3⟩			5 (30.3 ± 0.8) ⟨63.9 ± 0.0⟩
	04-09	5 (5.8 ± 1.8) ⟨5.8 ± 2.1⟩	5 (5.9 ± 2.4) ⟨7.8 ± 7.0⟩	5 (57.2 ± 76.5) ⟨133.1 ± 193.8⟩	5 (6.7 ± 3.8) ⟨8.0 ± 5.6⟩	5 (6.0 ± 3.3) ⟨8.0 ± 5.6⟩
	04-10	1 (1577.2±) ⟨11125.7±⟩	3 (1657.5 ± 68.6) ⟨14217.0 ± 471.0⟩			
	04-13	5 (61.1 ± 9.9) ⟨93.7 ± 21.7⟩	5 (62.2 ± 9.3) ⟨118.7 ± 31.2⟩	5 (380.7 ± 190.9) ⟨812.0 ± 427.1⟩	5 (50.3 ± 25.5) ⟨56.0 ± 23.0⟩	5 (44.3 ± 18.1) ⟨57.2 ± 16.9⟩

results for **ROC** continued on next page

	CG-iLAO* _{tied}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
□TW	08-01	5 (0.2 ± 0.0) (1.3 ± 0.0)	5 (0.2 ± 0.0) (1.6 ± 0.1)	5 (0.2 ± 0.0) (1.7 ± 0.1)	5 (0.2 ± 0.0) (1.4 ± 0.1)
	08-02	5 (0.2 ± 0.0) (0.2 ± 0.0)	5 (0.2 ± 0.0) (0.2 ± 0.0)	5 (0.2 ± 0.0) (0.4 ± 0.0)	5 (0.2 ± 0.0) (0.0 ± 0.0)
	08-03	5 (0.5 ± 0.0) (3.3 ± 0.1)	5 (0.5 ± 0.0) (4.1 ± 0.1)	5 (0.6 ± 0.1) (4.5 ± 0.2)	5 (0.6 ± 0.0) (3.7 ± 0.2)
	08-04	5 (0.4 ± 0.0) (0.0 ± 0.0)	5 (0.4 ± 0.0) (0.0 ± 0.0)	5 (0.5 ± 0.0) (0.0 ± 0.0)	5 (0.4 ± 0.0) (0.0 ± 0.0)
	08-05	5 (1.1 ± 0.0) (0.0 ± 0.0)	5 (1.1 ± 0.0) (0.0 ± 0.0)	5 (1.1 ± 0.0) (0.0 ± 0.0)	5 (1.1 ± 0.1) (0.0 ± 0.0)
	08-06	5 (4.6 ± 0.5) (19.5 ± 0.6)	5 (4.9 ± 0.7) (23.0 ± 0.6)	5 (4.6 ± 0.5) (34.8 ± 1.1)	5 (8.8 ± 2.0) (19.8 ± 1.0)
	08-07	5 (2.9 ± 0.4) (0.0 ± 0.0)	5 (2.7 ± 0.1) (0.0 ± 0.0)	5 (2.6 ± 0.1) (0.0 ± 0.0)	5 (2.7 ± 0.2) (0.0 ± 0.0)
	08-08	5 (2.6 ± 0.1) (0.0 ± 0.0)	5 (2.7 ± 0.1) (0.0 ± 0.0)	5 (2.6 ± 0.0) (0.0 ± 0.0)	5 (2.6 ± 0.1) (0.0 ± 0.0)
	08-09	5 (30.1 ± 6.4) (56.8 ± 0.9)	5 (31.2 ± 6.5) (78.2 ± 4.4)	5 (27.4 ± 3.0) (91.1 ± 3.1)	5 (63.0 ± 7.6) (79.5 ± 1.3)
	08-10	5 (14.1 ± 0.9) (285.6 ± 0.0)	5 (13.6 ± 0.9) (305.1 ± 34.6)	5 (13.6 ± 0.7) (381.2 ± 1.1)	5 (9.8 ± 0.1) (0.0 ± 0.0)
	08-11	5 (184.2 ± 20.2) (144.0 ± 0.9)	5 (175.1 ± 10.8) (160.1 ± 0.7)	5 (190.9 ± 18.3) (167.3 ± 5.7)	5 (667.7 ± 108.6) (152.9 ± 2.0)
	08-12	5 (225.5 ± 39.3) (41169.3 ± 444.2)	5 (210.5 ± 21.9) (42430.1 ± 539.4)	5 (209.7 ± 15.9) (44442.1 ± 652.1)	5 (59.3 ± 4.4) (0.0 ± 0.0)
S&R	08-01	5 (0.3 ± 0.0) (404.5 ± 1.8)	5 (0.3 ± 0.0) (405.7 ± 2.1)	5 (0.4 ± 0.0) (395.7 ± 3.2)	5 (1.2 ± 0.1) (414.6 ± 1.3)
	08-02	5 (1.2 ± 0.0) (1787.5 ± 9.1)	5 (1.2 ± 0.0) (1795.6 ± 4.6)	5 (1.5 ± 0.1) (1759.1 ± 7.6)	5 (4.2 ± 0.4) (1847.5 ± 5.0)
	08-03	5 (5.0 ± 0.3) (7423.1 ± 15.0)	5 (5.2 ± 0.2) (7468.7 ± 5.4)	5 (5.8 ± 0.4) (7337.7 ± 17.3)	5 (16.0 ± 0.9) (7621.2 ± 12.7)
	08-04	5 (89.2 ± 3.5) (114677.4 ± 113.5)	5 (100.4 ± 3.5) (115476.7 ± 51.9)	5 (93.5 ± 5.5) (113890.4 ± 67.3)	5 (241.7 ± 18.8) (116958.7 ± 41.5)
	08-05			1 (1751.4±) (1601114.0±)	

results for **ROC** continued on next page

		CG-iLAO [*] _{tied}	CG-iLAO [*] _{single}	CG-iLAO [*] _{trial}	CG-iLAO [*] _{FF-AO}	CG-iLAO [*] _{FF-MLO}
Sched	04-01	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.5 ± 0.0⟩
	04-02	5 (0.1 ± 0.0) ⟨1.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩
	04-03	5 (0.1 ± 0.0) ⟨2.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.6 ± 0.1⟩	5 (0.1 ± 0.0) ⟨1.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.5 ± 0.0⟩
	04-04	5 (25.5 ± 0.3) ⟨17655.9 ± 235.9⟩	5 (24.4 ± 0.5) ⟨14902.0 ± 245.5⟩	5 (28.6 ± 1.5) ⟨14900.2 ± 430.4⟩	5 (78.6 ± 4.2) ⟨16445.2 ± 217.3⟩	5 (107.3 ± 14.4) ⟨16437.5 ± 170.9⟩
	04-05	5 (1086.1 ± 17.5) ⟨431646.4 ± 320.3⟩	5 (1072.6 ± 44.1) ⟨392032.1 ± 744.0⟩	5 (1192.3 ± 93.7) ⟨380347.4 ± 318.6⟩		
Sys	08-01	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩
	08-02	5 (0.1 ± 0.0) ⟨1.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.9 ± 0.1⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩
	08-03	5 (0.1 ± 0.0) ⟨3.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.7 ± 0.2⟩	5 (0.1 ± 0.0) ⟨1.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.6 ± 0.0⟩
	08-04	5 (924.1 ± 49.8) ⟨389615.4 ± 2072.1⟩	5 (864.7 ± 31.4) ⟨322241.5 ± 1009.9⟩	5 (958.7 ± 54.1) ⟨320247.4 ± 1586.3⟩		
	08-01	5 (0.1 ± 0.0) ⟨18.9 ± 0.2⟩	5 (0.1 ± 0.0) ⟨19.4 ± 0.5⟩	5 (0.1 ± 0.0) ⟨10.8 ± 0.3⟩	5 (0.3 ± 0.0) ⟨9.6 ± 0.2⟩	5 (0.3 ± 0.0) ⟨9.6 ± 0.2⟩
ΔTW	08-02	5 (0.4 ± 0.0) ⟨117.4 ± 2.9⟩	5 (0.4 ± 0.0) ⟨117.6 ± 4.0⟩	5 (0.4 ± 0.0) ⟨49.7 ± 0.8⟩	5 (0.9 ± 0.2) ⟨45.1 ± 1.5⟩	5 (1.0 ± 0.2) ⟨45.1 ± 1.5⟩
	08-03	5 (1.4 ± 0.1) ⟨412.5 ± 7.1⟩	5 (1.4 ± 0.1) ⟨418.2 ± 8.8⟩	5 (1.0 ± 0.0) ⟨168.1 ± 1.6⟩		5 (2.5 ± 0.1) ⟨151.5 ± 1.2⟩
	08-04	5 (15.9 ± 3.8) ⟨5121.2 ± 54.4⟩	5 (15.2 ± 2.2) ⟨5164.4 ± 14.9⟩	5 (10.9 ± 0.9) ⟨2017.2 ± 13.4⟩		
	08-01	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.1 ± 0.0⟩
	08-02	5 (0.1 ± 0.0) ⟨2.8 ± 0.1⟩	5 (0.1 ± 0.0) ⟨2.8 ± 0.1⟩	5 (0.1 ± 0.0) ⟨2.7 ± 0.2⟩	5 (0.3 ± 0.0) ⟨2.5 ± 0.1⟩	5 (0.4 ± 0.0) ⟨2.5 ± 0.1⟩
	08-03	5 (0.8 ± 0.0) ⟨67.0 ± 0.9⟩	5 (0.8 ± 0.0) ⟨83.5 ± 8.6⟩	5 (0.9 ± 0.0) ⟨80.5 ± 8.1⟩	5 (3.6 ± 0.2) ⟨81.7 ± 5.5⟩	5 (4.1 ± 0.2) ⟨72.5 ± 17.1⟩
	08-04	5 (12.2 ± 0.5) ⟨1715.7 ± 131.9⟩	5 (13.0 ± 0.2) ⟨2203.1 ± 222.7⟩	5 (15.8 ± 0.9) ⟨2100.8 ± 206.5⟩	5 (47.9 ± 6.1) ⟨2398.0 ± 102.5⟩	5 (48.3 ± 3.8) ⟨2122.2 ± 554.7⟩
	08-05	5 (294.3 ± 5.2) ⟨37932.0 ± 731.8⟩	5 (309.4 ± 7.0) ⟨43569.5 ± 1506.0⟩	5 (371.0 ± 7.1) ⟨36846.5 ± 935.0⟩	5 (852.6 ± 54.1) ⟨51806.9 ± 1437.0⟩	5 (800.1 ± 61.6) ⟨45513.7 ± 11968.2⟩
	6, 8	5 (399.5 ± 10.3) ⟨32067.6 ± 2114.9⟩	5 (406.1 ± 9.7) ⟨34670.7 ± 1410.5⟩	5 (453.7 ± 8.5) ⟨26127.1 ± 1161.8⟩		1 (842.3±) ⟨17415.8±⟩
	6, 9	5 (949.9 ± 40.6) ⟨80591.8 ± 5432.3⟩	5 (950.6 ± 32.2) ⟨84234.9 ± 2912.5⟩	5 (1056.8 ± 56.0) ⟨67577.7 ± 4224.3⟩		

results for **ROC** continued on next page

	CG-iLAO [*] _{tied}	CG-iLAO [*] _{single}	CG-iLAO [*] _{trial}	CG-iLAO [*] _{FF-AO}	CG-iLAO [*] _{FF-MLO}	
Zeno	04-01	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	
	04-02	5 (135.1 ± 5.1) ⟨38273.4 ± 59.2⟩	5 (138.6 ± 5.1) ⟨41686.6 ± 127.7⟩	5 (138.1 ± 4.2) ⟨36301.5 ± 385.7⟩	5 (550.1 ± 17.7) ⟨38831.3 ± 62.3⟩	2 (865.4 ± 6.0) ⟨38800.3 ± 103.2⟩
	04-03	5 (1416.0 ± 55.1) ⟨491318.1 ± 445.1⟩	5 (1493.9 ± 57.3) ⟨517970.7 ± 769.8⟩	5 (1221.1 ± 37.9) ⟨435422.9 ± 1833.2⟩		
	04-04	5 (158.6 ± 5.9) ⟨37429.5 ± 52.0⟩	5 (154.5 ± 3.4) ⟨36948.9 ± 219.7⟩	5 (193.8 ± 4.7) ⟨23102.2 ± 155.9⟩	5 (584.0 ± 28.9) ⟨33429.5 ± 20.7⟩	4 (874.4 ± 62.6) ⟨33423.8 ± 22.6⟩
	04-05	5 (523.5 ± 14.5) ⟨147085.2 ± 230.8⟩	5 (551.8 ± 11.9) ⟨160930.7 ± 694.5⟩	5 (551.8 ± 16.9) ⟨122024.3 ± 699.3⟩		
	08-01	5 (5.8 ± 0.2) ⟨1664.9 ± 4.7⟩	5 (5.8 ± 0.1) ⟨1768.0 ± 21.7⟩	5 (7.4 ± 0.3) ⟨1431.0 ± 25.4⟩	5 (26.2 ± 1.7) ⟨1626.2 ± 5.4⟩	5 (44.4 ± 3.4) ⟨1626.2 ± 5.4⟩
	08-02	5 (3.8 ± 0.1) ⟨881.4 ± 9.1⟩	5 (3.8 ± 0.1) ⟨938.8 ± 8.6⟩	5 (6.0 ± 0.2) ⟨765.8 ± 18.7⟩	5 (19.9 ± 0.9) ⟨994.0 ± 8.4⟩	5 (34.4 ± 2.5) ⟨994.0 ± 8.4⟩
	08-04	5 (62.2 ± 1.2) ⟨17367.4 ± 42.4⟩	5 (64.5 ± 1.3) ⟨18443.1 ± 123.3⟩	5 (75.7 ± 1.9) ⟨14337.8 ± 154.4⟩	5 (287.7 ± 54.1) ⟨17036.8 ± 31.2⟩	5 (490.7 ± 64.7) ⟨17036.8 ± 31.2⟩

Table 1: Results for algorithms using **ROC**. We give X (Y) (Z) where X is the coverage, Y is the mean time (secs), and Z is the mean number of *thousands* of Q -values. Each mean is taken over the converged instances, and we report the 95% C.I.

	CG-iLAO [*] _{tied}	CG-iLAO [*] _{single}	CG-iLAO [*] _{trial}	CG-iLAO [*] _{FF-AO}	CG-iLAO [*] _{FF-MLO}	
BW	04-01	5 (0.5 ± 0.0) (4.4 ± 0.2)	5 (0.5 ± 0.0) (5.7 ± 1.1)	5 (0.5 ± 0.0) (5.8 ± 1.4)	5 (0.7 ± 0.1) (4.8 ± 0.1)	5 (0.6 ± 0.0) (3.9 ± 0.2)
	04-02	5 (0.5 ± 0.0) (2.8 ± 0.1)	5 (0.5 ± 0.0) (4.7 ± 0.7)	5 (0.5 ± 0.0) (4.1 ± 1.3)	5 (0.7 ± 0.0) (3.8 ± 0.2)	5 (0.6 ± 0.0) (3.5 ± 0.2)
	04-03	5 (0.5 ± 0.0) (0.4 ± 0.1)	5 (0.5 ± 0.0) (1.4 ± 0.5)	5 (0.5 ± 0.0) (1.6 ± 0.8)	5 (0.5 ± 0.0) (0.0 ± 0.0)	5 (0.5 ± 0.0) (0.0 ± 0.0)
	04-04	5 (0.5 ± 0.0) (7.0 ± 0.3)	5 (0.5 ± 0.0) (10.9 ± 1.7)	5 (0.5 ± 0.0) (10.5 ± 1.5)	5 (0.8 ± 0.1) (10.1 ± 0.5)	5 (0.7 ± 0.0) (8.2 ± 1.1)
	04-05	5 (0.5 ± 0.0) (1.6 ± 0.2)	5 (0.5 ± 0.0) (3.1 ± 0.7)	5 (0.5 ± 0.0) (3.3 ± 1.3)	5 (0.6 ± 0.0) (2.9 ± 0.1)	5 (0.5 ± 0.0) (2.2 ± 0.2)
	04-06	5 (18.7 ± 0.3) (830.2 ± 14.0)	5 (19.3 ± 0.8) (1053.2 ± 14.6)	5 (17.4 ± 0.2) (1068.6 ± 11.1)	5 (147.2 ± 2.2) (1044.0 ± 15.9)	5 (208.5 ± 1.4) (991.4 ± 4.9)
	04-07	5 (1223.6 ± 53.9) (84550.7 ± 751.0)	5 (1303.4 ± 79.0) (111665.8 ± 618.7)	5 (1144.3 ± 63.0) (114638.4 ± 613.0)		
	04-08	5 (9.3 ± 0.2) (257.8 ± 6.0)	5 (9.1 ± 0.2) (254.0 ± 6.9)	5 (9.0 ± 0.1) (285.3 ± 6.4)	5 (31.9 ± 2.2) (239.4 ± 2.6)	5 (21.9 ± 0.8) (224.2 ± 3.1)
	04-09	5 (8.5 ± 0.1) (171.2 ± 3.9)	5 (8.4 ± 0.1) (190.6 ± 5.2)	5 (8.5 ± 0.1) (222.5 ± 10.5)	5 (27.1 ± 1.4) (188.3 ± 4.3)	5 (63.2 ± 2.8) (182.2 ± 4.1)
	04-10	5 (39.6 ± 1.3) (2237.3 ± 11.6)	5 (40.6 ± 1.4) (2902.2 ± 49.3)	5 (36.5 ± 0.9) (3023.5 ± 31.4)	5 (688.5 ± 14.9) (2790.8 ± 16.0)	5 (1339.1 ± 12.4) (2766.3 ± 11.8)
	08-01	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)	5 (0.2 ± 0.0) (5.3 ± 2.1)	5 (0.3 ± 0.0) (2.6 ± 0.3)	5 (0.3 ± 0.0) (2.1 ± 0.3)
	08-02	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)	5 (0.2 ± 0.0) (6.2 ± 2.1)	5 (0.3 ± 0.0) (2.6 ± 0.3)	5 (0.3 ± 0.0) (2.1 ± 0.3)
	08-03	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)	5 (0.2 ± 0.0) (6.5 ± 1.7)	5 (0.3 ± 0.0) (2.6 ± 0.3)	5 (0.3 ± 0.0) (2.1 ± 0.3)
	08-04	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)	5 (0.2 ± 0.0) (6.3 ± 1.9)	5 (0.3 ± 0.0) (2.6 ± 0.3)	5 (0.3 ± 0.0) (2.1 ± 0.3)
	08-05	5 (91.9 ± 1.9) (7358.1 ± 9.8)	5 (92.2 ± 2.4) (8378.3 ± 110.0)	5 (81.3 ± 3.8) (9176.6 ± 28.4)	5 (982.0 ± 12.2) (8267.1 ± 71.2)	5 (530.9 ± 13.6) (8160.5 ± 61.8)
	08-06	5 (93.6 ± 2.0) (7358.1 ± 9.8)	5 (94.2 ± 5.0) (8378.3 ± 110.0)	5 (82.2 ± 3.4) (9176.6 ± 28.4)	5 (988.4 ± 9.6) (8267.1 ± 71.2)	5 (538.6 ± 18.0) (8160.5 ± 61.8)
	08-07	5 (91.3 ± 0.9) (7358.1 ± 9.8)	5 (92.0 ± 2.4) (8378.3 ± 110.0)	5 (84.2 ± 4.3) (9176.6 ± 28.4)	5 (995.6 ± 13.4) (8267.1 ± 71.2)	5 (545.2 ± 9.8) (8160.5 ± 61.8)
	08-08	5 (91.9 ± 1.2) (7358.1 ± 9.8)	5 (93.9 ± 3.2) (8378.3 ± 110.0)	5 (86.7 ± 1.4) (9137.7 ± 79.6)	5 (996.0 ± 20.6) (8267.1 ± 71.2)	5 (543.6 ± 8.8) (8160.5 ± 61.8)

results for **PDB** continued on next page

		CG-iLAO [*] _{tied}	CG-iLAO [*] _{single}	CG-iLAO [*] _{trial}	CG-iLAO [*] _{FF-AO}	CG-iLAO [*] _{FF-MLO}
Coresec	01	5 (0.1 ± 0.0) ⟨7.2 ± 0.2⟩	5 (0.1 ± 0.0) ⟨7.1 ± 0.1⟩	5 (0.1 ± 0.0) ⟨7.6 ± 0.3⟩	5 (0.8 ± 0.1) ⟨5.2 ± 0.0⟩	5 (1.1 ± 0.1) ⟨5.2 ± 0.0⟩
	02	5 (0.1 ± 0.0) ⟨7.2 ± 0.3⟩	5 (0.1 ± 0.0) ⟨7.3 ± 0.2⟩	5 (0.1 ± 0.0) ⟨7.8 ± 0.2⟩	5 (0.7 ± 0.1) ⟨5.2 ± 0.1⟩	5 (1.2 ± 0.2) ⟨5.2 ± 0.1⟩
	03	5 (0.3 ± 0.0) ⟨175.3 ± 1.2⟩	5 (0.3 ± 0.0) ⟨172.9 ± 2.0⟩	5 (0.4 ± 0.0) ⟨185.2 ± 1.2⟩	5 (9.9 ± 0.6) ⟨132.4 ± 1.6⟩	5 (13.7 ± 0.8) ⟨132.4 ± 1.6⟩
	04	5 (0.4 ± 0.0) ⟨180.7 ± 1.1⟩	5 (0.4 ± 0.0) ⟨179.3 ± 1.1⟩	5 (0.4 ± 0.0) ⟨187.5 ± 1.8⟩	5 (9.9 ± 0.5) ⟨132.4 ± 0.9⟩	5 (14.2 ± 0.6) ⟨132.4 ± 0.9⟩
	05	5 (1.1 ± 0.1) ⟨586.7 ± 4.7⟩	5 (1.1 ± 0.0) ⟨571.2 ± 1.5⟩	5 (1.2 ± 0.0) ⟨601.3 ± 3.4⟩	5 (25.0 ± 1.4) ⟨407.0 ± 1.2⟩	5 (34.8 ± 2.6) ⟨407.0 ± 1.2⟩
	06	5 (22.0 ± 1.3) ⟨9916.2 ± 26.3⟩	5 (20.2 ± 0.5) ⟨9781.3 ± 13.6⟩	5 (22.6 ± 1.0) ⟨10020.8 ± 22.3⟩	5 (354.3 ± 20.7) ⟨6949.2 ± 5.2⟩	5 (518.6 ± 33.7) ⟨6949.2 ± 5.2⟩
Elev	04-01	5 (0.1 ± 0.0) ⟨17.2 ± 1.8⟩	5 (0.1 ± 0.0) ⟨19.9 ± 2.5⟩	5 (0.1 ± 0.0) ⟨23.8 ± 1.9⟩	5 (0.5 ± 0.1) ⟨12.9 ± 0.4⟩	5 (0.5 ± 0.1) ⟨12.9 ± 0.4⟩
	04-02	5 (0.1 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.2 ± 0.1⟩	5 (0.1 ± 0.0) ⟨3.6 ± 4.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.1⟩	5 (0.1 ± 0.0) ⟨0.5 ± 0.0⟩
	04-03	5 (0.1 ± 0.0) ⟨18.0 ± 2.4⟩	5 (0.1 ± 0.0) ⟨20.8 ± 2.4⟩	5 (0.1 ± 0.0) ⟨25.0 ± 3.9⟩	5 (0.5 ± 0.1) ⟨15.5 ± 0.4⟩	5 (0.5 ± 0.1) ⟨15.4 ± 0.6⟩
	04-04	5 (0.1 ± 0.0) ⟨11.1 ± 0.1⟩	5 (0.1 ± 0.0) ⟨12.4 ± 0.6⟩	5 (0.1 ± 0.0) ⟨13.8 ± 0.7⟩	5 (0.3 ± 0.0) ⟨9.5 ± 0.2⟩	5 (0.3 ± 0.0) ⟨9.5 ± 0.1⟩
	04-05	5 (0.1 ± 0.0) ⟨5.4 ± 0.1⟩	5 (0.1 ± 0.0) ⟨6.3 ± 0.2⟩	5 (0.1 ± 0.0) ⟨7.9 ± 0.6⟩	5 (0.2 ± 0.0) ⟨5.1 ± 0.2⟩	5 (0.3 ± 0.0) ⟨5.2 ± 0.2⟩
	04-06	5 (0.2 ± 0.0) ⟨139.9 ± 15.4⟩	5 (0.3 ± 0.0) ⟨150.6 ± 12.8⟩	5 (0.3 ± 0.0) ⟨197.5 ± 9.9⟩	5 (3.7 ± 0.5) ⟨176.2 ± 2.3⟩	5 (3.6 ± 0.4) ⟨171.5 ± 1.9⟩
	04-07	5 (0.2 ± 0.0) ⟨144.5 ± 9.0⟩	5 (0.3 ± 0.0) ⟨160.2 ± 9.9⟩	5 (0.3 ± 0.0) ⟨209.3 ± 8.9⟩	5 (3.6 ± 0.3) ⟨147.4 ± 9.7⟩	5 (3.6 ± 0.3) ⟨146.8 ± 9.4⟩
	04-08	5 (1.4 ± 0.1) ⟨1348.2 ± 18.5⟩	5 (1.5 ± 0.0) ⟨1429.7 ± 22.1⟩	5 (1.8 ± 0.1) ⟨1713.8 ± 33.1⟩	5 (17.2 ± 2.1) ⟨1404.4 ± 11.4⟩	5 (21.8 ± 5.2) ⟨1402.6 ± 9.1⟩
	04-09	5 (0.8 ± 0.0) ⟨751.1 ± 9.1⟩	5 (0.9 ± 0.0) ⟨809.5 ± 15.2⟩	5 (1.2 ± 0.0) ⟨1021.5 ± 12.5⟩	5 (12.3 ± 1.1) ⟨765.1 ± 7.9⟩	5 (12.3 ± 1.1) ⟨744.8 ± 23.1⟩
	04-10	5 (3.2 ± 0.2) ⟨3875.3 ± 90.3⟩	5 (3.2 ± 0.1) ⟨3979.3 ± 112.9⟩	5 (4.6 ± 0.0) ⟨5107.4 ± 123.3⟩	5 (17.5 ± 2.4) ⟨4144.6 ± 22.5⟩	5 (23.8 ± 4.4) ⟨4139.8 ± 41.3⟩
	04-11	5 (286.3 ± 13.2) ⟨158909.8 ± 1767.3⟩	5 (297.1 ± 13.6) ⟨161907.7 ± 1430.2⟩	5 (361.9 ± 8.0) ⟨202289.0 ± 2238.1⟩	1 (900.2±) ⟨169620.8±⟩	1 (881.4±) ⟨170224.7±⟩
	04-12	5 (50.0 ± 1.3) ⟨21660.7 ± 267.2⟩	5 (51.9 ± 2.0) ⟨22498.8 ± 194.4⟩	5 (63.0 ± 1.9) ⟨27589.4 ± 204.0⟩	5 (330.5 ± 47.0) ⟨23807.9 ± 93.7⟩	5 (394.0 ± 71.1) ⟨23979.4 ± 412.2⟩
	04-13	5 (1406.7 ± 76.8) ⟨1077845.1 ± 6034.4⟩	5 (1467.7 ± 30.8) ⟨1069062.6 ± 6869.2⟩			
	04-14	5 (137.0 ± 2.9) ⟨44967.1 ± 1003.5⟩	5 (139.9 ± 2.3) ⟨47315.1 ± 1120.7⟩	5 (159.5 ± 3.5) ⟨56458.8 ± 763.0⟩	5 (585.2 ± 32.2) ⟨42008.4 ± 272.8⟩	5 (598.4 ± 25.8) ⟨41324.9 ± 618.8⟩
	04-15	5 (457.3 ± 82.4) ⟨198692.8 ± 837.6⟩	5 (408.2 ± 20.5) ⟨201766.7 ± 926.7⟩	5 (435.1 ± 11.2) ⟨230998.0 ± 850.2⟩		

results for **PDB** continued on next page

	CG-iLAO* _{tied}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
ExBW	04-01	5 (0.1 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.1⟩	5 (0.1 ± 0.0) ⟨0.1 ± 0.0⟩
	04-02	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩
	04-03	5 (0.1 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.2⟩	5 (0.1 ± 0.0) ⟨0.1 ± 0.0⟩
	04-04	5 (0.2 ± 0.0) ⟨144.2 ± 0.1⟩	5 (0.2 ± 0.0) ⟨141.2 ± 6.0⟩	5 (0.3 ± 0.0) ⟨128.5 ± 5.0⟩	5 (0.2 ± 0.0) ⟨1.7 ± 0.0⟩
	04-05	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩
	04-06	5 (0.3 ± 0.0) ⟨67.5 ± 0.1⟩	5 (0.3 ± 0.0) ⟨64.0 ± 4.6⟩	5 (0.5 ± 0.0) ⟨61.4 ± 20.6⟩	5 (0.8 ± 0.1) ⟨57.4 ± 0.4⟩
	04-07	5 (1.1 ± 0.0) ⟨380.5 ± 6.3⟩	5 (1.1 ± 0.0) ⟨367.3 ± 7.5⟩	5 (2.9 ± 0.2) ⟨586.0 ± 17.9⟩	5 (11.6 ± 0.8) ⟨358.0 ± 3.6⟩
	04-08	5 (0.3 ± 0.0) ⟨15.0 ± 0.2⟩	5 (0.3 ± 0.0) ⟨13.9 ± 0.5⟩	5 (0.8 ± 0.0) ⟨90.9 ± 3.5⟩	5 (0.9 ± 0.1) ⟨10.5 ± 0.7⟩
	04-09	5 (8.6 ± 0.3) ⟨4222.0 ± 85.9⟩	5 (8.4 ± 0.2) ⟨4035.8 ± 123.0⟩	5 (19.0 ± 0.4) ⟨6068.7 ± 238.4⟩	5 (80.4 ± 5.1) ⟨3925.3 ± 108.4⟩
	08-01	5 (0.5 ± 0.0) ⟨389.3 ± 3.9⟩	5 (0.4 ± 0.0) ⟨386.4 ± 9.2⟩	5 (0.7 ± 0.0) ⟨469.6 ± 29.7⟩	5 (0.1 ± 0.0) ⟨0.3 ± 0.0⟩
	08-02	5 (0.5 ± 0.0) ⟨498.7 ± 4.0⟩	5 (0.5 ± 0.0) ⟨503.1 ± 20.7⟩	5 (0.5 ± 0.0) ⟨377.2 ± 13.0⟩	5 (0.1 ± 0.0) ⟨0.5 ± 0.0⟩
	08-03	5 (0.8 ± 0.0) ⟨839.9 ± 7.4⟩	5 (0.8 ± 0.0) ⟨842.1 ± 17.5⟩	5 (1.2 ± 0.0) ⟨950.6 ± 21.1⟩	5 (0.4 ± 0.0) ⟨39.2 ± 0.9⟩
	08-04	5 (12.3 ± 0.5) ⟨12446.9 ± 25.2⟩	5 (12.1 ± 0.6) ⟨12179.5 ± 162.0⟩	5 (16.0 ± 0.4) ⟨11677.8 ± 343.5⟩	5 (1.0 ± 0.1) ⟨29.7 ± 8.7⟩
	08-05	5 (0.1 ± 0.0) ⟨0.7 ± 0.1⟩	5 (0.1 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.7 ± 0.6⟩	5 (0.2 ± 0.0) ⟨0.7 ± 0.0⟩
	08-06	5 (20.7 ± 0.3) ⟨20945.6 ± 33.6⟩	5 (20.7 ± 0.5) ⟨20595.2 ± 714.5⟩	5 (29.8 ± 1.0) ⟨22207.5 ± 1286.7⟩	5 (0.6 ± 0.0) ⟨33.9 ± 0.2⟩
	08-07	5 (0.2 ± 0.0) ⟨3.4 ± 0.5⟩	5 (0.2 ± 0.0) ⟨6.1 ± 1.4⟩	5 (0.3 ± 0.1) ⟨15.7 ± 2.5⟩	5 (0.5 ± 0.0) ⟨4.7 ± 0.6⟩
	08-08	5 (381.5 ± 67.2) ⟨179805.0 ± 299.1⟩	5 (353.5 ± 25.7) ⟨167553.3 ± 14611.0⟩	5 (546.0 ± 72.8) ⟨173778.9 ± 10000.0⟩	
	08-09	5 (20.7 ± 0.3) ⟨6031.9 ± 75.1⟩	5 (21.5 ± 0.3) ⟨6407.1 ± 77.5⟩	5 (73.1 ± 2.2) ⟨10639.6 ± 157.9⟩	5 (50.7 ± 1.4) ⟨1067.1 ± 18.6⟩
	08-10				5 (640.6 ± 23.4) ⟨84956.3 ± 231.7⟩
PARC-N					
	4, 1				5 (1537.8 ± 35.3) (2326.5 ± 12.9)
results for PDB continued on next page					

		CG-iLAO* _{tied}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
Rand	04-03	5 (1.6 ± 0.0) ⟨793.1 ± 21.6⟩	5 (1.1 ± 0.0) ⟨540.2 ± 15.1⟩	5 (1.8 ± 0.2) ⟨778.8 ± 70.6⟩	5 (6.9 ± 0.8) ⟨599.4 ± 10.5⟩	5 (7.8 ± 1.6) ⟨599.4 ± 10.5⟩
	04-04	5 (14.3 ± 1.3) ⟨26692.6 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.3 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.3 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.3 ± 0.0) ⟨0.0 ± 0.0⟩
	04-05	5 (5.3 ± 3.6) ⟨1698.0 ± 1358.1⟩	5 (1.0 ± 0.0) ⟨1.0 ± 0.3⟩	5 (1.0 ± 0.1) ⟨0.8 ± 0.1⟩	5 (2.0 ± 0.4) ⟨156.6 ± 48.7⟩	5 (2.3 ± 0.5) ⟨156.6 ± 48.7⟩
	04-07	5 (0.3 ± 0.0) ⟨0.3 ± 0.1⟩	5 (0.3 ± 0.0) ⟨0.5 ± 0.2⟩	5 (0.3 ± 0.0) ⟨1.3 ± 1.7⟩	5 (0.3 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.3 ± 0.0) ⟨0.9 ± 0.0⟩
	04-09	5 (1.0 ± 0.0) ⟨20.4 ± 5.7⟩	5 (1.0 ± 0.0) ⟨22.1 ± 13.3⟩	5 (2.1 ± 0.8) ⟨320.1 ± 224.0⟩	5 (1.2 ± 0.1) ⟨21.7 ± 5.3⟩	5 (1.2 ± 0.1) ⟨25.3 ± 6.2⟩
	04-13	5 (4.7 ± 0.5) ⟨526.0 ± 61.3⟩	5 (4.0 ± 0.6) ⟨457.5 ± 116.3⟩	5 (30.5 ± 13.8) ⟨4890.9 ± 2147.7⟩	5 (7.8 ± 1.9) ⟨542.6 ± 155.8⟩	5 (7.2 ± 2.0) ⟨522.3 ± 157.8⟩
□TW	08-01	5 (0.3 ± 0.0) ⟨1.2 ± 0.1⟩	5 (0.3 ± 0.0) ⟨1.3 ± 0.1⟩	5 (0.4 ± 0.0) ⟨1.4 ± 0.1⟩	5 (0.4 ± 0.0) ⟨1.2 ± 0.1⟩	5 (0.4 ± 0.0) ⟨1.0 ± 0.0⟩
	08-02	5 (0.4 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.0 ± 0.0⟩
	08-03	5 (0.9 ± 0.0) ⟨2.3 ± 0.1⟩	5 (0.9 ± 0.0) ⟨2.4 ± 0.2⟩	5 (0.8 ± 0.0) ⟨2.7 ± 0.1⟩	5 (0.9 ± 0.0) ⟨2.4 ± 0.1⟩	5 (0.9 ± 0.0) ⟨1.7 ± 0.1⟩
	08-04	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩	5 (1.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩
	08-05	5 (2.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.1) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.1) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.1) ⟨0.0 ± 0.0⟩
	08-06	5 (4.6 ± 0.1) ⟨11.7 ± 0.1⟩	5 (4.8 ± 0.3) ⟨11.8 ± 0.3⟩	5 (4.7 ± 0.1) ⟨25.1 ± 0.5⟩	5 (7.4 ± 0.3) ⟨10.7 ± 0.6⟩	5 (8.7 ± 0.5) ⟨11.4 ± 0.7⟩
	08-07	5 (4.4 ± 0.0) ⟨0.0 ± 0.0⟩	5 (4.4 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.4 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩
	08-08	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.6 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.5 ± 0.2) ⟨0.0 ± 0.0⟩	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩
	08-09	5 (20.1 ± 1.8) ⟨42.5 ± 7.0⟩	5 (23.0 ± 2.6) ⟨47.1 ± 5.6⟩	5 (23.5 ± 0.6) ⟨71.1 ± 4.4⟩	5 (59.4 ± 4.6) ⟨55.1 ± 1.4⟩	5 (78.9 ± 7.5) ⟨56.0 ± 1.9⟩
	08-10	5 (14.8 ± 0.8) ⟨314.3 ± 33.6⟩	5 (16.1 ± 1.0) ⟨324.1 ± 43.3⟩	5 (15.0 ± 0.9) ⟨383.9 ± 3.4⟩	5 (15.7 ± 0.1) ⟨0.0 ± 0.0⟩	5 (15.4 ± 0.3) ⟨0.0 ± 0.0⟩
	08-11	5 (98.5 ± 4.0) ⟨89.6 ± 1.3⟩	5 (100.1 ± 3.6) ⟨92.1 ± 1.1⟩	5 (104.0 ± 5.8) ⟨116.0 ± 5.4⟩	5 (632.8 ± 85.2) ⟨86.4 ± 0.6⟩	5 (1081.8 ± 137.3) ⟨92.1 ± 1.1⟩
	08-12	5 (128.4 ± 2.1) ⟨42652.0 ± 861.6⟩	5 (126.8 ± 4.0) ⟨42628.0 ± 997.4⟩	5 (142.8 ± 8.2) ⟨43815.0 ± 423.3⟩	5 (84.9 ± 13.2) ⟨0.0 ± 0.0⟩	5 (96.2 ± 10.2) ⟨0.0 ± 0.0⟩
S&R	08-01	5 (0.3 ± 0.0) ⟨446.9 ± 3.0⟩	5 (0.3 ± 0.0) ⟨449.0 ± 1.5⟩	5 (0.4 ± 0.0) ⟨440.9 ± 2.4⟩	5 (1.1 ± 0.1) ⟨412.4 ± 1.5⟩	5 (1.3 ± 0.2) ⟨412.4 ± 1.5⟩
	08-02	5 (1.0 ± 0.0) ⟨1841.4 ± 7.7⟩	5 (1.0 ± 0.0) ⟨1847.4 ± 4.6⟩	5 (1.2 ± 0.0) ⟨1814.4 ± 5.4⟩	5 (4.0 ± 0.4) ⟨1831.5 ± 3.2⟩	5 (4.3 ± 0.3) ⟨1831.5 ± 3.2⟩
	08-03	5 (4.0 ± 0.3) ⟨7452.9 ± 14.3⟩	5 (4.2 ± 0.4) ⟨7494.3 ± 10.7⟩	5 (4.5 ± 0.0) ⟨7377.5 ± 15.6⟩	5 (14.6 ± 1.3) ⟨7546.9 ± 7.9⟩	5 (15.6 ± 0.3) ⟨7546.9 ± 7.9⟩
	08-04	5 (75.0 ± 3.6) ⟨113676.4 ± 106.3⟩	5 (82.0 ± 5.4) ⟨114418.7 ± 52.8⟩	5 (79.9 ± 4.1) ⟨112863.5 ± 94.9⟩	5 (223.5 ± 10.5) ⟨115741.7 ± 41.6⟩	5 (227.7 ± 12.1) ⟨115741.7 ± 41.6⟩
	08-05	4 (1635.2 ± 100.4) ⟨1587933.7 ± 1602.7⟩		4 (1698.3 ± 63.5) ⟨1586947.3 ± 2064.9⟩		

results for **PDB** continued on next page

		CG-iLAO* _{tied}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
Sched	04-01	5 (0.1 ± 0.0) ⟨1.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.7 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩
	04-02	5 (0.1 ± 0.0) ⟨1.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.0 ± 0.0⟩
	04-03	5 (0.1 ± 0.0) ⟨3.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.3 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.3 ± 0.2⟩	5 (0.1 ± 0.0) ⟨2.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.1 ± 0.0⟩
	04-04				5 (94.2 ± 6.9) ⟨17968.8 ± 194.8⟩	5 (138.3 ± 9.9) ⟨18056.1 ± 286.7⟩
Sched	08-01	5 (0.1 ± 0.0) ⟨1.3 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.7 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.7 ± 0.0⟩
	08-02	5 (0.1 ± 0.0) ⟨1.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.3 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩
	08-03	5 (0.1 ± 0.0) ⟨4.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.4 ± 0.2⟩	5 (0.1 ± 0.0) ⟨2.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.2 ± 0.0⟩
	08-04					
Sys	08-01	5 (0.1 ± 0.0) ⟨24.7 ± 0.2⟩	5 (0.1 ± 0.0) ⟨28.2 ± 0.6⟩	5 (0.1 ± 0.0) ⟨13.6 ± 0.4⟩	5 (0.3 ± 0.1) ⟨11.5 ± 0.3⟩	5 (0.2 ± 0.0) ⟨11.5 ± 0.3⟩
	08-02	5 (0.2 ± 0.0) ⟨127.1 ± 0.8⟩	5 (0.2 ± 0.0) ⟨144.0 ± 2.7⟩	5 (0.2 ± 0.0) ⟨59.2 ± 0.9⟩	5 (0.7 ± 0.1) ⟨52.9 ± 0.4⟩	5 (0.7 ± 0.0) ⟨52.9 ± 0.4⟩
	08-03	5 (0.7 ± 0.0) ⟨477.7 ± 3.4⟩	5 (0.8 ± 0.0) ⟨550.6 ± 1.9⟩	5 (0.6 ± 0.0) ⟨201.3 ± 2.9⟩		5 (2.2 ± 0.1) ⟨179.4 ± 1.1⟩
	08-04	5 (36.1 ± 6.2) ⟨5621.9 ± 42.0⟩	5 (35.4 ± 2.2) ⟨6209.8 ± 52.3⟩	5 (30.1 ± 1.9) ⟨2306.3 ± 11.0⟩		
ΔTW	08-01	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.1 ± 0.0⟩
	08-02	5 (0.0 ± 0.0) ⟨2.6 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨2.4 ± 0.1⟩	5 (0.3 ± 0.0) ⟨2.0 ± 0.0⟩	5 (0.3 ± 0.0) ⟨2.1 ± 0.1⟩
	08-03	5 (0.2 ± 0.0) ⟨65.2 ± 5.2⟩	5 (0.2 ± 0.0) ⟨78.7 ± 2.7⟩	5 (0.3 ± 0.0) ⟨90.8 ± 13.3⟩	5 (3.3 ± 0.2) ⟨106.3 ± 5.5⟩	5 (3.8 ± 0.3) ⟨98.2 ± 19.3⟩
	08-04	5 (3.9 ± 1.1) ⟨1582.9 ± 37.1⟩	5 (5.5 ± 3.4) ⟨1994.1 ± 196.3⟩	5 (5.9 ± 0.6) ⟨2045.6 ± 185.9⟩	5 (47.1 ± 2.8) ⟨2507.1 ± 90.7⟩	5 (49.7 ± 7.5) ⟨2203.6 ± 617.7⟩
6, 10	08-05	5 (165.1 ± 19.2) ⟨37532.0 ± 1299.6⟩	5 (183.9 ± 19.3) ⟨41942.4 ± 2817.2⟩	5 (231.0 ± 19.2) ⟨36145.2 ± 2603.1⟩	1 (822.2±) ⟨46638.4±⟩	2 (813.6 ± 84.7) ⟨35542.1 ± 21748.7⟩
	6, 10	3 (1555.2 ± 76.3) ⟨274945.1 ± 872.2⟩	2 (1602.8 ± 69.4) ⟨278233.7 ± 9122.1⟩			
	6, 8	5 (206.3 ± 14.2) ⟨48453.0 ± 1606.4⟩	5 (201.1 ± 13.7) ⟨47970.6 ± 1070.0⟩	5 (282.4 ± 9.7) ⟨40408.9 ± 778.9⟩		
	6, 9	5 (576.9 ± 29.4) ⟨117320.6 ± 2268.4⟩	5 (701.9 ± 260.2) ⟨116958.3 ± 4077.7⟩	5 (766.4 ± 22.9) ⟨95611.5 ± 2928.7⟩		

results for **PDB** continued on next page

	CG-iLAO [*] _{tied}	CG-iLAO [*] _{single}	CG-iLAO [*] _{trial}	CG-iLAO [*] _{FF-AO}	CG-iLAO [*] _{FF-MLO}
Zeno	04-01	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	04-02	5 (26.8 ± 1.4) ⟨17300.6 ± 33.5⟩	5 (27.8 ± 1.2) ⟨18346.0 ± 73.5⟩	5 (30.6 ± 1.1) ⟨18419.7 ± 279.1⟩	5 (167.1 ± 2.9) ⟨18389.0 ± 42.3⟩
	04-03	5 (507.6 ± 44.4) ⟨310498.8 ± 268.3⟩	5 (527.6 ± 43.1) ⟨324647.6 ± 960.2⟩	5 (534.4 ± 25.1) ⟨312245.0 ± 2013.5⟩	
	04-04	5 (31.2 ± 0.9) ⟨19347.6 ± 47.6⟩	5 (30.7 ± 1.2) ⟨19111.9 ± 62.6⟩	5 (37.4 ± 5.9) ⟨9248.1 ± 102.0⟩	5 (173.4 ± 23.1) ⟨16321.1 ± 25.0⟩
	04-05	5 (102.1 ± 4.1) ⟨61198.8 ± 159.4⟩	5 (108.6 ± 5.3) ⟨64775.0 ± 329.9⟩	5 (144.5 ± 10.5) ⟨65660.7 ± 614.5⟩	5 (581.9 ± 22.3) ⟨71164.7 ± 122.1⟩
	08-01	5 (1.9 ± 0.1) ⟨1398.0 ± 10.8⟩	5 (2.1 ± 0.1) ⟨1575.9 ± 29.9⟩	5 (2.2 ± 0.0) ⟨1192.8 ± 14.4⟩	5 (16.7 ± 1.2) ⟨1604.2 ± 9.3⟩
	08-02	5 (1.4 ± 0.0) ⟨773.5 ± 3.3⟩	5 (1.5 ± 0.1) ⟨885.3 ± 10.0⟩	5 (1.9 ± 0.0) ⟨641.9 ± 18.2⟩	5 (13.4 ± 0.8) ⟨970.1 ± 5.1⟩
	08-04	5 (10.3 ± 0.5) ⟨7480.2 ± 19.3⟩	5 (10.7 ± 0.5) ⟨7678.4 ± 89.8⟩	5 (11.3 ± 0.2) ⟨5534.6 ± 107.5⟩	5 (60.4 ± 7.9) ⟨6115.6 ± 16.6⟩

Table 2: Results for algorithms using **PDB**. We give $X(Y\langle Z \rangle)$ where X is the coverage, Y is the mean time (secs), and Z is the mean number of *thousands* of Q -values. Each mean is taken over the converged instances, and we report the 95% C.I.

	CG-iLAO* _{tid}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}	
BW	04-01	5 (1.9 ± 0.2) ⟨44.6 ± 5.1⟩	5 (1.8 ± 0.2) ⟨49.1 ± 4.6⟩	5 (1.6 ± 0.1) ⟨49.4 ± 5.5⟩	5 (3.7 ± 0.4) ⟨56.9 ± 9.1⟩	5 (3.8 ± 0.4) ⟨52.9 ± 6.9⟩
	04-02	5 (1.2 ± 0.0) ⟨30.1 ± 0.5⟩	5 (1.2 ± 0.0) ⟨34.3 ± 0.8⟩	5 (1.1 ± 0.0) ⟨33.9 ± 0.6⟩	5 (2.2 ± 0.1) ⟨34.1 ± 1.3⟩	5 (2.2 ± 0.1) ⟨34.7 ± 1.1⟩
	04-03	5 (0.6 ± 0.0) ⟨10.0 ± 1.0⟩	5 (0.6 ± 0.0) ⟨11.9 ± 1.0⟩	5 (0.6 ± 0.0) ⟨12.0 ± 0.7⟩	5 (1.1 ± 0.2) ⟨13.4 ± 1.8⟩	5 (1.2 ± 0.1) ⟨15.1 ± 1.6⟩
	04-04	5 (2.8 ± 0.2) ⟨87.0 ± 11.9⟩	5 (2.8 ± 0.2) ⟨96.0 ± 12.4⟩	5 (1.9 ± 0.1) ⟨78.7 ± 7.7⟩	5 (5.3 ± 0.5) ⟨90.7 ± 9.2⟩	5 (5.7 ± 0.7) ⟨84.4 ± 8.7⟩
	04-05	5 (1.0 ± 0.1) ⟨26.8 ± 3.0⟩	5 (1.0 ± 0.1) ⟨31.6 ± 3.2⟩	5 (0.9 ± 0.0) ⟨29.4 ± 3.2⟩	5 (2.0 ± 0.2) ⟨32.1 ± 4.0⟩	5 (2.1 ± 0.3) ⟨31.9 ± 3.6⟩
	08-01	5 (0.6 ± 0.0) ⟨20.5 ± 2.2⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩	5 (0.6 ± 0.0) ⟨21.1 ± 2.0⟩	5 (1.3 ± 0.1) ⟨21.2 ± 1.3⟩	5 (1.2 ± 0.1) ⟨20.3 ± 1.4⟩
	08-02	5 (0.6 ± 0.0) ⟨20.5 ± 2.2⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩	5 (0.5 ± 0.0) ⟨20.9 ± 1.7⟩	5 (1.3 ± 0.1) ⟨21.2 ± 1.3⟩	5 (1.2 ± 0.1) ⟨20.3 ± 1.4⟩
	08-03	5 (0.6 ± 0.0) ⟨20.5 ± 2.2⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩	5 (0.5 ± 0.0) ⟨20.9 ± 2.0⟩	5 (1.4 ± 0.2) ⟨21.2 ± 1.3⟩	5 (1.2 ± 0.1) ⟨20.3 ± 1.4⟩
	08-04	5 (0.6 ± 0.0) ⟨20.5 ± 2.2⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩	5 (0.5 ± 0.0) ⟨21.0 ± 1.9⟩	5 (1.3 ± 0.1) ⟨21.2 ± 1.3⟩	5 (1.2 ± 0.0) ⟨20.3 ± 1.4⟩
Coresec	01	5 (0.4 ± 0.0) ⟨254.4 ± 1.8⟩	5 (0.4 ± 0.0) ⟨244.1 ± 1.9⟩	5 (0.5 ± 0.0) ⟨259.3 ± 2.6⟩	5 (12.5 ± 1.3) ⟨192.8 ± 1.5⟩	5 (25.1 ± 2.5) ⟨192.8 ± 1.5⟩
	02	5 (1.4 ± 0.1) ⟨836.7 ± 4.1⟩	5 (1.4 ± 0.1) ⟨800.9 ± 4.2⟩	5 (1.7 ± 0.1) ⟨866.9 ± 3.7⟩	5 (40.8 ± 5.7) ⟨651.2 ± 3.7⟩	5 (74.5 ± 4.6) ⟨651.2 ± 3.7⟩
	03	5 (4.4 ± 0.1) ⟨2507.5 ± 10.4⟩	5 (4.3 ± 0.1) ⟨2380.7 ± 9.2⟩	5 (5.0 ± 0.2) ⟨2612.0 ± 17.8⟩	5 (97.1 ± 10.0) ⟨2010.5 ± 5.7⟩	5 (201.1 ± 38.6) ⟨2010.5 ± 5.7⟩
	04	5 (13.6 ± 0.8) ⟨7337.4 ± 8.6⟩	5 (13.5 ± 0.8) ⟨7057.9 ± 12.5⟩	5 (15.7 ± 0.6) ⟨7709.8 ± 18.5⟩	5 (275.4 ± 20.8) ⟨5951.8 ± 28.9⟩	5 (530.3 ± 48.0) ⟨5951.8 ± 28.9⟩
	05	5 (653.2 ± 40.1) ⟨165985.5 ± 56.0⟩	5 (646.6 ± 47.9) ⟨160943.1 ± 101.6⟩	5 (740.3 ± 59.7) ⟨177006.9 ± 71.8⟩		

results for **LMCut** continued on next page

	CG-iLAO* _{tid}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
Elev	04-01	5 (0.0 ± 0.0) ⟨0.9 ± 0.2⟩	5 (0.0 ± 0.0) ⟨1.4 ± 0.4⟩	5 (0.0 ± 0.0) ⟨2.0 ± 0.4⟩	5 (0.1 ± 0.0) ⟨1.0 ± 0.0⟩
	04-02	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	04-03	5 (0.0 ± 0.0) ⟨1.9 ± 0.1⟩	5 (0.0 ± 0.0) ⟨2.4 ± 0.3⟩	5 (0.0 ± 0.0) ⟨3.1 ± 0.4⟩	5 (0.1 ± 0.0) ⟨1.8 ± 0.1⟩
	04-04	5 (0.0 ± 0.0) ⟨2.1 ± 0.1⟩	5 (0.0 ± 0.0) ⟨2.3 ± 0.1⟩	5 (0.0 ± 0.0) ⟨3.3 ± 0.1⟩	5 (0.1 ± 0.0) ⟨1.9 ± 0.1⟩
	04-05	5 (0.0 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.6 ± 0.1⟩	5 (0.0 ± 0.0) ⟨1.1 ± 0.5⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩
	04-06	5 (0.3 ± 0.0) ⟨16.9 ± 0.5⟩	5 (0.3 ± 0.0) ⟨17.8 ± 0.6⟩	5 (0.3 ± 0.0) ⟨23.1 ± 4.2⟩	5 (0.8 ± 0.0) ⟨14.4 ± 0.5⟩
	04-07	5 (0.2 ± 0.0) ⟨5.2 ± 0.7⟩	5 (0.2 ± 0.0) ⟨5.5 ± 0.7⟩	5 (0.3 ± 0.0) ⟨15.1 ± 2.1⟩	5 (0.4 ± 0.0) ⟨4.7 ± 0.2⟩
	04-08	5 (3.9 ± 0.1) ⟨587.5 ± 7.9⟩	5 (3.8 ± 0.1) ⟨590.9 ± 14.0⟩	5 (4.2 ± 0.0) ⟨755.7 ± 14.4⟩	5 (10.9 ± 0.4) ⟨582.0 ± 7.5⟩
	04-09	5 (1.6 ± 0.1) ⟨142.1 ± 3.5⟩	5 (1.7 ± 0.1) ⟨146.0 ± 6.4⟩	5 (1.9 ± 0.1) ⟨180.2 ± 8.2⟩	5 (5.7 ± 0.3) ⟨136.8 ± 1.3⟩
	04-10	5 (7.8 ± 0.2) ⟨2700.6 ± 34.2⟩	5 (7.8 ± 0.2) ⟨2737.9 ± 16.6⟩	5 (8.7 ± 0.1) ⟨3488.2 ± 76.9⟩	5 (22.1 ± 3.0) ⟨2836.5 ± 29.0⟩
	04-11	5 (627.7 ± 19.3) ⟨109462.0 ± 1124.1⟩	5 (623.0 ± 10.8) ⟨108232.1 ± 1333.8⟩	5 (661.7 ± 4.4) ⟨135429.2 ± 670.1⟩	3 (1098.1 ± 27.2) ⟨112566.7 ± 1898.4⟩
	04-12	5 (114.5 ± 3.1) ⟨8066.2 ± 104.7⟩	5 (115.1 ± 2.7) ⟨8383.7 ± 73.6⟩	5 (120.2 ± 2.2) ⟨10187.0 ± 137.9⟩	5 (247.0 ± 5.7) ⟨8127.6 ± 35.9⟩
	04-13		1 (1780.2±) ⟨926136.2±⟩		
	04-14	5 (65.8 ± 1.4) ⟨2016.7 ± 25.7⟩	5 (66.2 ± 1.6) ⟨2160.1 ± 20.6⟩	5 (91.0 ± 2.8) ⟨2715.3 ± 21.5⟩	5 (188.2 ± 9.3) ⟨2062.9 ± 14.3⟩
	04-15	5 (907.4 ± 16.8) ⟨121601.7 ± 488.0⟩	5 (935.8 ± 60.5) ⟨122326.7 ± 649.6⟩	5 (931.3 ± 13.6) ⟨145749.4 ± 1232.7⟩	

results for **LMCut** continued on next page

	CG-iLAO* _{tid}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
ExBW	04-01	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	04-02	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	04-03	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.3 ± 0.2⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	04-04	5 (0.1 ± 0.0) ⟨28.5 ± 6.9⟩	5 (0.1 ± 0.0) ⟨23.1 ± 8.1⟩	5 (0.1 ± 0.0) ⟨27.8 ± 1.7⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩
	04-05	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	04-06	5 (0.6 ± 0.0) ⟨63.7 ± 0.0⟩	5 (0.7 ± 0.1) ⟨60.8 ± 5.7⟩	5 (1.6 ± 0.2) ⟨20.3 ± 2.9⟩	5 (1.4 ± 0.1) ⟨57.1 ± 1.1⟩
	04-07	5 (5.3 ± 0.1) ⟨278.1 ± 0.4⟩	5 (5.3 ± 0.1) ⟨261.7 ± 6.1⟩	5 (7.8 ± 0.4) ⟨164.1 ± 71.9⟩	5 (9.8 ± 0.3) ⟨277.9 ± 4.0⟩
	04-08	5 (0.4 ± 0.0) ⟨2.0 ± 0.1⟩	5 (0.6 ± 0.2) ⟨2.9 ± 1.1⟩	5 (1.7 ± 0.2) ⟨12.3 ± 1.8⟩	5 (0.5 ± 0.0) ⟨1.3 ± 0.0⟩
	04-09	5 (64.2 ± 1.1) ⟨3307.2 ± 26.8⟩	5 (62.9 ± 0.9) ⟨3112.4 ± 65.8⟩	5 (96.5 ± 4.8) ⟨4403.7 ± 182.0⟩	5 (177.7 ± 2.8) ⟨4064.9 ± 24.2⟩
	04-10	5 (244.3 ± 7.0) ⟨22305.7 ± 210.1⟩	5 (242.1 ± 6.9) ⟨21378.8 ± 275.5⟩	5 (399.8 ± 8.5) ⟨28073.7 ± 362.6⟩	5 (517.2 ± 8.7) ⟨15494.2 ± 72.8⟩
	08-01	5 (0.0 ± 0.0) ⟨0.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.6 ± 0.3⟩	5 (0.1 ± 0.0) ⟨0.3 ± 0.0⟩
	08-02	5 (0.1 ± 0.0) ⟨14.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨16.8 ± 3.7⟩	5 (0.1 ± 0.0) ⟨34.7 ± 0.5⟩	5 (0.1 ± 0.0) ⟨0.7 ± 0.0⟩
	08-03	5 (0.1 ± 0.0) ⟨21.2 ± 1.5⟩	5 (0.1 ± 0.0) ⟨24.0 ± 5.2⟩	5 (0.1 ± 0.0) ⟨39.6 ± 1.5⟩	5 (0.3 ± 0.0) ⟨37.6 ± 0.5⟩
	08-04	5 (0.1 ± 0.0) ⟨25.1 ± 0.1⟩	5 (0.1 ± 0.0) ⟨25.1 ± 0.3⟩	5 (0.2 ± 0.0) ⟨21.1 ± 0.9⟩	5 (0.4 ± 0.0) ⟨4.3 ± 0.0⟩
	08-05	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.5 ± 0.2⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩
	08-06	5 (0.2 ± 0.0) ⟨5.5 ± 0.2⟩	5 (0.2 ± 0.0) ⟨5.5 ± 0.1⟩	5 (0.4 ± 0.0) ⟨20.9 ± 10.1⟩	5 (0.4 ± 0.0) ⟨4.2 ± 0.0⟩
	08-07	5 (0.3 ± 0.0) ⟨1.8 ± 0.2⟩	5 (0.5 ± 0.2) ⟨3.8 ± 1.4⟩	5 (1.5 ± 0.3) ⟨12.2 ± 2.4⟩	5 (0.8 ± 0.0) ⟨3.8 ± 0.5⟩
	08-08	5 (85.2 ± 2.9) ⟨8944.0 ± 71.5⟩	5 (91.2 ± 6.3) ⟨7975.7 ± 682.6⟩	5 (172.4 ± 3.7) ⟨8856.8 ± 227.5⟩	
	08-09	5 (16.5 ± 0.8) ⟨946.6 ± 29.2⟩	5 (17.1 ± 0.8) ⟨864.4 ± 112.3⟩	5 (25.8 ± 1.2) ⟨1055.4 ± 42.2⟩	5 (72.4 ± 1.4) ⟨936.3 ± 11.2⟩
	08-10	5 (339.7 ± 14.8) ⟨70581.3 ± 1891.8⟩	5 (335.3 ± 13.8) ⟨72276.6 ± 3571.6⟩	5 (479.9 ± 29.9) ⟨88905.2 ± 5014.5⟩	5 (1321.7 ± 20.7) ⟨65604.9 ± 327.5⟩
	08-11	1 (1792.6±) ⟨10366.8±⟩			

results for **LMCut** continued on next page

		CG-iLAO* _{tied}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
PARC-N	4, 1	5 (15.5 ± 0.3) ⟨1609.7 ± 33.8⟩	5 (15.4 ± 0.2) ⟨1209.5 ± 5.8⟩	5 (15.7 ± 0.4) ⟨1529.9 ± 23.3⟩	5 (36.3 ± 0.6) ⟨1332.5 ± 11.3⟩	5 (45.2 ± 2.4) ⟨1332.5 ± 11.3⟩
	4, 2	5 (26.4 ± 0.7) ⟨3328.1 ± 53.1⟩	5 (26.0 ± 0.8) ⟨2271.1 ± 19.5⟩	5 (26.9 ± 0.8) ⟨3063.9 ± 67.7⟩	5 (60.4 ± 1.7) ⟨2285.6 ± 23.4⟩	5 (83.0 ± 6.1) ⟨2285.6 ± 23.4⟩
	4, 3	5 (44.9 ± 0.6) ⟨5871.2 ± 77.2⟩	5 (45.6 ± 1.8) ⟨3716.2 ± 69.6⟩	5 (46.9 ± 1.2) ⟨5034.2 ± 93.8⟩	5 (119.2 ± 12.0) ⟨3845.7 ± 24.1⟩	5 (146.7 ± 17.4) ⟨3845.7 ± 24.1⟩
	5, 1	5 (208.2 ± 5.4) ⟨19432.1 ± 709.3⟩	5 (207.2 ± 4.2) ⟨13860.8 ± 135.6⟩	5 (207.2 ± 4.3) ⟨18184.9 ± 426.2⟩	5 (478.9 ± 49.6) ⟨15922.5 ± 57.8⟩	5 (543.5 ± 23.5) ⟨15922.5 ± 57.8⟩
	5, 2	5 (367.6 ± 9.2) ⟨43475.1 ± 1268.2⟩	5 (359.2 ± 9.1) ⟨27722.5 ± 133.8⟩	5 (354.0 ± 11.8) ⟨36355.7 ± 690.6⟩	5 (775.7 ± 27.2) ⟨27953.6 ± 86.6⟩	2 (903.1 ± 18.5) ⟨27860.1 ± 36.4⟩
	5, 3	5 (675.5 ± 48.1) ⟨75615.7 ± 2301.5⟩	5 (635.7 ± 17.9) ⟨44140.6 ± 335.0⟩	5 (630.0 ± 16.0) ⟨58728.2 ± 824.2⟩		
PARC-R	4, 1	5 (82.1 ± 2.3) ⟨9153.8 ± 54.6⟩	5 (81.9 ± 2.4) ⟨8768.6 ± 16.5⟩	5 (81.7 ± 1.1) ⟨9869.2 ± 62.1⟩	5 (166.2 ± 4.7) ⟨9589.6 ± 17.7⟩	5 (183.3 ± 5.0) ⟨9589.6 ± 17.7⟩
	4, 2	5 (208.4 ± 5.1) ⟨26611.8 ± 79.4⟩	5 (225.6 ± 21.9) ⟨26155.2 ± 67.1⟩	5 (234.3 ± 7.0) ⟨31546.5 ± 307.0⟩	5 (445.9 ± 16.8) ⟨28966.6 ± 36.6⟩	5 (543.1 ± 20.9) ⟨28966.6 ± 36.6⟩
	4, 3	5 (327.1 ± 7.8) ⟨38188.8 ± 155.5⟩	5 (332.6 ± 3.0) ⟨38037.8 ± 157.3⟩	5 (359.4 ± 2.0) ⟨45156.3 ± 184.2⟩	5 (694.4 ± 21.2) ⟨40985.9 ± 130.4⟩	5 (829.4 ± 8.0) ⟨40985.9 ± 130.4⟩
	5, 1	5 (1219.7 ± 25.4) ⟨105687.8 ± 937.6⟩	5 (1218.6 ± 21.5) ⟨99770.9 ± 558.6⟩	5 (1233.9 ± 18.7) ⟨112842.5 ± 311.3⟩		
Rand	04-03	5 (1.8 ± 0.1) ⟨28.6 ± 0.3⟩	5 (1.8 ± 0.0) ⟨31.9 ± 0.3⟩	5 (1.7 ± 0.0) ⟨32.1 ± 1.0⟩	5 (1.8 ± 0.0) ⟨27.0 ± 0.2⟩	5 (1.8 ± 0.0) ⟨27.0 ± 0.2⟩
	04-06	5 (5.5 ± 0.8) ⟨2.4 ± 0.0⟩	5 (5.4 ± 0.6) ⟨1.3 ± 0.1⟩	5 (5.4 ± 0.6) ⟨1.1 ± 0.1⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	04-07	5 (0.2 ± 0.0) ⟨0.3 ± 0.1⟩	5 (0.2 ± 0.1) ⟨0.3 ± 0.1⟩	5 (0.4 ± 0.4) ⟨1.1 ± 1.3⟩	5 (0.4 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.9 ± 0.0⟩
	04-09				5 (1551.0 ± 72.1) ⟨2987.1 ± 67.7⟩	5 (1529.9 ± 57.3) ⟨2965.0 ± 55.1⟩
	04-13	5 (23.4 ± 1.2) ⟨21.2 ± 0.1⟩	5 (24.4 ± 1.8) ⟨21.2 ± 0.1⟩	5 (33.7 ± 14.1) ⟨47.0 ± 28.4⟩	5 (22.2 ± 3.5) ⟨15.6 ± 4.7⟩	5 (21.5 ± 2.8) ⟨15.7 ± 4.8⟩

results for **LMCut** continued on next page

	CG-iLAO [*] _{ti}	CG-iLAO [*] _{single}	CG-iLAO [*] _{trial}	CG-iLAO [*] _{FF-AO}	CG-iLAO [*] _{FF-MLO}	
□TW	08-01	5 (0.0 ± 0.0) ⟨1.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.5 ± 0.1⟩	5 (0.0 ± 0.0) ⟨2.0 ± 0.9⟩	5 (0.1 ± 0.0) ⟨1.2 ± 0.1⟩	5 (0.0 ± 0.0) ⟨1.1 ± 0.1⟩
	08-02	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	08-03	5 (0.2 ± 0.0) ⟨3.1 ± 0.1⟩	5 (0.2 ± 0.0) ⟨3.8 ± 0.2⟩	5 (0.2 ± 0.0) ⟨4.1 ± 0.1⟩	5 (0.2 ± 0.0) ⟨3.1 ± 0.2⟩	5 (0.2 ± 0.0) ⟨2.8 ± 0.1⟩
	08-04	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	08-05	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩
	08-06	5 (4.3 ± 0.8) ⟨17.8 ± 0.4⟩	5 (4.0 ± 0.4) ⟨21.3 ± 0.4⟩	5 (3.4 ± 0.3) ⟨33.0 ± 0.7⟩	5 (8.8 ± 2.7) ⟨18.4 ± 0.7⟩	5 (9.0 ± 0.6) ⟨19.1 ± 0.5⟩
	08-07	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	08-08	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	08-09	5 (33.2 ± 1.3) ⟨53.5 ± 0.5⟩	5 (35.8 ± 4.8) ⟨74.7 ± 4.4⟩	5 (36.8 ± 4.2) ⟨88.7 ± 3.3⟩	5 (67.2 ± 12.0) ⟨75.8 ± 1.5⟩	5 (97.5 ± 14.0) ⟨78.1 ± 0.9⟩
	08-10	5 (7.8 ± 1.2) ⟨285.6 ± 0.0⟩	5 (8.2 ± 2.5) ⟨305.1 ± 34.6⟩	5 (7.3 ± 0.9) ⟨381.2 ± 1.1⟩	5 (1.1 ± 0.3) ⟨0.0 ± 0.0⟩	5 (1.3 ± 0.3) ⟨0.0 ± 0.0⟩
08-11	5 (444.1 ± 214.6) ⟨138.1 ± 1.0⟩	5 (333.5 ± 54.7) ⟨154.5 ± 1.3⟩	5 (331.0 ± 18.6) ⟨162.4 ± 5.1⟩	5 (848.6 ± 102.0) ⟨147.3 ± 1.7⟩	5 (1245.3 ± 185.9) ⟨147.8 ± 1.4⟩	
08-12	5 (331.9 ± 86.6) ⟨40969.9 ± 542.7⟩	5 (305.3 ± 33.1) ⟨42206.2 ± 315.7⟩	5 (315.1 ± 21.2) ⟨44615.4 ± 738.9⟩	5 (32.1 ± 6.0) ⟨0.0 ± 0.0⟩	5 (42.9 ± 5.2) ⟨0.0 ± 0.0⟩	
08-14	5 (15.8 ± 3.3) ⟨0.0 ± 0.0⟩	5 (13.9 ± 0.5) ⟨0.0 ± 0.0⟩	5 (13.9 ± 0.6) ⟨0.0 ± 0.0⟩	5 (12.4 ± 1.9) ⟨0.0 ± 0.0⟩	5 (12.9 ± 1.1) ⟨0.0 ± 0.0⟩	
S&R	08-01	5 (0.2 ± 0.0) ⟨401.9 ± 1.0⟩	5 (0.2 ± 0.0) ⟨400.6 ± 1.7⟩	5 (0.3 ± 0.0) ⟨387.8 ± 3.6⟩	5 (1.1 ± 0.1) ⟨408.1 ± 0.6⟩	5 (1.4 ± 0.2) ⟨408.1 ± 0.6⟩
	08-02	5 (1.1 ± 0.1) ⟨1773.9 ± 4.4⟩	5 (1.0 ± 0.0) ⟨1768.8 ± 3.9⟩	5 (1.2 ± 0.1) ⟨1719.6 ± 2.7⟩	5 (3.9 ± 0.3) ⟨1809.9 ± 3.8⟩	5 (4.3 ± 0.3) ⟨1809.9 ± 3.8⟩
	08-03	5 (4.3 ± 0.3) ⟨7374.6 ± 10.3⟩	5 (4.3 ± 0.2) ⟨7340.2 ± 10.1⟩	5 (4.9 ± 0.3) ⟨7183.1 ± 13.6⟩	5 (15.6 ± 1.1) ⟨7470.3 ± 6.6⟩	5 (16.3 ± 1.1) ⟨7470.3 ± 6.6⟩
	08-04	5 (83.7 ± 2.2) ⟨113816.3 ± 30.9⟩	5 (108.4 ± 36.8) ⟨113537.7 ± 45.0⟩	5 (88.6 ± 4.4) ⟨111609.7 ± 122.6⟩	5 (232.8 ± 12.7) ⟨114671.1 ± 41.7⟩	5 (239.3 ± 11.2) ⟨114671.1 ± 41.7⟩
	08-05	4 (1713.5 ± 33.7) ⟨1593746.8 ± 335.4⟩		2 (1703.4 ± 5.8) ⟨1569075.1 ± 1616.3⟩		

results for **LMCut** continued on next page

	CG-iLAO* _{ti}	CG-iLAO* _{single}	CG-iLAO* _{trial}	CG-iLAO* _{FF-AO}	CG-iLAO* _{FF-MLO}
Sched	04-01	5 (0.0 ± 0.0) ⟨1.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.8 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.7 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.6 ± 0.0⟩
	04-02	5 (0.0 ± 0.0) ⟨1.5 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.0 ± 0.0⟩
	04-03	5 (0.0 ± 0.0) ⟨3.5 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.1 ± 0.0⟩
	04-04	5 (18.7 ± 2.5) ⟨9565.0 ± 163.9⟩	5 (18.2 ± 1.1) ⟨8396.7 ± 183.9⟩	5 (20.1 ± 0.7) ⟨9896.6 ± 210.0⟩	5 (61.0 ± 2.1) ⟨13730.4 ± 173.9⟩
	04-05	5 (1590.1 ± 101.6) ⟨440820.6 ± 717.5⟩	5 (1558.6 ± 104.4) ⟨383534.0 ± 1971.1⟩	5 (1625.1 ± 80.4) ⟨374023.0 ± 435.5⟩	
Sys	08-01	5 (0.0 ± 0.0) ⟨1.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.8 ± 0.1⟩	5 (0.0 ± 0.0) ⟨0.7 ± 0.0⟩
	08-02	5 (0.0 ± 0.0) ⟨1.9 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.2 ± 0.1⟩	5 (0.0 ± 0.0) ⟨1.1 ± 0.0⟩
	08-03	5 (0.0 ± 0.0) ⟨4.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.5 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.3 ± 0.2⟩	5 (0.0 ± 0.0) ⟨2.2 ± 0.0⟩
	08-04	5 (757.3 ± 16.6) ⟨211614.3 ± 1062.1⟩	5 (794.6 ± 29.2) ⟨181461.8 ± 1424.1⟩	5 (842.0 ± 40.2) ⟨206097.9 ± 1958.1⟩	
	08-01	5 (0.0 ± 0.0) ⟨19.2 ± 0.4⟩	5 (0.0 ± 0.0) ⟨19.2 ± 0.3⟩	5 (0.0 ± 0.0) ⟨11.0 ± 0.1⟩	5 (0.2 ± 0.0) ⟨10.0 ± 0.1⟩
ΔTW	08-02	5 (0.2 ± 0.0) ⟨110.3 ± 1.0⟩	5 (0.1 ± 0.0) ⟨110.7 ± 1.9⟩	5 (0.1 ± 0.0) ⟨51.1 ± 0.5⟩	5 (0.6 ± 0.1) ⟨46.1 ± 0.6⟩
	08-03	5 (0.6 ± 0.0) ⟨412.1 ± 3.8⟩	5 (0.7 ± 0.0) ⟨424.8 ± 5.6⟩	5 (0.5 ± 0.0) ⟨172.5 ± 1.8⟩	5 (2.0 ± 0.1) ⟨154.4 ± 0.8⟩
	08-04	5 (9.0 ± 0.6) ⟨5085.1 ± 34.7⟩	5 (8.8 ± 0.4) ⟨5173.8 ± 28.8⟩	5 (7.8 ± 0.9) ⟨2032.0 ± 16.1⟩	
	08-01	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩
	08-02	5 (0.0 ± 0.0) ⟨2.6 ± 0.2⟩	5 (0.0 ± 0.0) ⟨2.5 ± 0.1⟩	5 (0.0 ± 0.0) ⟨3.7 ± 0.4⟩	5 (0.2 ± 0.0) ⟨3.5 ± 0.0⟩
6, 8	08-03	5 (0.5 ± 0.0) ⟨70.8 ± 1.3⟩	5 (0.5 ± 0.0) ⟨84.5 ± 3.8⟩	5 (0.7 ± 0.0) ⟨78.4 ± 6.2⟩	5 (4.5 ± 0.8) ⟨77.7 ± 1.7⟩
	08-04	5 (15.9 ± 0.6) ⟨2106.3 ± 30.8⟩	5 (17.0 ± 1.0) ⟨2318.3 ± 73.8⟩	5 (21.8 ± 0.5) ⟨1981.9 ± 64.9⟩	5 (81.9 ± 6.5) ⟨1672.5 ± 12.2⟩
	08-05	5 (554.4 ± 14.6) ⟨58626.0 ± 514.2⟩	5 (547.3 ± 9.0) ⟨59966.7 ± 1387.6⟩	5 (714.7 ± 30.4) ⟨46823.8 ± 1933.7⟩	5 (82.9 ± 5.8) ⟨1618.7 ± 112.0⟩
	6, 8	5 (1059.0 ± 20.5) ⟨91451.9 ± 2297.3⟩	5 (1112.7 ± 41.4) ⟨95547.7 ± 3063.1⟩	5 (1399.7 ± 28.9) ⟨69665.7 ± 2248.4⟩	

results for **LMCut** continued on next page

	CG-iLAO [*] _{tied}	CG-iLAO [*] _{single}	CG-iLAO [*] _{trial}	CG-iLAO [*] _{FF-AO}	CG-iLAO [*] _{FF-MLO}
Zeno	04-01	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	04-02	5 (1201.3 ± 83.9) ⟨47914.9 ± 290.1⟩	5 (1177.5 ± 66.6) ⟨48882.4 ± 274.2⟩	5 (1001.9 ± 37.8) ⟨42064.5 ± 121.3⟩	
	08-01	5 (24.3 ± 0.6) ⟨1941.0 ± 11.9⟩	5 (24.6 ± 0.8) ⟨1988.7 ± 15.3⟩	5 (30.6 ± 0.7) ⟨1591.2 ± 11.8⟩	5 (61.1 ± 1.6) ⟨1820.9 ± 9.9⟩
	08-02	5 (22.4 ± 0.4) ⟨1086.1 ± 17.7⟩	5 (22.4 ± 0.2) ⟨1106.8 ± 18.3⟩	5 (30.4 ± 0.7) ⟨877.1 ± 13.2⟩	5 (59.1 ± 3.2) ⟨1092.0 ± 6.2⟩
	08-04	5 (654.5 ± 39.5) ⟨23235.9 ± 247.3⟩	5 (631.9 ± 18.3) ⟨23718.0 ± 262.2⟩	5 (633.8 ± 15.1) ⟨17847.1 ± 230.5⟩	5 (1141.0 ± 23.4) ⟨20401.4 ± 289.7⟩

Table 3: Results for algorithms using **LMCut**. We give X (Y) (Z) where X is the coverage, Y is the mean time (secs), and Z is the mean number of *thousands* of Q -values. Each mean is taken over the converged instances, and we report the 95% C.I.

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}	
BW	04-01	5 (0.4 ± 0.0) (7.8 ± 1.1)	5 (0.2 ± 0.0) (6.8 ± 0.4)	5 (0.2 ± 0.0) (4.1 ± 0.2)	5 (0.2 ± 0.0) (4.4 ± 0.4)	5 (0.2 ± 0.0) (5.4 ± 0.8)
	04-02	5 (0.3 ± 0.0) (5.1 ± 1.0)	5 (0.2 ± 0.0) (4.8 ± 0.2)	5 (0.2 ± 0.0) (2.9 ± 0.1)	5 (0.2 ± 0.0) (2.9 ± 0.2)	5 (0.2 ± 0.0) (4.5 ± 0.5)
	04-03	5 (0.2 ± 0.0) (2.5 ± 1.3)	5 (0.1 ± 0.0) (1.4 ± 0.4)	5 (0.1 ± 0.0) (0.8 ± 0.2)	5 (0.2 ± 0.0) (0.5 ± 0.2)	5 (0.1 ± 0.0) (1.4 ± 0.4)
	04-04	5 (0.5 ± 0.0) (11.8 ± 1.3)	5 (0.3 ± 0.0) (14.0 ± 0.1)	5 (0.3 ± 0.0) (8.3 ± 0.1)	5 (0.2 ± 0.0) (7.4 ± 0.1)	5 (0.2 ± 0.0) (10.8 ± 1.7)
	04-05	5 (0.3 ± 0.0) (5.2 ± 0.6)	5 (0.2 ± 0.0) (3.1 ± 0.1)	5 (0.2 ± 0.0) (1.9 ± 0.0)	5 (0.2 ± 0.0) (2.0 ± 0.1)	5 (0.2 ± 0.0) (3.3 ± 0.7)
	04-06		5 (79.4 ± 4.5) (1356.9 ± 75.2)	5 (86.9 ± 4.5) (868.6 ± 44.0)	5 (67.5 ± 1.3) (834.4 ± 8.8)	5 (70.1 ± 5.3) (1041.7 ± 10.9)
	04-08	5 (477.5 ± 53.1) (2057.1 ± 175.1)	5 (23.1 ± 0.5) (697.7 ± 21.2)	5 (21.9 ± 0.7) (447.7 ± 10.8)	5 (17.8 ± 1.3) (266.3 ± 7.4)	5 (16.2 ± 1.0) (266.2 ± 12.6)
	04-09	5 (561.2 ± 100.2) (2229.9 ± 381.6)	5 (16.7 ± 1.1) (373.8 ± 5.6)	5 (18.3 ± 3.1) (251.6 ± 6.1)	5 (14.4 ± 1.2) (173.8 ± 4.4)	5 (13.8 ± 1.4) (194.2 ± 6.0)
	04-10		5 (202.4 ± 6.8) (3880.9 ± 39.5)	5 (204.3 ± 9.1) (2301.8 ± 20.4)	5 (170.9 ± 9.9) (2250.5 ± 21.5)	5 (172.7 ± 6.9) (2906.7 ± 34.8)
	08-01	5 (0.2 ± 0.0) (4.1 ± 0.6)	5 (0.2 ± 0.0) (3.5 ± 0.5)	5 (0.2 ± 0.0) (2.0 ± 0.3)	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)
	08-02	5 (0.2 ± 0.0) (4.3 ± 0.5)	5 (0.2 ± 0.0) (3.5 ± 0.5)	5 (0.2 ± 0.0) (2.0 ± 0.3)	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)
	08-03	5 (0.2 ± 0.0) (4.1 ± 0.6)	5 (0.2 ± 0.0) (3.5 ± 0.5)	5 (0.2 ± 0.0) (2.0 ± 0.3)	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)
08-04	5 (0.2 ± 0.0) (4.4 ± 0.5)	5 (0.2 ± 0.0) (3.5 ± 0.5)	5 (0.2 ± 0.0) (2.0 ± 0.3)	5 (0.2 ± 0.0) (1.8 ± 0.1)	5 (0.2 ± 0.0) (3.4 ± 0.8)	
08-05		5 (427.4 ± 17.5) (9657.7 ± 120.0)	5 (406.8 ± 29.1) (6027.1 ± 75.8)	5 (361.1 ± 19.4) (6615.8 ± 53.7)	5 (351.0 ± 6.5) (8267.2 ± 152.7)	
08-06		5 (427.0 ± 18.1) (9657.7 ± 120.0)	5 (383.6 ± 6.7) (6027.1 ± 75.8)	5 (354.8 ± 18.0) (6615.8 ± 53.7)	5 (348.2 ± 11.2) (8267.2 ± 152.7)	
08-07		5 (423.3 ± 16.8) (9657.7 ± 120.0)	5 (418.2 ± 28.7) (6027.1 ± 75.8)	5 (359.9 ± 24.3) (6615.8 ± 53.7)	5 (370.4 ± 20.8) (8267.2 ± 152.7)	
08-08		5 (436.0 ± 29.8) (9657.7 ± 120.0)	5 (394.0 ± 13.5) (6027.1 ± 75.8)	5 (358.6 ± 9.0) (6615.8 ± 53.7)	5 (363.2 ± 25.2) (8267.2 ± 152.7)	
08-09		5 (352.2 ± 31.4) (1468.6 ± 104.0)	5 (353.8 ± 25.8) (868.0 ± 65.0)	5 (321.0 ± 36.5) (708.9 ± 26.8)	5 (293.9 ± 31.8) (912.6 ± 31.9)	
08-10		5 (366.9 ± 49.7) (1468.6 ± 104.0)	5 (336.0 ± 50.7) (868.0 ± 65.0)	5 (338.1 ± 30.5) (708.9 ± 26.8)	5 (327.6 ± 44.6) (912.6 ± 31.9)	
08-11		5 (353.2 ± 35.2) (1468.6 ± 104.0)	5 (383.0 ± 13.9) (868.0 ± 65.0)	5 (352.6 ± 17.8) (708.9 ± 26.8)	5 (334.3 ± 20.5) (912.6 ± 31.9)	
08-12		5 (318.3 ± 32.6) (1468.6 ± 104.0)	5 (361.0 ± 21.4) (868.0 ± 65.0)	5 (297.5 ± 40.9) (708.9 ± 26.8)	5 (270.0 ± 19.7) (912.6 ± 31.9)	

results for **ROC** continued on next page

		LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tid}	CG-iLAO* _{single}
Coresec	01	5 (0.8 ± 0.0) ⟨212.7 ± 2.7⟩	5 (0.7 ± 0.0) ⟨144.4 ± 0.9⟩	5 (0.7 ± 0.0) ⟨114.8 ± 1.3⟩	5 (0.7 ± 0.0) ⟨100.9 ± 0.7⟩	5 (0.7 ± 0.0) ⟨94.6 ± 1.1⟩
	02	5 (2.2 ± 0.1) ⟨688.2 ± 13.9⟩	5 (2.1 ± 0.0) ⟨483.1 ± 2.1⟩	5 (2.2 ± 0.1) ⟨394.5 ± 4.3⟩	5 (2.1 ± 0.1) ⟨331.1 ± 3.8⟩	5 (2.1 ± 0.1) ⟨310.4 ± 3.1⟩
	03	5 (7.5 ± 0.2) ⟨3432.0 ± 34.5⟩	5 (6.3 ± 0.3) ⟨1902.2 ± 1.7⟩	5 (6.2 ± 0.2) ⟨1427.1 ± 11.3⟩	5 (5.9 ± 0.1) ⟨1161.1 ± 8.6⟩	5 (5.9 ± 0.1) ⟨1099.9 ± 4.8⟩
	04	5 (19.9 ± 0.3) ⟨8788.2 ± 32.2⟩	5 (16.5 ± 0.3) ⟨4926.9 ± 19.6⟩	5 (17.1 ± 0.3) ⟨3746.2 ± 12.2⟩	5 (16.2 ± 0.4) ⟨3024.4 ± 12.0⟩	5 (16.0 ± 0.2) ⟨2871.6 ± 13.2⟩
	05	5 (467.1 ± 16.6) ⟨222353.7 ± 681.8⟩	5 (370.5 ± 8.1) ⟨104245.2 ± 152.0⟩	5 (394.1 ± 3.3) ⟨82256.4 ± 115.2⟩	5 (397.4 ± 14.3) ⟨67262.5 ± 60.6⟩	5 (413.1 ± 15.9) ⟨64252.3 ± 112.7⟩
Elev	04-01	5 (0.2 ± 0.0) ⟨23.0 ± 0.6⟩	5 (0.2 ± 0.0) ⟨61.1 ± 5.1⟩	5 (0.2 ± 0.0) ⟨39.2 ± 2.6⟩	5 (0.2 ± 0.0) ⟨31.0 ± 1.9⟩	5 (0.2 ± 0.0) ⟨34.2 ± 2.2⟩
	04-02	5 (0.1 ± 0.0) ⟨3.4 ± 0.7⟩	5 (0.1 ± 0.0) ⟨3.2 ± 0.2⟩	5 (0.1 ± 0.0) ⟨2.1 ± 0.1⟩	5 (0.1 ± 0.0) ⟨1.5 ± 0.1⟩	5 (0.1 ± 0.0) ⟨2.1 ± 0.2⟩
	04-03	5 (0.2 ± 0.0) ⟨23.8 ± 0.4⟩	5 (0.2 ± 0.0) ⟨69.3 ± 3.6⟩	5 (0.2 ± 0.0) ⟨45.2 ± 2.9⟩	5 (0.2 ± 0.0) ⟨34.9 ± 2.2⟩	5 (0.2 ± 0.0) ⟨38.0 ± 1.8⟩
	04-04	5 (0.2 ± 0.0) ⟨16.7 ± 0.7⟩	5 (0.2 ± 0.0) ⟨39.9 ± 1.8⟩	5 (0.1 ± 0.0) ⟨27.0 ± 1.1⟩	5 (0.1 ± 0.0) ⟨20.7 ± 0.1⟩	5 (0.1 ± 0.0) ⟨22.8 ± 0.4⟩
	04-05	5 (0.2 ± 0.0) ⟨14.9 ± 0.6⟩	5 (0.1 ± 0.0) ⟨27.4 ± 3.5⟩	5 (0.1 ± 0.0) ⟨18.4 ± 2.3⟩	5 (0.1 ± 0.0) ⟨12.8 ± 0.2⟩	5 (0.1 ± 0.0) ⟨14.3 ± 0.5⟩
	04-06	5 (2.7 ± 0.1) ⟨477.6 ± 11.1⟩	5 (2.0 ± 0.1) ⟨773.9 ± 68.6⟩	5 (1.3 ± 0.1) ⟨518.8 ± 56.3⟩	5 (1.1 ± 0.0) ⟨323.6 ± 23.7⟩	5 (1.2 ± 0.0) ⟨339.7 ± 18.4⟩
	04-07	5 (2.8 ± 0.0) ⟨568.0 ± 9.5⟩	5 (2.1 ± 0.1) ⟨826.7 ± 40.7⟩	5 (1.3 ± 0.1) ⟨536.5 ± 41.9⟩	5 (1.2 ± 0.1) ⟨359.7 ± 7.7⟩	5 (1.2 ± 0.1) ⟨380.5 ± 13.1⟩
	04-08	5 (4.1 ± 0.1) ⟨1264.2 ± 21.1⟩	5 (8.5 ± 0.1) ⟨3862.5 ± 59.0⟩	5 (4.0 ± 0.2) ⟨2782.4 ± 174.3⟩	5 (3.5 ± 0.1) ⟨1847.7 ± 22.4⟩	5 (3.7 ± 0.1) ⟨1950.9 ± 16.2⟩
	04-09	5 (3.5 ± 0.0) ⟨962.5 ± 19.1⟩	5 (5.9 ± 0.1) ⟨2704.5 ± 46.3⟩	5 (3.1 ± 0.1) ⟨1958.9 ± 153.0⟩	5 (2.8 ± 0.1) ⟨1212.9 ± 21.4⟩	5 (2.9 ± 0.1) ⟨1290.6 ± 17.5⟩
	04-10	5 (5.8 ± 0.1) ⟨2238.8 ± 25.3⟩	5 (22.5 ± 0.3) ⟨11472.6 ± 280.3⟩	5 (7.1 ± 0.5) ⟨8392.6 ± 925.9⟩	5 (5.5 ± 0.2) ⟨4483.0 ± 74.3⟩	5 (5.7 ± 0.2) ⟨4588.2 ± 99.1⟩
	04-11	5 (375.5 ± 7.0) ⟨119315.6 ± 944.1⟩	5 (918.2 ± 30.2) ⟨351303.2 ± 9568.8⟩	5 (418.0 ± 35.5) ⟨353258.0 ± 46252.2⟩	5 (324.5 ± 30.9) ⟨181799.6 ± 2298.6⟩	5 (347.0 ± 35.3) ⟨184142.2 ± 1767.0⟩
	04-12	5 (206.1 ± 4.6) ⟨48556.8 ± 452.0⟩	5 (179.3 ± 8.1) ⟨70529.0 ± 2041.0⟩	5 (92.9 ± 1.8) ⟨60415.2 ± 3465.3⟩	5 (75.0 ± 2.1) ⟨35569.6 ± 539.2⟩	5 (79.8 ± 1.9) ⟨36463.0 ± 550.7⟩
	04-13	5 (1122.2 ± 21.9) ⟨348879.6 ± 851.2⟩			5 (1471.3 ± 112.9) ⟨1142941.6 ± 10195.4⟩	5 (1505.3 ± 110.0) ⟨1124113.1 ± 5931.4⟩
	04-14	5 (220.9 ± 3.2) ⟨53849.9 ± 726.4⟩	5 (527.3 ± 17.4) ⟨202878.8 ± 6359.2⟩	5 (203.8 ± 6.9) ⟨157605.9 ± 11361.7⟩	5 (163.5 ± 5.5) ⟨84851.3 ± 1605.0⟩	5 (174.5 ± 5.8) ⟨86241.8 ± 1638.0⟩
	04-15	5 (429.7 ± 13.8) ⟨119575.2 ± 580.9⟩	4 (1537.3 ± 69.7) ⟨493072.2 ± 14866.6⟩	5 (520.7 ± 46.1) ⟨422234.2 ± 37882.4⟩	5 (407.8 ± 23.7) ⟨233426.6 ± 1388.6⟩	5 (415.2 ± 16.4) ⟨235101.4 ± 1406.1⟩

results for **ROC** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
ExBW	04-01	5 (0.1 ± 0.0) (0.1 ± 0.0)	5 (0.1 ± 0.0) (0.4 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.1 ± 0.0)
	04-02	5 (0.1 ± 0.0) (0.1 ± 0.1)	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.1 ± 0.0)	5 (0.1 ± 0.0) (0.1 ± 0.0)
	04-03	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.5 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)
	04-04	5 (0.2 ± 0.0) (13.1 ± 1.3)	5 (0.3 ± 0.0) (97.4 ± 5.4)	5 (0.2 ± 0.0) (45.6 ± 1.1)	5 (0.2 ± 0.0) (35.8 ± 3.3)
	04-05	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)
	04-06	5 (1.9 ± 0.3) (27.6 ± 5.2)	5 (0.6 ± 0.0) (19.9 ± 1.2)	5 (0.5 ± 0.0) (9.9 ± 0.7)	5 (0.6 ± 0.0) (63.0 ± 0.4)
	04-07	5 (21.0 ± 1.1) (774.9 ± 34.5)	5 (16.8 ± 0.4) (1908.7 ± 16.3)	5 (11.3 ± 0.4) (923.7 ± 11.4)	5 (11.0 ± 0.3) (453.4 ± 8.5)
	04-08	5 (1.8 ± 0.4) (12.1 ± 3.5)	5 (1.0 ± 0.0) (14.9 ± 0.4)	5 (0.9 ± 0.0) (8.4 ± 0.2)	5 (0.9 ± 0.0) (4.7 ± 0.1)
	04-09	5 (60.6 ± 2.9) (2081.8 ± 83.9)	5 (90.4 ± 7.5) (16004.9 ± 2516.3)	5 (52.6 ± 5.0) (9783.0 ± 341.4)	5 (41.1 ± 0.9) (3535.5 ± 25.5)
	04-10	5 (511.5 ± 17.9) (44146.3 ± 1346.7)	5 (881.1 ± 28.9) (112737.0 ± 725.3)	5 (279.0 ± 12.7) (62256.7 ± 929.1)	5 (239.9 ± 6.1) (27692.1 ± 291.6)
	08-01	5 (0.1 ± 0.0) (1.2 ± 0.2)	5 (0.1 ± 0.0) (2.0 ± 0.0)	5 (0.1 ± 0.0) (1.0 ± 0.0)	5 (0.1 ± 0.0) (0.7 ± 0.0)
	08-02	5 (0.2 ± 0.0) (28.7 ± 2.1)	5 (0.3 ± 0.0) (94.2 ± 1.9)	5 (0.2 ± 0.0) (50.6 ± 3.1)	5 (0.2 ± 0.0) (17.1 ± 0.4)
	08-03	5 (0.3 ± 0.0) (26.6 ± 0.5)	5 (0.4 ± 0.0) (120.1 ± 23.3)	5 (0.2 ± 0.0) (68.9 ± 3.1)	5 (0.2 ± 0.0) (19.3 ± 0.3)
	08-04	5 (0.5 ± 0.0) (31.8 ± 1.1)	5 (0.6 ± 0.0) (93.3 ± 1.7)	5 (0.3 ± 0.0) (45.5 ± 0.5)	5 (0.3 ± 0.0) (27.3 ± 0.2)
	08-05	5 (0.1 ± 0.0) (1.6 ± 0.5)	5 (0.1 ± 0.0) (0.6 ± 0.1)	5 (0.1 ± 0.0) (0.3 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)
	08-06	5 (0.5 ± 0.0) (14.5 ± 1.2)	5 (0.4 ± 0.0) (35.5 ± 0.2)	5 (0.3 ± 0.0) (18.6 ± 0.4)	5 (0.3 ± 0.0) (37.8 ± 0.7)
	08-07	5 (1.2 ± 0.3) (13.8 ± 4.7)	5 (0.5 ± 0.1) (11.5 ± 1.3)	5 (0.4 ± 0.1) (6.1 ± 0.8)	5 (0.4 ± 0.1) (3.1 ± 0.5)
	08-08	5 (63.6 ± 2.9) (2856.8 ± 174.9)	5 (263.7 ± 5.8) (31845.9 ± 148.5)	5 (57.6 ± 2.9) (17206.1 ± 123.5)	5 (50.7 ± 4.1) (8422.5 ± 75.0)
	08-09	5 (21.1 ± 1.2) (1678.6 ± 45.0)	5 (25.3 ± 1.0) (2437.7 ± 90.5)	5 (8.0 ± 0.1) (1893.4 ± 12.7)	5 (5.6 ± 0.2) (457.4 ± 9.4)
	08-10	5 (148.3 ± 5.6) (8833.7 ± 225.7)		5 (296.3 ± 44.7) (216254.3 ± 10276.0)	5 (165.5 ± 13.7) (79741.2 ± 2529.6)
	08-11	5 (1272.1 ± 41.3) (20785.6 ± 745.2)	5 (519.3 ± 4.7) (26664.8 ± 112.9)	5 (429.1 ± 9.5) (14022.9 ± 103.2)	5 (386.5 ± 14.2) (10930.6 ± 90.2)

results for **ROC** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}	
PARC-N	4, 1	5 (9.2 ± 0.5) ⟨437.4 ± 10.3⟩	5 (14.3 ± 0.6) ⟨2945.1 ± 230.2⟩	5 (10.9 ± 1.4) ⟨2180.0 ± 155.0⟩	5 (9.3 ± 0.3) ⟨1027.5 ± 10.0⟩	5 (9.2 ± 0.4) ⟨719.2 ± 9.4⟩
	4, 2	5 (13.4 ± 0.4) ⟨584.6 ± 4.8⟩	5 (24.3 ± 1.6) ⟨5126.7 ± 415.5⟩	5 (15.2 ± 0.6) ⟨3342.1 ± 227.1⟩	5 (14.0 ± 0.3) ⟨1678.7 ± 36.8⟩	5 (13.8 ± 0.5) ⟨1094.9 ± 4.3⟩
	4, 3	5 (16.8 ± 1.0) ⟨683.3 ± 10.1⟩	5 (41.3 ± 3.3) ⟨9346.1 ± 767.8⟩	5 (21.2 ± 1.9) ⟨5435.0 ± 468.2⟩	5 (19.2 ± 0.8) ⟨2902.4 ± 82.8⟩	5 (18.6 ± 1.0) ⟨1676.8 ± 23.2⟩
	5, 1	5 (102.2 ± 4.4) ⟨5270.6 ± 107.0⟩	5 (194.5 ± 13.1) ⟨41577.4 ± 2390.9⟩	5 (123.1 ± 4.0) ⟨29435.2 ± 899.4⟩	5 (104.9 ± 3.9) ⟨12478.9 ± 325.1⟩	5 (104.3 ± 2.3) ⟨7946.2 ± 53.6⟩
	5, 2	5 (150.0 ± 7.8) ⟨7136.7 ± 138.0⟩	5 (325.5 ± 24.3) ⟨69568.1 ± 4558.8⟩	5 (182.8 ± 12.0) ⟨44381.1 ± 1562.3⟩	5 (161.0 ± 8.7) ⟨20373.5 ± 406.7⟩	5 (155.9 ± 7.8) ⟨12420.7 ± 58.6⟩
	5, 3	5 (185.7 ± 10.9) ⟨8356.2 ± 148.1⟩	5 (542.3 ± 41.0) ⟨121371.9 ± 6434.1⟩	5 (249.5 ± 18.9) ⟨74093.5 ± 3047.3⟩	5 (218.8 ± 12.9) ⟨36113.3 ± 1005.8⟩	5 (204.7 ± 13.4) ⟨18934.2 ± 96.6⟩
PARC-R	4, 1	5 (93.8 ± 1.8) ⟨40838.6 ± 631.2⟩	5 (62.6 ± 2.2) ⟨15601.4 ± 245.3⟩	5 (35.2 ± 2.5) ⟨8251.3 ± 13.3⟩	5 (33.6 ± 2.0) ⟨6146.0 ± 49.9⟩	5 (33.5 ± 2.3) ⟨5817.6 ± 33.8⟩
	4, 2	5 (684.2 ± 16.5) ⟨353312.3 ± 6318.3⟩	5 (136.8 ± 5.2) ⟨32341.6 ± 405.9⟩	5 (82.6 ± 1.4) ⟨18467.2 ± 98.7⟩	5 (84.4 ± 1.4) ⟨14664.1 ± 69.2⟩	5 (85.2 ± 4.4) ⟨14490.7 ± 71.7⟩
	4, 3	5 (1517.2 ± 60.1) ⟨786303.3 ± 12156.1⟩	5 (172.7 ± 5.9) ⟨41427.0 ± 374.3⟩	5 (113.3 ± 3.1) ⟨25911.2 ± 406.4⟩	5 (119.3 ± 4.3) ⟨21937.1 ± 56.3⟩	5 (119.2 ± 4.1) ⟨21833.5 ± 144.0⟩
	5, 1	5 (1327.8 ± 35.0) ⟨549475.6 ± 5442.2⟩	5 (690.2 ± 15.7) ⟨150805.6 ± 2429.8⟩	5 (474.0 ± 29.9) ⟨87942.5 ± 1224.0⟩	5 (498.8 ± 107.5) ⟨62262.2 ± 641.7⟩	5 (450.7 ± 14.3) ⟨58383.3 ± 118.4⟩
	5, 2		5 (1502.5 ± 11.5) ⟨291331.5 ± 3214.3⟩	5 (1156.9 ± 22.9) ⟨212992.1 ± 3218.2⟩	5 (1227.0 ± 23.2) ⟨159572.7 ± 498.4⟩	5 (1225.0 ± 31.1) ⟨157581.7 ± 403.4⟩
	5, 3			2 (1708.6 ± 77.7) ⟨310888.5 ± 3794.1⟩		
Rand	04-03	5 (0.3 ± 0.0) ⟨0.7 ± 0.1⟩	5 (0.3 ± 0.0) ⟨3.6 ± 0.0⟩	5 (0.3 ± 0.0) ⟨2.3 ± 0.4⟩	5 (0.3 ± 0.0) ⟨0.6 ± 0.1⟩	5 (0.3 ± 0.0) ⟨0.6 ± 0.1⟩
	04-04	5 (0.5 ± 0.0) ⟨3.5 ± 2.7⟩	5 (0.5 ± 0.0) ⟨3.3 ± 0.0⟩	5 (0.5 ± 0.0) ⟨1.4 ± 0.0⟩	5 (0.5 ± 0.0) ⟨0.6 ± 0.0⟩	5 (0.5 ± 0.0) ⟨0.6 ± 0.0⟩
	04-05		5 (45.5 ± 15.9) ⟨41.3 ± 3.1⟩	5 (35.6 ± 8.1) ⟨34.5 ± 3.4⟩	5 (13.3 ± 4.1) ⟨9.8 ± 4.9⟩	5 (15.5 ± 7.8) ⟨11.0 ± 6.0⟩
	04-06	5 (1.5 ± 0.1) ⟨2.3 ± 0.1⟩	5 (1.6 ± 0.1) ⟨21.9 ± 1.1⟩	5 (1.6 ± 0.2) ⟨10.4 ± 0.5⟩	5 (1.5 ± 0.1) ⟨2.4 ± 0.0⟩	5 (1.5 ± 0.1) ⟨1.3 ± 0.1⟩
	04-07	5 (20.9 ± 31.8) ⟨144.0 ± 249.4⟩	5 (0.6 ± 0.3) ⟨3.2 ± 2.2⟩	5 (0.6 ± 0.3) ⟨2.0 ± 1.7⟩	5 (0.6 ± 0.3) ⟨1.3 ± 1.2⟩	5 (0.6 ± 0.3) ⟨1.8 ± 1.5⟩
	04-08	5 (1365.9 ± 176.0) ⟨762.6 ± 3.5⟩				5 (1471.1 ± 122.8) ⟨811.2 ± 5.3⟩
	04-09	5 (25.3 ± 27.3) ⟨48.3 ± 58.3⟩	5 (5.4 ± 2.2) ⟨24.3 ± 5.1⟩	5 (6.0 ± 1.7) ⟨11.3 ± 3.4⟩	5 (5.8 ± 1.8) ⟨5.8 ± 2.1⟩	5 (5.9 ± 2.4) ⟨7.8 ± 7.0⟩
	04-10		3 (1713.1 ± 49.4) ⟨75076.2 ± 11365.8⟩		1 (1577.2±) ⟨11125.7±⟩	3 (1657.5 ± 68.6) ⟨14217.0 ± 471.0⟩
	04-13	3 (545.5 ± 291.4) ⟨1178.3 ± 579.9⟩	5 (59.5 ± 12.6) ⟨237.5 ± 36.7⟩	5 (66.0 ± 11.8) ⟨158.8 ± 25.7⟩	5 (61.1 ± 9.9) ⟨93.7 ± 21.7⟩	5 (62.2 ± 9.3) ⟨118.7 ± 31.2⟩

results for **ROC** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
□TW	08-01	5 (0.2 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.2 ± 0.0) ⟨3.3 ± 0.5⟩	5 (0.2 ± 0.0) ⟨2.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨1.3 ± 0.0⟩
	08-02	5 (0.2 ± 0.1) ⟨0.5 ± 0.0⟩	5 (0.2 ± 0.0) ⟨1.7 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.2 ± 0.0⟩
	08-03	5 (0.5 ± 0.0) ⟨3.0 ± 0.0⟩	5 (0.6 ± 0.0) ⟨7.9 ± 0.1⟩	5 (0.5 ± 0.0) ⟨4.1 ± 0.1⟩	5 (0.5 ± 0.0) ⟨3.3 ± 0.1⟩
	08-04	5 (0.4 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.0 ± 0.0⟩
	08-05	5 (1.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (1.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (1.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (1.1 ± 0.0) ⟨0.0 ± 0.0⟩
	08-06	5 (6.9 ± 1.7) ⟨12.5 ± 0.2⟩	5 (12.5 ± 0.9) ⟨42.3 ± 2.1⟩	5 (5.3 ± 1.0) ⟨22.3 ± 1.3⟩	5 (4.6 ± 0.5) ⟨19.5 ± 0.6⟩
	08-07	5 (2.6 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.6 ± 0.1) ⟨0.0 ± 0.0⟩	5 (2.7 ± 0.2) ⟨0.0 ± 0.0⟩	5 (2.9 ± 0.4) ⟨0.0 ± 0.0⟩
	08-08	5 (2.6 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.6 ± 0.1) ⟨0.0 ± 0.0⟩	5 (2.7 ± 0.1) ⟨0.0 ± 0.0⟩	5 (2.6 ± 0.1) ⟨0.0 ± 0.0⟩
	08-09	5 (187.5 ± 71.4) ⟨84.2 ± 0.3⟩	5 (347.2 ± 136.2) ⟨296.1 ± 6.8⟩	5 (31.8 ± 3.3) ⟨92.3 ± 0.6⟩	5 (30.1 ± 6.4) ⟨56.8 ± 0.9⟩
	08-10	5 (466.6 ± 158.7) ⟨306.4 ± 0.4⟩	3 (1087.1 ± 137.2) ⟨4660.8 ± 884.1⟩	5 (14.3 ± 1.2) ⟨377.6 ± 64.7⟩	5 (14.1 ± 0.9) ⟨285.6 ± 0.0⟩
	08-11	5 (782.1 ± 117.7) ⟨88.8 ± 0.6⟩		5 (196.4 ± 5.4) ⟨165.8 ± 1.9⟩	5 (184.2 ± 20.2) ⟨144.0 ± 0.9⟩
	08-12			5 (235.2 ± 45.1) ⟨44655.9 ± 796.9⟩	5 (225.5 ± 39.3) ⟨41169.3 ± 444.2⟩
S&R	08-01	5 (0.2 ± 0.0) ⟨213.8 ± 2.7⟩	5 (0.5 ± 0.0) ⟨612.8 ± 8.3⟩	5 (0.3 ± 0.0) ⟨438.5 ± 3.9⟩	5 (0.3 ± 0.0) ⟨404.5 ± 1.8⟩
	08-02	5 (0.9 ± 0.0) ⟨982.8 ± 19.2⟩	5 (2.0 ± 0.0) ⟨2523.2 ± 28.4⟩	5 (1.2 ± 0.0) ⟨1914.1 ± 20.3⟩	5 (1.2 ± 0.0) ⟨1787.5 ± 9.1⟩
	08-03	5 (3.8 ± 0.1) ⟨4405.4 ± 110.7⟩	5 (7.8 ± 0.4) ⟨9922.3 ± 23.5⟩	5 (4.9 ± 0.5) ⟨7805.2 ± 67.8⟩	5 (5.0 ± 0.3) ⟨7423.1 ± 15.0⟩
	08-04	5 (67.1 ± 1.4) ⟨77155.4 ± 659.7⟩	5 (127.8 ± 2.8) ⟨143528.2 ± 785.1⟩	5 (82.9 ± 2.1) ⟨119164.4 ± 570.0⟩	5 (89.2 ± 3.5) ⟨114677.4 ± 113.5⟩
	08-05	5 (1306.0 ± 28.1) ⟨1336086.9 ± 5907.0⟩		5 (1524.5 ± 40.2) ⟨1646775.3 ± 10809.1⟩	

results for **ROC** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
Sched	04-01	5 (0.1 ± 0.0) ⟨5.3 ± 0.2⟩	5 (0.1 ± 0.0) ⟨8.1 ± 1.0⟩	5 (0.1 ± 0.0) ⟨1.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.6 ± 0.0⟩
	04-02	5 (0.1 ± 0.0) ⟨7.9 ± 0.1⟩	5 (0.1 ± 0.0) ⟨13.0 ± 1.7⟩	5 (0.1 ± 0.0) ⟨1.2 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.9 ± 0.0⟩
	04-03	5 (0.1 ± 0.0) ⟨21.2 ± 0.5⟩	5 (0.1 ± 0.0) ⟨37.8 ± 3.7⟩	5 (0.1 ± 0.0) ⟨2.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.6 ± 0.0⟩
	04-04	5 (31.1 ± 0.4) ⟨7759.4 ± 35.2⟩	5 (57.3 ± 1.8) ⟨23782.5 ± 1098.8⟩	5 (30.1 ± 0.7) ⟨24610.7 ± 826.6⟩	5 (25.5 ± 0.3) ⟨17655.9 ± 235.9⟩
	04-05	5 (1578.5 ± 29.1) ⟨326777.6 ± 1071.9⟩		5 (1455.9 ± 61.3) ⟨639484.0 ± 1212.4⟩	5 (1086.1 ± 17.5) ⟨431646.4 ± 320.3⟩
Sys	08-01	5 (0.1 ± 0.0) ⟨8.2 ± 0.2⟩	5 (0.1 ± 0.0) ⟨12.6 ± 1.2⟩	5 (0.1 ± 0.0) ⟨1.4 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.8 ± 0.0⟩
	08-02	5 (0.1 ± 0.0) ⟨12.4 ± 0.4⟩	5 (0.1 ± 0.0) ⟨20.2 ± 1.9⟩	5 (0.1 ± 0.0) ⟨1.8 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.0 ± 0.0⟩
	08-03	5 (0.1 ± 0.0) ⟨30.1 ± 0.7⟩	5 (0.2 ± 0.0) ⟨54.3 ± 2.3⟩	5 (0.1 ± 0.0) ⟨3.7 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.8 ± 0.0⟩
	08-04	5 (1247.5 ± 27.5) ⟨218057.1 ± 1801.2⟩		5 (1183.2 ± 59.0) ⟨557641.9 ± 14828.3⟩	5 (924.1 ± 49.8) ⟨389615.4 ± 2072.1⟩
	08-05				5 (1072.6 ± 44.1) ⟨392032.1 ± 744.0⟩
ΔTW	08-01	5 (0.2 ± 0.0) ⟨105.4 ± 6.6⟩	5 (0.2 ± 0.0) ⟨56.4 ± 0.6⟩	5 (0.1 ± 0.0) ⟨18.7 ± 0.3⟩	5 (0.1 ± 0.0) ⟨19.4 ± 0.5⟩
	08-02	5 (1.2 ± 0.0) ⟨751.4 ± 25.7⟩	5 (0.8 ± 0.1) ⟨326.8 ± 48.2⟩	5 (0.4 ± 0.0) ⟨111.9 ± 1.3⟩	5 (0.4 ± 0.0) ⟨117.6 ± 4.0⟩
	08-03	5 (5.1 ± 0.2) ⟨3063.0 ± 38.2⟩	5 (3.4 ± 0.5) ⟨1399.7 ± 256.7⟩	5 (1.4 ± 0.1) ⟨390.5 ± 3.2⟩	5 (1.4 ± 0.1) ⟨418.2 ± 8.8⟩
	08-04	5 (105.8 ± 11.2) ⟨56801.2 ± 1415.8⟩	5 (59.8 ± 19.3) ⟨19326.6 ± 3939.1⟩	5 (16.9 ± 3.6) ⟨4967.5 ± 29.8⟩	5 (15.9 ± 3.8) ⟨5121.2 ± 54.4⟩
	08-05				5 (15.2 ± 2.2) ⟨5164.4 ± 14.9⟩
6, 8	08-01	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.5 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩
	08-02	5 (0.1 ± 0.0) ⟨7.2 ± 0.8⟩	5 (0.1 ± 0.0) ⟨6.5 ± 0.1⟩	5 (0.1 ± 0.0) ⟨3.5 ± 0.1⟩	5 (0.1 ± 0.0) ⟨2.8 ± 0.1⟩
	08-03	5 (1.5 ± 0.0) ⟨118.7 ± 2.1⟩	5 (1.1 ± 0.0) ⟨112.3 ± 2.1⟩	5 (0.8 ± 0.0) ⟨57.3 ± 0.7⟩	5 (0.8 ± 0.0) ⟨67.0 ± 0.9⟩
	08-04	5 (23.1 ± 0.4) ⟨2098.6 ± 23.9⟩	5 (23.3 ± 1.2) ⟨2945.6 ± 68.5⟩	5 (12.1 ± 0.3) ⟨1386.0 ± 33.8⟩	5 (12.2 ± 0.5) ⟨1715.7 ± 131.9⟩
	08-05	5 (479.1 ± 13.8) ⟨50548.2 ± 898.4⟩	5 (697.1 ± 10.7) ⟨75017.7 ± 231.1⟩	5 (277.6 ± 7.4) ⟨35772.9 ± 253.7⟩	5 (294.3 ± 5.2) ⟨37932.0 ± 731.8⟩
6, 9	6, 8	5 (719.1 ± 22.5) ⟨58287.2 ± 517.8⟩	5 (790.1 ± 23.2) ⟨64206.3 ± 3625.7⟩	5 (393.5 ± 12.2) ⟨32699.6 ± 2063.4⟩	5 (399.5 ± 10.3) ⟨32067.6 ± 2114.9⟩
	6, 9	5 (1684.2 ± 64.3) ⟨140848.4 ± 1672.0⟩		5 (935.6 ± 29.8) ⟨83453.7 ± 3472.4⟩	5 (950.6 ± 32.2) ⟨84234.9 ± 2912.5⟩

results for **ROC** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
Zeno	04-01	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	04-02		5 (384.5 ± 3.8) ⟨58904.1 ± 34.4⟩	5 (149.9 ± 5.2) ⟨37832.3 ± 65.6⟩	5 (135.1 ± 5.1) ⟨38273.4 ± 59.2⟩
	04-03		4 (1557.4 ± 18.8) ⟨528699.4 ± 407.7⟩	5 (1416.0 ± 55.1) ⟨491318.1 ± 445.1⟩	5 (138.6 ± 5.1) ⟨41686.6 ± 127.7⟩
	04-04		5 (314.6 ± 11.3) ⟨57121.6 ± 84.1⟩	5 (158.6 ± 5.9) ⟨37429.5 ± 52.0⟩	5 (1493.9 ± 57.3) ⟨517970.7 ± 769.8⟩
	04-05		5 (1664.2 ± 31.9) ⟨241579.7 ± 217.6⟩	5 (625.3 ± 23.3) ⟨148524.6 ± 303.6⟩	5 (154.5 ± 3.4) ⟨36948.9 ± 219.7⟩
	08-01	5 (65.1 ± 2.2) ⟨7366.3 ± 239.1⟩	5 (11.3 ± 0.3) ⟨2662.6 ± 4.8⟩	5 (6.3 ± 0.1) ⟨1648.0 ± 2.6⟩	5 (523.5 ± 14.5) ⟨147085.2 ± 230.8⟩
	08-02	5 (49.9 ± 1.4) ⟨3715.0 ± 74.0⟩	5 (6.9 ± 0.1) ⟨1379.8 ± 2.5⟩	5 (5.8 ± 0.2) ⟨1664.9 ± 4.7⟩	5 (5.8 ± 0.1) ⟨1768.0 ± 21.7⟩
	08-04	5 (857.7 ± 12.8) ⟨107645.8 ± 590.1⟩	5 (4.2 ± 0.1) ⟨863.5 ± 2.0⟩	5 (3.8 ± 0.1) ⟨881.4 ± 9.1⟩	5 (3.8 ± 0.1) ⟨938.8 ± 8.6⟩
		5 (136.5 ± 1.8) ⟨27849.2 ± 40.5⟩	5 (69.5 ± 1.5) ⟨17357.0 ± 37.1⟩	5 (62.2 ± 1.2) ⟨17367.4 ± 42.4⟩	5 (64.5 ± 1.3) ⟨18443.1 ± 123.3⟩

Table 4: Results for algorithms using **ROC**. We give X (Y) (Z) where X is the coverage, Y is the mean time (secs), and Z is the mean number of *thousands* of Q -values. Each mean is taken over the converged instances, and we report the 95% C.I.

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
04-01	5 (0.5 ± 0.0) (7.4 ± 0.6)	5 (0.5 ± 0.0) (7.4 ± 1.1)	5 (0.6 ± 0.1) (4.4 ± 0.5)	5 (0.5 ± 0.0) (4.4 ± 0.2)	5 (0.5 ± 0.0) (5.7 ± 1.1)
04-02	5 (0.5 ± 0.0) (6.8 ± 1.4)	5 (0.5 ± 0.0) (4.6 ± 0.2)	5 (0.5 ± 0.0) (2.9 ± 0.1)	5 (0.5 ± 0.0) (2.8 ± 0.1)	5 (0.5 ± 0.0) (4.7 ± 0.7)
04-03	5 (0.5 ± 0.0) (2.4 ± 1.2)	5 (0.5 ± 0.0) (1.3 ± 0.3)	5 (0.5 ± 0.0) (0.7 ± 0.2)	5 (0.5 ± 0.0) (0.4 ± 0.1)	5 (0.5 ± 0.0) (1.4 ± 0.5)
04-04	5 (0.5 ± 0.0) (11.6 ± 1.3)	5 (0.5 ± 0.0) (13.9 ± 0.3)	5 (0.5 ± 0.0) (8.2 ± 0.2)	5 (0.5 ± 0.0) (7.0 ± 0.3)	5 (0.5 ± 0.0) (10.9 ± 1.7)
BW 04-05	5 (0.5 ± 0.0) (4.8 ± 0.7)	5 (0.5 ± 0.0) (3.1 ± 0.3)	5 (0.5 ± 0.0) (1.8 ± 0.2)	5 (0.5 ± 0.0) (1.6 ± 0.2)	5 (0.5 ± 0.0) (3.1 ± 0.7)
04-06	5 (454.6 ± 39.1) (12137.3 ± 1168.2)	5 (31.7 ± 1.6) (1356.8 ± 73.7)	5 (22.2 ± 0.4) (864.7 ± 44.4)	5 (18.7 ± 0.3) (830.2 ± 14.0)	5 (19.3 ± 0.8) (1053.2 ± 14.6)
04-07				5 (1223.6 ± 53.9) (84550.7 ± 751.0)	5 (1303.4 ± 79.0) (111665.8 ± 618.7)
04-08	5 (93.2 ± 11.2) (2015.4 ± 168.1)	5 (17.3 ± 0.9) (706.9 ± 18.0)	5 (10.6 ± 0.3) (456.1 ± 16.9)	5 (9.3 ± 0.2) (257.8 ± 6.0)	5 (9.1 ± 0.2) (254.0 ± 6.9)
04-09	5 (104.4 ± 17.4) (2249.4 ± 341.8)	5 (13.1 ± 0.6) (406.3 ± 18.0)	5 (9.2 ± 0.3) (282.3 ± 19.5)	5 (8.5 ± 0.1) (171.2 ± 3.9)	5 (8.4 ± 0.1) (190.6 ± 5.2)
04-10	5 (1334.6 ± 122.6) (36469.3 ± 3151.1)	5 (77.9 ± 1.7) (3904.8 ± 44.2)	5 (47.2 ± 1.3) (2313.5 ± 21.7)	5 (39.6 ± 1.3) (2237.3 ± 11.6)	5 (40.6 ± 1.4) (2902.2 ± 49.3)
08-01	5 (0.2 ± 0.0) (4.9 ± 0.4)	5 (0.2 ± 0.0) (5.8 ± 0.3)	5 (0.2 ± 0.0) (3.7 ± 0.1)	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)
08-02	5 (0.2 ± 0.0) (5.0 ± 0.5)	5 (0.2 ± 0.0) (5.8 ± 0.3)	5 (0.2 ± 0.0) (3.7 ± 0.1)	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)
08-03	5 (0.2 ± 0.0) (4.7 ± 0.5)	5 (0.2 ± 0.0) (5.8 ± 0.3)	5 (0.2 ± 0.0) (3.7 ± 0.1)	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)
08-04	5 (0.2 ± 0.0) (4.7 ± 0.6)	5 (0.2 ± 0.0) (5.8 ± 0.3)	5 (0.2 ± 0.0) (3.7 ± 0.1)	5 (0.2 ± 0.0) (3.2 ± 0.3)	5 (0.2 ± 0.0) (3.2 ± 0.6)
08-05	5 (1623.9 ± 66.6) (40338.9 ± 1030.2)	5 (269.1 ± 14.9) (15072.4 ± 752.4)	5 (112.0 ± 10.0) (8338.7 ± 314.0)	5 (91.9 ± 1.9) (7358.1 ± 9.8)	5 (92.2 ± 2.4) (8378.3 ± 110.0)
08-06	5 (1620.2 ± 52.7) (40341.9 ± 1029.8)	5 (293.2 ± 58.4) (15072.4 ± 752.4)	5 (108.3 ± 2.1) (8338.7 ± 314.0)	5 (93.6 ± 2.0) (7358.1 ± 9.8)	5 (94.2 ± 5.0) (8378.3 ± 110.0)
08-07	5 (1618.5 ± 70.3) (40338.9 ± 1030.2)	5 (266.9 ± 14.7) (15072.4 ± 752.4)	5 (105.9 ± 1.0) (8338.7 ± 314.0)	5 (91.3 ± 0.9) (7358.1 ± 9.8)	5 (92.0 ± 2.4) (8378.3 ± 110.0)
08-08	4 (1622.3 ± 73.8) (40464.6 ± 1303.8)	5 (263.4 ± 9.0) (15072.4 ± 752.4)	5 (106.9 ± 2.5) (8338.7 ± 314.0)	5 (91.9 ± 1.2) (7358.1 ± 9.8)	5 (93.9 ± 3.2) (8378.3 ± 110.0)

results for **PDB** continued on next page

		LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
Coresec	01	5 (0.1 ± 0.0) ⟨30.4 ± 2.5⟩	5 (0.1 ± 0.0) ⟨17.2 ± 0.3⟩	5 (0.1 ± 0.0) ⟨10.3 ± 0.2⟩	5 (0.1 ± 0.0) ⟨7.2 ± 0.2⟩	5 (0.1 ± 0.0) ⟨7.1 ± 0.1⟩
	02	5 (0.1 ± 0.0) ⟨31.7 ± 1.4⟩	5 (0.1 ± 0.0) ⟨18.5 ± 0.1⟩	5 (0.1 ± 0.0) ⟨10.8 ± 0.2⟩	5 (0.1 ± 0.0) ⟨7.2 ± 0.3⟩	5 (0.1 ± 0.0) ⟨7.3 ± 0.2⟩
	03	5 (1.0 ± 0.0) ⟨1122.8 ± 40.8⟩	5 (0.5 ± 0.0) ⟨460.0 ± 3.7⟩	5 (0.4 ± 0.0) ⟨264.6 ± 2.3⟩	5 (0.3 ± 0.0) ⟨175.3 ± 1.2⟩	5 (0.3 ± 0.0) ⟨172.9 ± 2.0⟩
	04	5 (1.1 ± 0.0) ⟨1161.4 ± 35.7⟩	5 (0.6 ± 0.0) ⟨491.3 ± 5.3⟩	5 (0.4 ± 0.0) ⟨277.4 ± 3.9⟩	5 (0.4 ± 0.0) ⟨180.7 ± 1.1⟩	5 (0.4 ± 0.0) ⟨179.3 ± 1.1⟩
	05	5 (3.8 ± 0.1) ⟨4271.7 ± 164.4⟩	5 (1.8 ± 0.1) ⟨1687.4 ± 18.4⟩	5 (1.5 ± 0.1) ⟨993.1 ± 7.8⟩	5 (1.1 ± 0.1) ⟨586.7 ± 4.7⟩	5 (1.1 ± 0.0) ⟨571.2 ± 1.5⟩
	06	5 (130.5 ± 2.3) ⟨150177.3 ± 1704.3⟩	5 (29.1 ± 1.3) ⟨23016.8 ± 154.1⟩	5 (26.5 ± 1.6) ⟨15193.0 ± 63.6⟩	5 (22.0 ± 1.3) ⟨9916.2 ± 26.3⟩	5 (20.2 ± 0.5) ⟨9781.3 ± 13.6⟩
	07	5 (1182.1 ± 53.4) ⟨378289.3 ± 6943.3⟩				
Elev	04-01	5 (0.1 ± 0.0) ⟨17.5 ± 1.1⟩	5 (0.1 ± 0.0) ⟨35.3 ± 4.4⟩	5 (0.1 ± 0.0) ⟨22.3 ± 2.4⟩	5 (0.1 ± 0.0) ⟨17.2 ± 1.8⟩	5 (0.1 ± 0.0) ⟨19.9 ± 2.5⟩
	04-02	5 (0.1 ± 0.0) ⟨1.4 ± 0.2⟩	5 (0.1 ± 0.0) ⟨1.5 ± 0.1⟩	5 (0.1 ± 0.0) ⟨1.1 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.9 ± 0.0⟩	5 (0.1 ± 0.0) ⟨1.2 ± 0.1⟩
	04-03	5 (0.1 ± 0.0) ⟨18.0 ± 0.7⟩	5 (0.1 ± 0.0) ⟨36.2 ± 4.3⟩	5 (0.1 ± 0.0) ⟨23.9 ± 2.9⟩	5 (0.1 ± 0.0) ⟨18.0 ± 2.4⟩	5 (0.1 ± 0.0) ⟨20.8 ± 2.4⟩
	04-04	5 (0.1 ± 0.0) ⟨10.0 ± 0.8⟩	5 (0.1 ± 0.0) ⟨20.3 ± 0.9⟩	5 (0.1 ± 0.0) ⟨14.2 ± 0.7⟩	5 (0.1 ± 0.0) ⟨11.1 ± 0.1⟩	5 (0.1 ± 0.0) ⟨12.4 ± 0.6⟩
	04-05	5 (0.1 ± 0.0) ⟨9.7 ± 1.0⟩	5 (0.1 ± 0.0) ⟨12.8 ± 2.1⟩	5 (0.1 ± 0.0) ⟨8.3 ± 1.3⟩	5 (0.1 ± 0.0) ⟨5.4 ± 0.1⟩	5 (0.1 ± 0.0) ⟨6.3 ± 0.2⟩
	04-06	5 (0.6 ± 0.0) ⟨260.7 ± 14.2⟩	5 (0.6 ± 0.0) ⟨349.2 ± 35.4⟩	5 (0.3 ± 0.0) ⟨230.2 ± 31.9⟩	5 (0.2 ± 0.0) ⟨139.9 ± 15.4⟩	5 (0.3 ± 0.0) ⟨150.6 ± 12.8⟩
	04-07	5 (0.9 ± 0.0) ⟨397.8 ± 11.9⟩	5 (0.6 ± 0.1) ⟨339.8 ± 22.8⟩	5 (0.3 ± 0.0) ⟨214.8 ± 19.6⟩	5 (0.2 ± 0.0) ⟨144.5 ± 9.0⟩	5 (0.3 ± 0.0) ⟨160.2 ± 9.9⟩
	04-08	5 (2.1 ± 0.1) ⟨1059.3 ± 21.9⟩	5 (4.6 ± 0.2) ⟨2734.6 ± 104.9⟩	5 (1.7 ± 0.1) ⟨2121.0 ± 133.6⟩	5 (1.4 ± 0.1) ⟨1348.2 ± 18.5⟩	5 (1.5 ± 0.0) ⟨1429.7 ± 22.1⟩
	04-09	5 (1.6 ± 0.1) ⟨784.6 ± 17.3⟩	5 (2.7 ± 0.1) ⟨1676.2 ± 30.6⟩	5 (1.0 ± 0.0) ⟨1190.6 ± 84.9⟩	5 (0.8 ± 0.0) ⟨751.1 ± 9.1⟩	5 (0.9 ± 0.0) ⟨809.5 ± 15.2⟩
	04-10	5 (3.7 ± 0.1) ⟨2009.6 ± 26.9⟩	5 (17.8 ± 0.4) ⟨10080.8 ± 343.4⟩	5 (4.5 ± 0.4) ⟨7303.2 ± 759.7⟩	5 (3.2 ± 0.2) ⟨3875.3 ± 90.3⟩	5 (3.2 ± 0.1) ⟨3979.3 ± 112.9⟩
	04-11	5 (351.6 ± 3.7) ⟨107973.8 ± 894.9⟩	5 (816.3 ± 19.7) ⟨313730.5 ± 10015.9⟩	5 (384.2 ± 32.6) ⟨325193.4 ± 45370.3⟩	5 (286.3 ± 13.2) ⟨158909.8 ± 1767.3⟩	5 (297.1 ± 13.6) ⟨161907.7 ± 1430.2⟩
	04-12	5 (186.0 ± 5.0) ⟨37656.5 ± 468.4⟩	5 (113.5 ± 5.0) ⟨45082.1 ± 2069.5⟩	5 (66.4 ± 2.3) ⟨41161.8 ± 2615.4⟩	5 (50.0 ± 1.3) ⟨21660.7 ± 267.2⟩	5 (51.9 ± 2.0) ⟨22498.8 ± 194.4⟩
	04-13	5 (1104.8 ± 26.6) ⟨334594.5 ± 927.5⟩				5 (1406.7 ± 76.8) ⟨1077845.1 ± 6034.4⟩
	04-14	5 (233.0 ± 2.6) ⟨40572.5 ± 755.7⟩	5 (311.2 ± 6.4) ⟨107585.0 ± 2501.1⟩	5 (152.1 ± 4.3) ⟨80439.4 ± 4791.4⟩	5 (137.0 ± 2.9) ⟨44967.1 ± 1003.5⟩	5 (139.9 ± 2.3) ⟨47315.1 ± 1120.7⟩
	04-15	5 (418.7 ± 7.3) ⟨105228.4 ± 470.2⟩	5 (1390.5 ± 64.5) ⟨428631.9 ± 15262.2⟩	5 (553.9 ± 85.2) ⟨366651.5 ± 32602.1⟩	5 (457.3 ± 82.4) ⟨198692.8 ± 837.6⟩	5 (408.2 ± 20.5) ⟨201766.7 ± 926.7⟩

results for **PDB** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
ExBW	04-01	5 (0.1 ± 0.0) (0.5 ± 0.2)	5 (0.1 ± 0.0) (0.4 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)
	04-02	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.1 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)
	04-03	5 (0.1 ± 0.0) (1.0 ± 0.3)	5 (0.1 ± 0.0) (0.6 ± 0.0)	5 (0.1 ± 0.0) (0.3 ± 0.0)	5 (0.1 ± 0.0) (0.2 ± 0.0)
	04-04	5 (0.3 ± 0.0) (100.9 ± 5.1)	5 (1.0 ± 0.0) (517.7 ± 7.2)	5 (0.2 ± 0.0) (242.0 ± 1.8)	5 (0.2 ± 0.0) (144.2 ± 0.1)
	04-05	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)	5 (0.1 ± 0.0) (0.0 ± 0.0)
	04-06	5 (1.8 ± 0.1) (186.4 ± 4.2)	5 (0.3 ± 0.0) (26.0 ± 0.8)	5 (0.2 ± 0.0) (13.6 ± 0.5)	5 (0.3 ± 0.0) (67.5 ± 0.1)
	04-07	5 (5.3 ± 0.2) (997.4 ± 22.0)	5 (4.8 ± 0.1) (1373.4 ± 32.0)	5 (1.5 ± 0.1) (802.6 ± 8.2)	5 (1.1 ± 0.0) (380.5 ± 6.3)
	04-08	5 (2.9 ± 0.4) (351.9 ± 55.7)	5 (0.4 ± 0.0) (39.0 ± 0.8)	5 (0.3 ± 0.0) (23.8 ± 0.6)	5 (0.3 ± 0.0) (15.0 ± 0.2)
	04-09	5 (37.8 ± 0.6) (6326.9 ± 86.6)	5 (63.5 ± 3.4) (18886.7 ± 745.4)	5 (14.1 ± 0.5) (10961.6 ± 487.5)	5 (8.6 ± 0.3) (4222.0 ± 85.9)
	08-01	5 (0.2 ± 0.0) (83.7 ± 3.1)	5 (2.7 ± 0.1) (1622.7 ± 47.8)	5 (0.6 ± 0.0) (807.8 ± 22.9)	5 (0.5 ± 0.0) (389.3 ± 3.9)
	08-02	5 (0.4 ± 0.0) (155.4 ± 3.2)	5 (2.9 ± 0.1) (1659.2 ± 15.6)	5 (0.6 ± 0.0) (777.1 ± 7.3)	5 (0.5 ± 0.0) (498.7 ± 4.0)
	08-03	5 (1.4 ± 0.0) (577.1 ± 17.7)	5 (6.0 ± 0.1) (2971.7 ± 36.2)	5 (1.1 ± 0.0) (1469.2 ± 22.2)	5 (0.8 ± 0.0) (839.9 ± 7.4)
	08-04	5 (27.9 ± 0.7) (9866.0 ± 169.9)	5 (130.9 ± 2.7) (40373.3 ± 308.2)	5 (14.8 ± 0.7) (18816.9 ± 41.6)	5 (12.3 ± 0.5) (12446.9 ± 25.2)
	08-05	5 (0.2 ± 0.0) (13.4 ± 2.5)	5 (0.1 ± 0.0) (1.7 ± 0.2)	5 (0.1 ± 0.0) (1.1 ± 0.1)	5 (0.1 ± 0.0) (0.7 ± 0.1)
	08-06	5 (63.9 ± 2.8) (12787.6 ± 240.9)	5 (525.7 ± 22.6) (100241.7 ± 380.1)	5 (32.7 ± 1.0) (47762.4 ± 607.4)	5 (20.7 ± 0.3) (20945.6 ± 33.6)
	08-07	5 (0.4 ± 0.0) (28.5 ± 6.0)	5 (0.2 ± 0.0) (12.9 ± 1.4)	5 (0.2 ± 0.0) (6.9 ± 0.8)	5 (0.2 ± 0.0) (3.4 ± 0.5)
Rand	08-08	5 (393.1 ± 8.0) (62326.3 ± 535.0)		5 (700.1 ± 123.1) (374674.8 ± 1107.5)	5 (381.5 ± 67.2) (179805.0 ± 299.1)
	08-09	5 (131.2 ± 8.5) (16361.7 ± 450.2)	5 (113.1 ± 1.3) (16184.5 ± 67.4)	5 (30.0 ± 1.2) (10293.1 ± 50.5)	5 (20.7 ± 0.3) (6031.9 ± 75.1)
	04-03	5 (15.6 ± 4.8) (6827.1 ± 2231.4)	5 (3.3 ± 0.3) (1646.7 ± 64.9)	5 (2.0 ± 0.1) (1053.3 ± 37.3)	5 (1.6 ± 0.0) (793.1 ± 21.6)
	04-04	5 (0.3 ± 0.0) (6.5 ± 4.0)	5 (221.4 ± 3.8) (338688.2 ± 2291.7)	5 (123.5 ± 4.8) (245909.8 ± 0.0)	5 (14.3 ± 1.3) (26692.6 ± 0.0)
	04-05	5 (1.1 ± 0.1) (2.2 ± 0.9)	5 (194.4 ± 41.1) (137249.8 ± 25658.6)	5 (41.6 ± 34.6) (50135.5 ± 40116.4)	5 (5.3 ± 3.6) (1698.0 ± 1358.1)
	04-07	5 (0.3 ± 0.0) (1.8 ± 1.7)	5 (0.3 ± 0.0) (1.4 ± 0.1)	5 (0.3 ± 0.0) (0.6 ± 0.1)	5 (0.3 ± 0.0) (0.3 ± 0.1)
	04-09	5 (1.9 ± 0.6) (297.4 ± 186.2)	5 (1.0 ± 0.0) (42.7 ± 9.0)	5 (1.1 ± 0.0) (26.3 ± 6.7)	5 (1.0 ± 0.0) (20.4 ± 5.7)
	04-13	5 (74.4 ± 63.7) (12814.9 ± 10307.5)	5 (6.4 ± 0.7) (1175.2 ± 197.2)	5 (6.6 ± 0.8) (906.5 ± 146.8)	5 (4.7 ± 0.5) (526.0 ± 61.3)

results for **PDB** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tid}	CG-iLAO* _{single}	
□TW	08-01	5 (0.3 ± 0.0) ⟨0.9 ± 0.1⟩	5 (0.3 ± 0.0) ⟨2.7 ± 0.6⟩	5 (0.3 ± 0.0) ⟨1.8 ± 0.0⟩	5 (0.3 ± 0.0) ⟨1.2 ± 0.1⟩	5 (0.3 ± 0.0) ⟨1.3 ± 0.1⟩
	08-02	5 (0.4 ± 0.0) ⟨0.5 ± 0.1⟩	5 (0.4 ± 0.0) ⟨1.7 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.4 ± 0.0) ⟨0.2 ± 0.0⟩
	08-03	5 (0.9 ± 0.0) ⟨2.0 ± 0.1⟩	5 (0.9 ± 0.0) ⟨4.4 ± 0.5⟩	5 (0.9 ± 0.0) ⟨2.5 ± 0.0⟩	5 (0.9 ± 0.0) ⟨2.3 ± 0.1⟩	5 (0.9 ± 0.0) ⟨2.4 ± 0.2⟩
	08-04	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩	5 (1.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.9 ± 0.0) ⟨0.0 ± 0.0⟩
	08-05	5 (2.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (2.1 ± 0.0) ⟨0.0 ± 0.0⟩
	08-06	5 (6.1 ± 0.1) ⟨8.9 ± 0.1⟩	5 (9.4 ± 0.5) ⟨25.4 ± 2.2⟩	5 (5.0 ± 0.6) ⟨12.8 ± 1.3⟩	5 (4.6 ± 0.1) ⟨11.7 ± 0.1⟩	5 (4.8 ± 0.3) ⟨11.8 ± 0.3⟩
	08-07	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.4 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.4 ± 0.0) ⟨0.0 ± 0.0⟩	5 (4.4 ± 0.1) ⟨0.0 ± 0.0⟩
	08-08	5 (4.6 ± 0.2) ⟨0.0 ± 0.0⟩	5 (4.7 ± 0.5) ⟨0.0 ± 0.0⟩	5 (4.4 ± 0.0) ⟨0.0 ± 0.0⟩	5 (4.5 ± 0.1) ⟨0.0 ± 0.0⟩	5 (4.6 ± 0.1) ⟨0.0 ± 0.0⟩
	08-09	5 (153.5 ± 40.4) ⟨75.8 ± 0.4⟩	5 (279.3 ± 68.1) ⟨205.1 ± 8.5⟩	5 (27.3 ± 9.3) ⟨65.9 ± 0.5⟩	5 (20.1 ± 1.8) ⟨42.5 ± 7.0⟩	5 (23.0 ± 2.6) ⟨47.1 ± 5.6⟩
	08-10	5 (388.6 ± 150.7) ⟨305.8 ± 0.1⟩	2 (1328.4 ± 520.9) ⟨3619.1 ± 1228.2⟩	5 (14.8 ± 0.3) ⟨377.2 ± 64.7⟩	5 (14.8 ± 0.8) ⟨314.3 ± 33.6⟩	5 (16.1 ± 1.0) ⟨324.1 ± 43.3⟩
	08-11	5 (488.7 ± 40.3) ⟨67.1 ± 0.4⟩	5 (1421.8 ± 157.2) ⟨206.0 ± 2.2⟩	5 (122.2 ± 11.3) ⟨98.5 ± 0.6⟩	5 (98.5 ± 4.0) ⟨89.6 ± 1.3⟩	5 (100.1 ± 3.6) ⟨92.1 ± 1.1⟩
	08-12			5 (136.7 ± 3.8) ⟨44526.2 ± 750.1⟩	5 (128.4 ± 2.1) ⟨42652.0 ± 861.6⟩	5 (126.8 ± 4.0) ⟨42628.0 ± 997.4⟩
	08-14	5 (288.7 ± 14.9) ⟨0.0 ± 0.0⟩	5 (291.6 ± 14.8) ⟨0.0 ± 0.0⟩			
S&R	08-01	5 (0.2 ± 0.0) ⟨220.7 ± 2.7⟩	5 (0.5 ± 0.0) ⟨686.2 ± 11.9⟩	5 (0.3 ± 0.0) ⟨495.2 ± 9.8⟩	5 (0.3 ± 0.0) ⟨446.9 ± 3.0⟩	5 (0.3 ± 0.0) ⟨449.0 ± 1.5⟩
	08-02	5 (0.6 ± 0.0) ⟨996.7 ± 16.0⟩	5 (1.8 ± 0.0) ⟨2627.6 ± 18.5⟩	5 (1.0 ± 0.0) ⟨2005.1 ± 109.1⟩	5 (1.0 ± 0.0) ⟨1841.4 ± 7.7⟩	5 (1.0 ± 0.0) ⟨1847.4 ± 4.6⟩
	08-03	5 (2.7 ± 0.0) ⟨4400.8 ± 44.8⟩	5 (6.8 ± 0.2) ⟨10029.1 ± 21.4⟩	5 (3.9 ± 0.3) ⟨7900.3 ± 207.4⟩	5 (4.0 ± 0.3) ⟨7452.9 ± 14.3⟩	5 (4.2 ± 0.4) ⟨7494.3 ± 10.7⟩
	08-04	5 (52.9 ± 0.9) ⟨77293.1 ± 479.7⟩	5 (112.8 ± 3.9) ⟨142243.7 ± 716.9⟩	5 (66.2 ± 3.7) ⟨118443.8 ± 1612.3⟩	5 (75.0 ± 3.6) ⟨113676.4 ± 106.3⟩	5 (82.0 ± 5.4) ⟨114418.7 ± 52.8⟩
	08-05	5 (1150.8 ± 21.4) ⟨1350761.4 ± 5240.7⟩		5 (1382.6 ± 164.1) ⟨1614095.2 ± 36753.9⟩	4 (1635.2 ± 100.4) ⟨1587933.7 ± 1602.7⟩	

results for **PDB** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
Sched	04-01	5 (0.1 ± 0.0) (7.2 ± 0.2)	5 (0.1 ± 0.0) (10.5 ± 1.4)	5 (0.1 ± 0.0) (1.3 ± 0.0)	5 (0.1 ± 0.0) (0.8 ± 0.0)
	04-02	5 (0.1 ± 0.0) (11.0 ± 0.2)	5 (0.1 ± 0.0) (17.8 ± 2.4)	5 (0.1 ± 0.0) (2.0 ± 0.0)	5 (0.1 ± 0.0) (1.1 ± 0.0)
	04-03	5 (0.1 ± 0.0) (33.4 ± 0.5)	5 (0.2 ± 0.0) (56.4 ± 5.7)	5 (0.1 ± 0.0) (4.4 ± 0.0)	5 (0.1 ± 0.0) (3.5 ± 0.0)
	04-04	5 (413.7 ± 53.7) (162494.4 ± 14251.9)			
Sys	08-01	5 (0.1 ± 0.0) (11.0 ± 0.2)	5 (0.1 ± 0.0) (16.4 ± 1.6)	5 (0.1 ± 0.0) (1.6 ± 0.0)	5 (0.1 ± 0.0) (0.9 ± 0.0)
	08-02	5 (0.1 ± 0.0) (17.3 ± 0.2)	5 (0.1 ± 0.0) (27.7 ± 2.7)	5 (0.1 ± 0.0) (2.4 ± 0.0)	5 (0.1 ± 0.0) (1.3 ± 0.0)
	08-03	5 (0.2 ± 0.0) (48.2 ± 1.1)	5 (0.2 ± 0.0) (80.2 ± 3.3)	5 (0.1 ± 0.0) (5.1 ± 0.0)	5 (0.1 ± 0.0) (4.2 ± 0.0)
	08-01	5 (0.2 ± 0.0) (116.4 ± 6.2)	5 (0.2 ± 0.0) (82.2 ± 1.1)	5 (0.1 ± 0.0) (24.7 ± 0.2)	5 (0.1 ± 0.0) (24.7 ± 0.2)
	08-02	5 (1.1 ± 0.0) (797.9 ± 16.3)	5 (0.7 ± 0.1) (410.5 ± 68.1)	5 (0.2 ± 0.0) (127.1 ± 0.8)	5 (0.2 ± 0.0) (127.1 ± 0.8)
	08-03	5 (4.9 ± 0.1) (3297.5 ± 56.3)	5 (2.8 ± 0.2) (1439.7 ± 15.9)	5 (0.7 ± 0.0) (477.7 ± 3.4)	5 (0.7 ± 0.0) (477.7 ± 3.4)
	08-04	5 (127.7 ± 8.9) (60470.3 ± 637.0)	5 (74.0 ± 8.6) (21602.6 ± 3721.8)	5 (35.6 ± 5.2) (5621.9 ± 42.0)	5 (36.1 ± 6.2) (5621.9 ± 42.0)
	08-01	5 (0.0 ± 0.0) (0.2 ± 0.0)	5 (0.0 ± 0.0) (0.5 ± 0.0)	5 (0.0 ± 0.0) (0.3 ± 0.0)	5 (0.0 ± 0.0) (0.2 ± 0.0)
	08-02	5 (0.1 ± 0.0) (5.6 ± 0.6)	5 (0.1 ± 0.0) (6.0 ± 0.0)	5 (0.0 ± 0.0) (3.1 ± 0.0)	5 (0.0 ± 0.0) (2.6 ± 0.0)
	08-03	5 (0.3 ± 0.0) (96.5 ± 2.6)	5 (0.3 ± 0.0) (107.7 ± 2.0)	5 (0.1 ± 0.0) (54.3 ± 0.6)	5 (0.2 ± 0.0) (65.2 ± 5.2)
	08-04	5 (5.1 ± 0.1) (1800.8 ± 11.6)	5 (12.0 ± 0.4) (2994.2 ± 34.4)	5 (3.8 ± 1.7) (1411.7 ± 19.1)	5 (3.9 ± 1.1) (1582.9 ± 37.1)
	08-05	5 (195.4 ± 17.9) (40251.9 ± 262.2)	5 (594.8 ± 9.9) (83364.7 ± 472.9)	5 (141.4 ± 9.0) (38359.1 ± 462.7)	5 (165.1 ± 19.2) (37532.0 ± 1299.6)
ΔTW	6, 10		2 (1652.1 ± 74.1) (301273.3 ± 11969.7)	3 (1555.2 ± 76.3) (274945.1 ± 872.2)	2 (1602.8 ± 69.4) (278233.7 ± 9122.1)
	6, 8	5 (289.6 ± 17.7) (55267.5 ± 271.2)	5 (847.4 ± 15.3) (114898.7 ± 2209.9)	5 (201.5 ± 10.8) (53108.2 ± 1128.4)	5 (206.3 ± 14.2) (48453.0 ± 1606.4)
	6, 9	5 (873.6 ± 57.6) (125517.9 ± 888.1)		5 (582.9 ± 40.6) (129205.6 ± 2910.9)	5 (701.9 ± 260.2) (116958.3 ± 4077.7)
				5 (576.9 ± 29.4) (117320.6 ± 2268.4)	

results for **PDB** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
04-01	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
04-02	5 (557.5 ± 18.4) ⟨88820.0 ± 2565.8⟩	5 (147.8 ± 3.0) ⟨28269.2 ± 24.6⟩	5 (30.8 ± 1.1) ⟨18738.4 ± 34.9⟩	5 (26.8 ± 1.4) ⟨17300.6 ± 33.5⟩	5 (27.8 ± 1.2) ⟨18346.0 ± 73.5⟩
04-03			5 (526.0 ± 28.1) ⟨340854.8 ± 307.3⟩	5 (507.6 ± 44.4) ⟨310498.8 ± 268.3⟩	5 (527.6 ± 43.1) ⟨324647.6 ± 960.2⟩
04-04	5 (587.6 ± 19.1) ⟨76756.1 ± 1711.9⟩	5 (101.6 ± 1.4) ⟨25828.1 ± 43.7⟩	5 (33.2 ± 1.1) ⟨18547.1 ± 16.5⟩	5 (31.2 ± 0.9) ⟨19347.6 ± 47.6⟩	5 (30.7 ± 1.2) ⟨19111.9 ± 62.6⟩
04-05		5 (640.7 ± 10.9) ⟨111407.2 ± 59.7⟩	5 (123.5 ± 4.7) ⟨70481.4 ± 200.7⟩	5 (102.1 ± 4.1) ⟨61198.8 ± 159.4⟩	5 (108.6 ± 5.3) ⟨64775.0 ± 329.9⟩
08-01	5 (25.2 ± 1.0) ⟨6787.4 ± 158.9⟩	5 (6.4 ± 0.1) ⟨2208.0 ± 3.7⟩	5 (2.2 ± 0.1) ⟨1400.7 ± 2.6⟩	5 (1.9 ± 0.1) ⟨1398.0 ± 10.8⟩	5 (2.1 ± 0.1) ⟨1575.9 ± 29.9⟩
08-02	5 (41.4 ± 0.9) ⟨7910.0 ± 170.8⟩	5 (4.1 ± 0.0) ⟨1199.0 ± 4.3⟩	5 (1.6 ± 0.1) ⟨761.8 ± 3.8⟩	5 (1.4 ± 0.0) ⟨773.5 ± 3.3⟩	5 (1.5 ± 0.1) ⟨885.3 ± 10.0⟩
08-04	5 (221.5 ± 3.9) ⟨39821.5 ± 269.7⟩	5 (40.0 ± 0.8) ⟨10970.9 ± 24.6⟩	5 (11.5 ± 0.6) ⟨7770.8 ± 24.1⟩	5 (10.3 ± 0.5) ⟨7480.2 ± 19.3⟩	5 (10.7 ± 0.5) ⟨7678.4 ± 89.8⟩

Table 5: Results for algorithms using **PDB**. We give $X(Y)\langle Z \rangle$ where X is the coverage, Y is the mean time (secs), and Z is the mean number of *thousands* of Q -values. Each mean is taken over the converged instances, and we report the 95% C.I.

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
BW	04-01	5 (24.5 ± 1.8) ⟨303.4 ± 34.7⟩	5 (1.7 ± 0.2) ⟨45.9 ± 4.9⟩	5 (1.9 ± 0.2) ⟨30.6 ± 3.5⟩	5 (1.8 ± 0.2) ⟨49.1 ± 4.6⟩
	04-02	5 (17.3 ± 1.3) ⟨178.2 ± 17.3⟩	5 (1.1 ± 0.0) ⟨37.7 ± 2.1⟩	5 (1.2 ± 0.0) ⟨25.2 ± 0.6⟩	5 (1.2 ± 0.0) ⟨34.3 ± 0.8⟩
	04-03	5 (11.6 ± 0.8) ⟨104.4 ± 9.4⟩	5 (0.6 ± 0.0) ⟨11.6 ± 1.1⟩	5 (0.6 ± 0.0) ⟨7.4 ± 0.6⟩	5 (0.6 ± 0.0) ⟨11.9 ± 1.0⟩
	04-04	5 (27.7 ± 1.5) ⟨365.4 ± 36.4⟩	5 (2.5 ± 0.1) ⟨84.0 ± 6.9⟩	5 (2.8 ± 0.2) ⟨57.9 ± 6.8⟩	5 (2.8 ± 0.2) ⟨96.0 ± 12.4⟩
	04-05	5 (15.6 ± 1.2) ⟨155.5 ± 21.8⟩	5 (1.0 ± 0.1) ⟨35.9 ± 3.5⟩	5 (1.0 ± 0.1) ⟨22.0 ± 2.3⟩	5 (1.0 ± 0.1) ⟨31.6 ± 3.2⟩
	08-01	5 (0.9 ± 0.0) ⟨20.9 ± 0.8⟩	5 (0.6 ± 0.0) ⟨28.4 ± 3.7⟩	5 (0.6 ± 0.0) ⟨14.7 ± 1.3⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩
	08-02	5 (0.9 ± 0.0) ⟨20.6 ± 0.9⟩	5 (0.6 ± 0.0) ⟨28.4 ± 3.7⟩	5 (0.6 ± 0.0) ⟨14.7 ± 1.3⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩
	08-03	5 (0.9 ± 0.0) ⟨20.4 ± 1.2⟩	5 (0.6 ± 0.0) ⟨28.4 ± 3.7⟩	5 (0.6 ± 0.0) ⟨14.7 ± 1.3⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩
	08-04	5 (0.9 ± 0.0) ⟨20.8 ± 0.9⟩	5 (0.6 ± 0.0) ⟨28.4 ± 3.7⟩	5 (0.6 ± 0.1) ⟨14.7 ± 1.3⟩	5 (0.6 ± 0.0) ⟨22.6 ± 3.0⟩
Coresec	01	5 (0.8 ± 0.0) ⟨1094.4 ± 16.4⟩	5 (0.4 ± 0.0) ⟨450.4 ± 5.1⟩	5 (0.4 ± 0.0) ⟨273.2 ± 3.2⟩	5 (0.4 ± 0.0) ⟨254.4 ± 1.8⟩
	02	5 (3.1 ± 0.1) ⟨4102.2 ± 22.0⟩	5 (1.4 ± 0.1) ⟨1456.5 ± 14.0⟩	5 (1.4 ± 0.1) ⟨877.7 ± 5.7⟩	5 (1.4 ± 0.1) ⟨836.7 ± 4.1⟩
	03	5 (11.2 ± 0.1) ⟨14446.5 ± 50.1⟩	5 (5.2 ± 0.2) ⟨5355.6 ± 29.7⟩	5 (4.6 ± 0.1) ⟨2961.3 ± 16.3⟩	5 (4.4 ± 0.1) ⟨2507.5 ± 10.4⟩
	04	5 (39.0 ± 0.4) ⟨46929.6 ± 152.6⟩	5 (15.5 ± 0.3) ⟨14794.4 ± 82.1⟩	5 (13.8 ± 0.6) ⟨8363.1 ± 26.3⟩	5 (13.6 ± 0.8) ⟨7337.4 ± 8.6⟩
	05		5 (566.2 ± 17.3) ⟨310675.1 ± 713.8⟩	5 (474.6 ± 8.3) ⟨180804.8 ± 119.1⟩	5 (653.2 ± 40.1) ⟨165985.5 ± 56.0⟩
				5 (646.6 ± 47.9) ⟨160943.1 ± 101.6⟩	

results for **LMCut** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
04-01	5 (0.0 ± 0.0) ⟨3.0 ± 0.9⟩	5 (0.0 ± 0.0) ⟨4.0 ± 0.7⟩	5 (0.0 ± 0.0) ⟨2.6 ± 0.4⟩	5 (0.0 ± 0.0) ⟨0.9 ± 0.2⟩	5 (0.0 ± 0.0) ⟨1.4 ± 0.4⟩
04-02	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩
04-03	5 (0.1 ± 0.0) ⟨3.0 ± 0.3⟩	5 (0.0 ± 0.0) ⟨4.3 ± 0.4⟩	5 (0.0 ± 0.0) ⟨2.9 ± 0.3⟩	5 (0.0 ± 0.0) ⟨1.9 ± 0.1⟩	5 (0.0 ± 0.0) ⟨2.4 ± 0.3⟩
04-04	5 (0.0 ± 0.0) ⟨2.8 ± 0.2⟩	5 (0.0 ± 0.0) ⟨4.3 ± 0.3⟩	5 (0.0 ± 0.0) ⟨2.9 ± 0.2⟩	5 (0.0 ± 0.0) ⟨2.1 ± 0.1⟩	5 (0.0 ± 0.0) ⟨2.3 ± 0.1⟩
04-05	5 (0.0 ± 0.0) ⟨1.2 ± 0.3⟩	5 (0.0 ± 0.0) ⟨1.4 ± 0.2⟩	5 (0.0 ± 0.0) ⟨0.9 ± 0.1⟩	5 (0.0 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.6 ± 0.1⟩
04-06	5 (0.9 ± 0.0) ⟨18.7 ± 1.5⟩	5 (0.3 ± 0.0) ⟨33.3 ± 5.9⟩	5 (0.3 ± 0.0) ⟨20.1 ± 4.7⟩	5 (0.3 ± 0.0) ⟨16.9 ± 0.5⟩	5 (0.3 ± 0.0) ⟨17.8 ± 0.6⟩
04-07	5 (0.8 ± 0.1) ⟨11.7 ± 1.5⟩	5 (0.2 ± 0.0) ⟨13.6 ± 1.6⟩	5 (0.2 ± 0.0) ⟨8.3 ± 1.3⟩	5 (0.2 ± 0.0) ⟨5.2 ± 0.7⟩	5 (0.2 ± 0.0) ⟨5.5 ± 0.7⟩
04-08	5 (6.8 ± 0.1) ⟨488.2 ± 11.7⟩	5 (5.7 ± 0.1) ⟨1606.0 ± 71.7⟩	5 (4.0 ± 0.1) ⟨1053.2 ± 55.8⟩	5 (3.9 ± 0.1) ⟨587.5 ± 7.9⟩	5 (3.8 ± 0.1) ⟨590.9 ± 14.0⟩
04-09	5 (5.5 ± 0.2) ⟨210.9 ± 5.8⟩	5 (2.0 ± 0.1) ⟨346.0 ± 10.1⟩	5 (1.8 ± 0.0) ⟨218.1 ± 11.9⟩	5 (1.6 ± 0.1) ⟨142.1 ± 3.5⟩	5 (1.7 ± 0.1) ⟨146.0 ± 6.4⟩
04-10	5 (9.2 ± 0.1) ⟨1464.8 ± 25.7⟩	5 (19.8 ± 0.3) ⟨7881.7 ± 241.2⟩	5 (8.9 ± 0.5) ⟨5571.8 ± 595.0⟩	5 (7.8 ± 0.2) ⟨2700.6 ± 34.2⟩	5 (7.8 ± 0.2) ⟨2737.9 ± 16.6⟩
04-11	5 (943.9 ± 13.7) ⟨68808.4 ± 683.6⟩	5 (1276.5 ± 37.5) ⟨293792.6 ± 16867.3⟩	5 (737.3 ± 17.6) ⟨272222.4 ± 42602.9⟩	5 (627.7 ± 19.3) ⟨109462.0 ± 1124.1⟩	5 (623.0 ± 10.8) ⟨108232.1 ± 1333.8⟩
04-12	5 (383.0 ± 32.3) ⟨8252.6 ± 265.0⟩	5 (133.7 ± 2.1) ⟨18245.5 ± 647.2⟩	5 (119.2 ± 4.1) ⟨13651.1 ± 575.5⟩	5 (114.5 ± 3.1) ⟨8066.2 ± 104.7⟩	5 (115.1 ± 2.7) ⟨8383.7 ± 73.6⟩
04-13	5 (1757.0 ± 13.2) ⟨292526.1 ± 684.0⟩				1 (1780.2±) ⟨926136.2±⟩
04-14	5 (423.4 ± 19.8) ⟨4303.8 ± 195.8⟩	5 (64.3 ± 1.7) ⟨4279.6 ± 89.1⟩	5 (68.1 ± 2.3) ⟨3048.4 ± 142.0⟩	5 (65.8 ± 1.4) ⟨2016.7 ± 25.7⟩	5 (66.2 ± 1.6) ⟨2160.1 ± 20.6⟩
04-15	5 (1018.0 ± 20.4) ⟨70213.2 ± 439.7⟩	5 (1577.5 ± 56.8) ⟨288180.1 ± 13346.4⟩	5 (1078.9 ± 112.4) ⟨244175.3 ± 21326.4⟩	5 (907.4 ± 16.8) ⟨121601.7 ± 488.0⟩	5 (935.8 ± 60.5) ⟨122326.7 ± 649.6⟩

results for **LMCut** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
ExBW	04-01	5 (0.0 ± 0.0) (0.1 ± 0.0)	5 (0.0 ± 0.0) (0.4 ± 0.0)	5 (0.0 ± 0.0) (0.2 ± 0.0)	5 (0.0 ± 0.0) (0.1 ± 0.0)
	04-02	5 (0.0 ± 0.0) (0.0 ± 0.0)	5 (0.0 ± 0.0) (0.1 ± 0.0)	5 (0.0 ± 0.0) (0.1 ± 0.0)	5 (0.0 ± 0.0) (0.0 ± 0.0)
	04-03	5 (0.0 ± 0.0) (0.1 ± 0.0)	5 (0.0 ± 0.0) (0.5 ± 0.0)	5 (0.0 ± 0.0) (0.2 ± 0.0)	5 (0.0 ± 0.0) (0.1 ± 0.0)
	04-04	5 (0.1 ± 0.0) (11.4 ± 0.5)	5 (0.2 ± 0.0) (94.1 ± 4.9)	5 (0.1 ± 0.0) (44.0 ± 1.7)	5 (0.1 ± 0.0) (28.5 ± 6.9)
	04-05	5 (0.0 ± 0.0) (0.0 ± 0.0)	5 (0.0 ± 0.0) (0.0 ± 0.0)	5 (0.0 ± 0.0) (0.0 ± 0.0)	5 (0.0 ± 0.0) (0.0 ± 0.0)
	04-06	5 (2.6 ± 0.0) (19.8 ± 3.5)	5 (0.6 ± 0.0) (16.9 ± 0.5)	5 (0.6 ± 0.0) (8.3 ± 0.3)	5 (0.6 ± 0.0) (63.7 ± 0.0)
	04-07	5 (8.1 ± 0.1) (208.8 ± 10.7)	5 (5.3 ± 0.1) (274.6 ± 2.4)	5 (5.2 ± 0.2) (146.2 ± 0.8)	5 (5.3 ± 0.1) (278.1 ± 0.4)
	04-08	5 (1.5 ± 0.5) (9.1 ± 3.7)	5 (0.7 ± 0.0) (10.1 ± 0.4)	5 (0.7 ± 0.0) (5.3 ± 0.2)	5 (0.4 ± 0.0) (2.0 ± 0.1)
	04-09	5 (88.9 ± 4.3) (1687.8 ± 65.4)	5 (111.3 ± 7.5) (15931.6 ± 2356.6)	5 (72.6 ± 0.9) (9999.8 ± 128.3)	5 (64.2 ± 1.1) (3307.2 ± 26.8)
	04-10	5 (597.9 ± 13.9) (46707.1 ± 1413.1)	5 (759.6 ± 22.2) (90100.1 ± 842.6)	5 (277.7 ± 13.7) (51226.2 ± 705.5)	5 (244.3 ± 7.0) (22305.7 ± 210.1)
	08-01	5 (0.0 ± 0.0) (0.4 ± 0.0)	5 (0.0 ± 0.0) (1.1 ± 0.0)	5 (0.0 ± 0.0) (0.5 ± 0.0)	5 (0.0 ± 0.0) (0.3 ± 0.0)
	08-02	5 (0.1 ± 0.0) (30.4 ± 2.8)	5 (0.2 ± 0.0) (94.1 ± 1.9)	5 (0.1 ± 0.0) (46.6 ± 2.5)	5 (0.1 ± 0.0) (14.9 ± 0.0)
	08-03	5 (0.2 ± 0.0) (28.1 ± 3.1)	5 (0.3 ± 0.0) (120.5 ± 23.0)	5 (0.1 ± 0.0) (73.2 ± 0.0)	5 (0.1 ± 0.0) (21.2 ± 1.5)
	08-04	5 (0.2 ± 0.0) (21.8 ± 1.1)	5 (0.4 ± 0.0) (93.9 ± 1.4)	5 (0.1 ± 0.0) (44.4 ± 0.2)	5 (0.1 ± 0.0) (25.1 ± 0.1)
	08-05	5 (0.0 ± 0.0) (0.4 ± 0.3)	5 (0.0 ± 0.0) (0.5 ± 0.0)	5 (0.0 ± 0.0) (0.3 ± 0.0)	5 (0.0 ± 0.0) (0.2 ± 0.0)
	08-06	5 (0.4 ± 0.1) (11.8 ± 2.5)	5 (0.3 ± 0.0) (30.4 ± 0.3)	5 (0.2 ± 0.0) (15.1 ± 0.1)	5 (0.2 ± 0.0) (5.5 ± 0.2)
	08-07	5 (1.1 ± 0.2) (6.4 ± 2.0)	5 (0.5 ± 0.1) (9.3 ± 1.0)	5 (0.4 ± 0.1) (4.8 ± 0.7)	5 (0.3 ± 0.0) (1.8 ± 0.2)
	08-08	5 (169.4 ± 8.1) (2825.7 ± 159.0)	5 (329.9 ± 2.5) (35346.6 ± 136.1)	5 (91.1 ± 4.8) (18292.5 ± 85.8)	5 (85.2 ± 2.9) (8944.0 ± 71.5)
	08-09	5 (71.6 ± 4.5) (1739.8 ± 58.8)	5 (35.9 ± 0.7) (2461.3 ± 93.0)	5 (18.9 ± 0.2) (1874.4 ± 4.2)	5 (16.5 ± 0.8) (946.6 ± 29.2)
	08-10	5 (741.3 ± 17.4) (9799.4 ± 123.5)		5 (428.1 ± 30.2) (207405.7 ± 8482.5)	5 (339.7 ± 14.8) (70581.3 ± 1891.8)
	08-11			1 (1792.6±) (10366.8±)	5 (335.3 ± 13.8) (72276.6 ± 3571.6)

results for **LMCut** continued on next page

		LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
PARC-N	4, 1	5 (19.2 ± 0.4) (925.3 ± 16.1)	5 (22.7 ± 0.4) (4483.3 ± 424.8)	5 (16.8 ± 0.3) (3288.9 ± 156.4)	5 (15.5 ± 0.3) (1609.7 ± 33.8)	5 (15.4 ± 0.2) (1209.5 ± 5.8)
	4, 2	5 (33.8 ± 0.8) (1501.0 ± 16.2)	5 (47.4 ± 2.4) (10520.7 ± 812.6)	5 (28.7 ± 0.5) (6156.1 ± 394.5)	5 (26.4 ± 0.7) (3328.1 ± 53.1)	5 (26.0 ± 0.8) (2271.1 ± 19.5)
	4, 3	5 (53.2 ± 1.4) (2116.4 ± 17.7)	5 (79.5 ± 4.2) (16678.6 ± 1678.1)	5 (48.9 ± 0.6) (10220.4 ± 848.3)	5 (44.9 ± 0.6) (5871.2 ± 77.2)	5 (45.6 ± 1.8) (3716.2 ± 69.6)
	5, 1	5 (263.6 ± 13.7) (11608.8 ± 190.7)	5 (311.1 ± 11.1) (57987.5 ± 3279.3)	5 (235.3 ± 8.6) (44350.4 ± 2212.0)	5 (208.2 ± 5.4) (19432.1 ± 709.3)	5 (207.2 ± 4.2) (13860.8 ± 135.6)
	5, 2	5 (469.9 ± 33.9) (18931.7 ± 320.8)	5 (688.8 ± 35.2) (141891.2 ± 5484.5)	5 (420.3 ± 12.6) (86084.5 ± 3493.3)	5 (367.6 ± 9.2) (43475.1 ± 1268.2)	5 (359.2 ± 9.1) (27722.5 ± 133.8)
	5, 3	5 (749.8 ± 53.8) (26521.3 ± 406.5)	5 (1188.2 ± 68.7) (228332.8 ± 10938.8)	5 (761.1 ± 21.5) (142659.6 ± 7680.7)	5 (675.5 ± 48.1) (75615.7 ± 2301.5)	5 (635.7 ± 17.9) (44140.6 ± 335.0)
PARC-R	4, 1	5 (385.5 ± 18.1) (118983.5 ± 1926.0)	5 (118.9 ± 4.1) (20566.4 ± 298.7)	5 (83.4 ± 1.8) (12066.2 ± 90.1)	5 (82.1 ± 2.3) (9153.8 ± 54.6)	5 (81.9 ± 2.4) (8768.6 ± 16.5)
	4, 2		5 (295.4 ± 9.6) (49469.5 ± 643.6)	5 (221.1 ± 20.6) (32892.8 ± 347.8)	5 (208.4 ± 5.1) (26611.8 ± 79.4)	5 (225.6 ± 21.9) (26155.2 ± 67.1)
	4, 3		5 (406.4 ± 9.9) (66662.3 ± 641.6)	5 (316.2 ± 7.1) (46503.5 ± 1041.8)	5 (327.1 ± 7.8) (38188.8 ± 155.5)	5 (332.6 ± 3.0) (38037.8 ± 157.3)
	5, 1		5 (1584.7 ± 32.7) (210133.0 ± 1969.1)	5 (1245.5 ± 18.1) (144582.1 ± 1411.6)	5 (1219.7 ± 25.4) (105687.8 ± 937.6)	5 (1218.6 ± 21.5) (99770.9 ± 558.6)
Rand	04-03	5 (2.0 ± 0.0) (167.6 ± 14.4)	5 (2.1 ± 0.0) (210.1 ± 4.4)	5 (1.9 ± 0.1) (111.6 ± 3.0)	5 (1.8 ± 0.1) (28.6 ± 0.3)	5 (1.8 ± 0.0) (31.9 ± 0.3)
	04-06	5 (5.5 ± 0.5) (2.3 ± 0.1)	5 (5.5 ± 0.5) (21.9 ± 1.1)	5 (5.8 ± 0.7) (10.4 ± 0.5)	5 (5.5 ± 0.8) (2.4 ± 0.0)	5 (5.4 ± 0.6) (1.3 ± 0.1)
	04-07	5 (0.5 ± 0.5) (1.5 ± 1.5)	5 (0.2 ± 0.0) (1.4 ± 0.1)	5 (0.2 ± 0.1) (0.6 ± 0.1)	5 (0.2 ± 0.0) (0.3 ± 0.1)	5 (0.2 ± 0.1) (0.3 ± 0.1)
	04-13	5 (90.4 ± 55.9) (133.6 ± 92.6)	5 (23.0 ± 0.9) (63.5 ± 0.3)	5 (22.7 ± 1.7) (35.7 ± 0.1)	5 (23.4 ± 1.2) (21.2 ± 0.1)	5 (24.4 ± 1.8) (21.2 ± 0.1)

results for **LMCut** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}	
□TW	08-01	5 (0.0 ± 0.0) ⟨1.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.9 ± 0.5⟩	5 (0.0 ± 0.0) ⟨1.9 ± 0.1⟩	5 (0.0 ± 0.0) ⟨1.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.5 ± 0.1⟩
	08-02	5 (0.0 ± 0.0) ⟨0.5 ± 0.1⟩	5 (0.0 ± 0.0) ⟨1.7 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩
	08-03	5 (0.2 ± 0.0) ⟨2.6 ± 0.1⟩	5 (0.3 ± 0.0) ⟨6.8 ± 0.3⟩	5 (0.2 ± 0.0) ⟨3.8 ± 0.1⟩	5 (0.2 ± 0.0) ⟨3.1 ± 0.1⟩	5 (0.2 ± 0.0) ⟨3.8 ± 0.2⟩
	08-04	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	08-05	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.1 ± 0.0) ⟨0.0 ± 0.0⟩
	08-06	5 (7.2 ± 2.9) ⟨12.1 ± 0.3⟩	5 (15.7 ± 5.5) ⟨39.2 ± 1.9⟩	5 (4.3 ± 0.7) ⟨20.7 ± 1.3⟩	5 (4.3 ± 0.8) ⟨17.8 ± 0.4⟩	5 (4.0 ± 0.4) ⟨21.3 ± 0.4⟩
	08-07	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	08-08	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.2 ± 0.0) ⟨0.0 ± 0.0⟩
	08-09	5 (162.2 ± 31.9) ⟨83.8 ± 0.3⟩	5 (407.1 ± 80.0) ⟨283.8 ± 7.9⟩	5 (39.3 ± 4.1) ⟨88.7 ± 0.3⟩	5 (33.2 ± 1.3) ⟨53.5 ± 0.5⟩	5 (35.8 ± 4.8) ⟨74.7 ± 4.4⟩
	08-10	5 (425.6 ± 189.6) ⟨306.3 ± 0.4⟩	5 (1157.8 ± 417.2) ⟨4357.6 ± 643.7⟩	5 (7.7 ± 0.9) ⟨377.6 ± 64.7⟩	5 (7.8 ± 1.2) ⟨285.6 ± 0.0⟩	5 (8.2 ± 2.5) ⟨305.1 ± 34.6⟩
	08-11	4 (868.9 ± 43.5) ⟨86.7 ± 0.4⟩		5 (366.2 ± 41.1) ⟨159.6 ± 1.8⟩	5 (444.1 ± 214.6) ⟨138.1 ± 1.0⟩	5 (333.5 ± 54.7) ⟨154.5 ± 1.3⟩
	08-12			5 (319.2 ± 24.4) ⟨44608.9 ± 853.2⟩	5 (331.9 ± 86.6) ⟨40969.9 ± 542.7⟩	5 (305.3 ± 33.1) ⟨42206.2 ± 315.7⟩
	08-14	5 (13.7 ± 0.3) ⟨0.0 ± 0.0⟩	5 (14.8 ± 0.7) ⟨0.0 ± 0.0⟩	5 (14.1 ± 0.6) ⟨0.0 ± 0.0⟩	5 (15.8 ± 3.3) ⟨0.0 ± 0.0⟩	5 (13.9 ± 0.5) ⟨0.0 ± 0.0⟩
	S&R	08-01	5 (0.1 ± 0.0) ⟨208.8 ± 1.8⟩	5 (0.4 ± 0.0) ⟨602.4 ± 9.3⟩	5 (0.2 ± 0.0) ⟨429.8 ± 5.1⟩	5 (0.2 ± 0.0) ⟨401.9 ± 1.0⟩
08-02		5 (0.7 ± 0.0) ⟨970.1 ± 11.5⟩	5 (1.7 ± 0.1) ⟨2480.3 ± 28.4⟩	5 (1.0 ± 0.0) ⟨1873.0 ± 16.2⟩	5 (1.1 ± 0.1) ⟨1773.9 ± 4.4⟩	5 (1.0 ± 0.0) ⟨1768.8 ± 3.9⟩
08-03		5 (3.0 ± 0.1) ⟨4251.7 ± 43.3⟩	5 (7.0 ± 0.3) ⟨9732.0 ± 18.1⟩	5 (4.1 ± 0.2) ⟨7646.6 ± 57.3⟩	5 (4.3 ± 0.3) ⟨7374.6 ± 10.3⟩	5 (4.3 ± 0.2) ⟨7340.2 ± 10.1⟩
08-04		5 (60.8 ± 0.9) ⟨76491.4 ± 674.0⟩	5 (119.2 ± 1.5) ⟨140710.5 ± 623.2⟩	5 (80.6 ± 10.4) ⟨116715.0 ± 578.6⟩	5 (83.7 ± 2.2) ⟨113816.3 ± 30.9⟩	5 (108.4 ± 36.8) ⟨113537.7 ± 45.0⟩
08-05		5 (1250.3 ± 10.2) ⟨1328128.5 ± 3188.8⟩		5 (1433.8 ± 53.0) ⟨1615243.8 ± 7191.8⟩	4 (1713.5 ± 33.7) ⟨1593746.8 ± 335.4⟩	

results for **LMCut** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}	
Sched	04-01	5 (0.0 ± 0.0) ⟨7.0 ± 0.2⟩	5 (0.0 ± 0.0) ⟨10.4 ± 1.4⟩	5 (0.0 ± 0.0) ⟨1.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.8 ± 0.0⟩	
	04-02	5 (0.0 ± 0.0) ⟨10.9 ± 0.3⟩	5 (0.0 ± 0.0) ⟨17.7 ± 2.4⟩	5 (0.0 ± 0.0) ⟨2.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.1 ± 0.0⟩	
	04-03	5 (0.1 ± 0.0) ⟨33.2 ± 0.6⟩	5 (0.1 ± 0.0) ⟨56.2 ± 5.6⟩	5 (0.0 ± 0.0) ⟨4.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.3 ± 0.0⟩	
	04-04	5 (31.7 ± 0.2) ⟨8394.6 ± 37.7⟩	5 (56.7 ± 1.3) ⟨25994.1 ± 786.2⟩	5 (25.1 ± 0.4) ⟨25398.0 ± 883.8⟩	5 (18.7 ± 2.5) ⟨9565.0 ± 163.9⟩	5 (18.2 ± 1.1) ⟨8396.7 ± 183.9⟩
	04-05				5 (1590.1 ± 101.6) ⟨440820.6 ± 717.5⟩	5 (1558.6 ± 104.4) ⟨383534.0 ± 1971.1⟩
	08-01	5 (0.0 ± 0.0) ⟨11.1 ± 0.2⟩	5 (0.0 ± 0.0) ⟨16.2 ± 1.5⟩	5 (0.0 ± 0.0) ⟨1.6 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.3 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.9 ± 0.0⟩
	08-02	5 (0.0 ± 0.0) ⟨17.4 ± 0.3⟩	5 (0.1 ± 0.0) ⟨27.7 ± 2.7⟩	5 (0.0 ± 0.0) ⟨2.4 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.9 ± 0.0⟩	5 (0.0 ± 0.0) ⟨1.3 ± 0.0⟩
	08-03	5 (0.1 ± 0.0) ⟨48.0 ± 1.2⟩	5 (0.2 ± 0.0) ⟨80.2 ± 3.3⟩	5 (0.0 ± 0.0) ⟨5.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨4.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨2.5 ± 0.0⟩
	08-04	5 (1494.9 ± 34.8) ⟨236185.9 ± 2145.7⟩		5 (1221.3 ± 64.8) ⟨577721.4 ± 14311.7⟩	5 (757.3 ± 16.6) ⟨211614.3 ± 1062.1⟩	5 (794.6 ± 29.2) ⟨181461.8 ± 1424.1⟩
	Sys	08-01	5 (0.1 ± 0.0) ⟨104.8 ± 4.0⟩	5 (0.1 ± 0.0) ⟨57.9 ± 0.8⟩	5 (0.0 ± 0.0) ⟨18.7 ± 0.1⟩	5 (0.0 ± 0.0) ⟨19.2 ± 0.4⟩
08-02		5 (1.0 ± 0.0) ⟨740.5 ± 9.8⟩	5 (0.6 ± 0.1) ⟨333.2 ± 59.0⟩	5 (0.1 ± 0.0) ⟨106.1 ± 0.7⟩	5 (0.2 ± 0.0) ⟨110.3 ± 1.0⟩	5 (0.1 ± 0.0) ⟨110.7 ± 1.9⟩
08-03		5 (4.6 ± 0.0) ⟨3140.8 ± 58.7⟩	5 (2.7 ± 0.5) ⟨1430.7 ± 255.3⟩	5 (0.6 ± 0.0) ⟨397.6 ± 2.6⟩	5 (0.6 ± 0.0) ⟨412.1 ± 3.8⟩	5 (0.7 ± 0.0) ⟨424.8 ± 5.6⟩
08-04		5 (101.1 ± 4.2) ⟨58318.0 ± 1383.3⟩	5 (60.0 ± 9.6) ⟨23327.9 ± 2572.4⟩	5 (8.7 ± 0.5) ⟨4999.2 ± 24.9⟩	5 (9.0 ± 0.6) ⟨5085.1 ± 34.7⟩	5 (8.8 ± 0.4) ⟨5173.8 ± 28.8⟩
ΔTW	08-01	5 (0.0 ± 0.0) ⟨0.1 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.5 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.2 ± 0.0⟩
	08-02	5 (0.0 ± 0.0) ⟨5.2 ± 0.3⟩	5 (0.0 ± 0.0) ⟨5.6 ± 0.2⟩	5 (0.0 ± 0.0) ⟨3.0 ± 0.1⟩	5 (0.0 ± 0.0) ⟨2.6 ± 0.2⟩	5 (0.0 ± 0.0) ⟨2.5 ± 0.1⟩
	08-03	5 (1.0 ± 0.0) ⟨129.2 ± 2.1⟩	5 (0.7 ± 0.0) ⟨146.5 ± 2.5⟩	5 (0.5 ± 0.0) ⟨77.6 ± 1.9⟩	5 (0.5 ± 0.0) ⟨70.8 ± 1.3⟩	5 (0.5 ± 0.0) ⟨84.5 ± 3.8⟩
	08-04	5 (30.1 ± 0.4) ⟨3040.9 ± 34.9⟩	5 (29.7 ± 0.8) ⟨4488.3 ± 56.1⟩	5 (17.2 ± 1.4) ⟨2227.7 ± 11.3⟩	5 (15.9 ± 0.6) ⟨2106.3 ± 30.8⟩	5 (17.0 ± 1.0) ⟨2318.3 ± 73.8⟩
	08-05	5 (909.6 ± 32.6) ⟨67492.8 ± 599.6⟩	5 (1221.0 ± 25.3) ⟨129550.1 ± 1968.0⟩	5 (560.7 ± 21.3) ⟨62669.5 ± 1087.0⟩	5 (554.4 ± 14.6) ⟨58626.0 ± 514.2⟩	5 (547.3 ± 9.0) ⟨59966.7 ± 1387.6⟩
6, 8			5 (1079.1 ± 35.6) ⟨95912.4 ± 965.9⟩	5 (1059.0 ± 20.5) ⟨91451.9 ± 2297.3⟩	5 (1112.7 ± 41.4) ⟨95547.7 ± 3063.1⟩	

results for **LMCut** continued on next page

	LRTDP	iLAO*	CG-iLAO* _{all}	CG-iLAO* _{tied}	CG-iLAO* _{single}
Zeno	04-01	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩	5 (0.0 ± 0.0) ⟨0.0 ± 0.0⟩
	04-02		5 (1378.9 ± 41.3) ⟨72394.3 ± 155.2⟩	5 (1371.0 ± 75.4) ⟨48355.5 ± 122.8⟩	5 (1201.3 ± 83.9) ⟨47914.9 ± 290.1⟩
	08-01	5 (309.3 ± 1.0) ⟨9665.2 ± 163.8⟩	5 (28.0 ± 0.2) ⟨2928.0 ± 6.9⟩	5 (26.9 ± 0.8) ⟨1837.2 ± 8.2⟩	5 (24.3 ± 0.6) ⟨1941.0 ± 11.9⟩
	08-02	5 (220.0 ± 5.0) ⟨2778.7 ± 51.3⟩	5 (22.7 ± 0.1) ⟨1520.7 ± 8.6⟩	5 (24.3 ± 0.3) ⟨989.7 ± 8.7⟩	5 (22.4 ± 0.4) ⟨1086.1 ± 17.7⟩
	08-04		5 (640.6 ± 22.9) ⟨34651.4 ± 369.3⟩	5 (683.6 ± 18.2) ⟨22103.7 ± 170.7⟩	5 (654.5 ± 39.5) ⟨23235.9 ± 247.3⟩

Table 6: Results for algorithms using **LMCut**. We give $X(Y)\langle Z \rangle$ where X is the coverage, Y is the mean time (secs), and Z is the mean number of *thousands* of Q -values. Each mean is taken over the converged instances, and we report the 95% C.I.